

LINEAR REGRESSION AND CORRELATION



A BEGINNER'S GUIDE

SCOTT HARTSHORN

Linear Regression And Correlation

A Beginner's Guide

By Scott Hartshorn

What Is In This Book

Thank you for getting this book! This book contains examples of how to do linear regression in order to turn a scatter plot of data into a single equation. It is intended to be direct and to give easy to follow example problems that you can duplicate. In addition to information about simple linear regression, this book contains a worked example for these types of problems

- Multiple Linear Regression – How to do regression with more than one variable
- Exponential Regression – Regression where the data is increasing at a faster rate, such as Moore's law predicts for computer chips
- R-Squared and Adjusted R-Squared – A metric for determining how good your regression was
- Correlation – A way of determining how much two sets of data change together, which has used in investments

Every example has been worked by hand showing the appropriate equations. There is no reliance on a software package to do the solutions, even for the more complicated parts such as multiple regression. This book shows how everything is done in a way you can duplicate. Additionally all the examples have all been solved using those equations in an Excel spreadsheet that you can download for free.

If you want to help us produce more material like this, then please leave a [positive review for this book on Amazon](#). It really does make a difference!

If you spot any errors in this book, think of topics that we should include, or have any suggestions for future books then I would love to hear from you.

Please email me at **ImFairlyNerdy (at) gmail (dot) com**

~ Scott Hartshorn

Your Free Gift

As a way of saying thank you for your purchase, I'm offering this free Linear Regression cheat sheet that's exclusive to my readers.

This cheat sheet contains all the equations required to do linear regression, with explanations of how they work. This is a PDF document that I encourage you to print, save, and share. [You can download it by going here](http://www.fairlynerdy.com/linear-regression-cheat-sheet/)

Linear Regression

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With linear regression, you are trying to reduce a set of data into a line. The equation for a line is $y = a + bx$

a is the Intercept, b is the Slope

The first thing to calculate is the slope of the regression line. That value is the correlation of the two data sets, multiplied by the ratio of their standard deviations. Note that those are sample standard deviations, not population

$b = r * \frac{s_y}{s_x}$

r is Pearson's Correlation, s_y is Sample Standard Deviation of y, s_x is Sample Standard Deviation of x

Correlation shows how much two sets of data change together. Correlation is always between -1 and 1, and is unit-less. Correlation is frequently around the average x, average y, but if you want to force the line through a specific point, you can use that point instead of average x, y in all the

$r = \frac{\sum((x - \bar{x}) * (y - \bar{y}))}{(n - 1) * s_x * s_y}$

$\sum((x - \bar{x}) * (y - \bar{y}))$ is Sum Over All Data Points of x & y values of each point minus x & y mean values

$(n - 1)$ is # of Data Points, s_x is Standard Deviation of x, s_y is Standard Deviation of y

$s_x = \sqrt{\frac{\sum (x - \bar{x})^2}{(n - 1)}}$ $s_y = \sqrt{\frac{\sum (y - \bar{y})^2}{(n - 1)}}$

The sample standard deviation of x and y measure how spread out the x and y values are around their mean. They have units

Once you have done those calculations you have the slope. With the slope and a point the line goes through you can calculate the intercept. If you used the average x, y before use it here again. Otherwise use the same x, y that you specified before

$a = \bar{y} - b\bar{x}$

$\bar{y}$ is Y & X Average, b is Slope

When you have done the regression, one way of evaluating it's quality is R-squared. R-squared is a measure of the summed squared error in the regression vs the error if you didn't do a regression. Summed Squared error is shown below

$R^2 = 1 - \frac{SS_{Regression}}{SS_{Total}}$

$SS_{Regression}$ is Sum Squared Regression Error, SS_{Total} is Sum Squared Total Error

$SS_{Total} = \sum (y_i - \bar{y})^2$

$\sum (y_i - \bar{y})^2$ is Sum Over All Data Points of Each Data Point minus Mean Value, squared

$SS_{Regression} = \sum (y_i - y_{Regression})^2$

$\sum (y_i - y_{Regression})^2$ is Sum Over All Data Points of Each Data Point minus Regression Value, squared

<http://www.fairlynerdy.com/linear-regression-cheat-sheet/>

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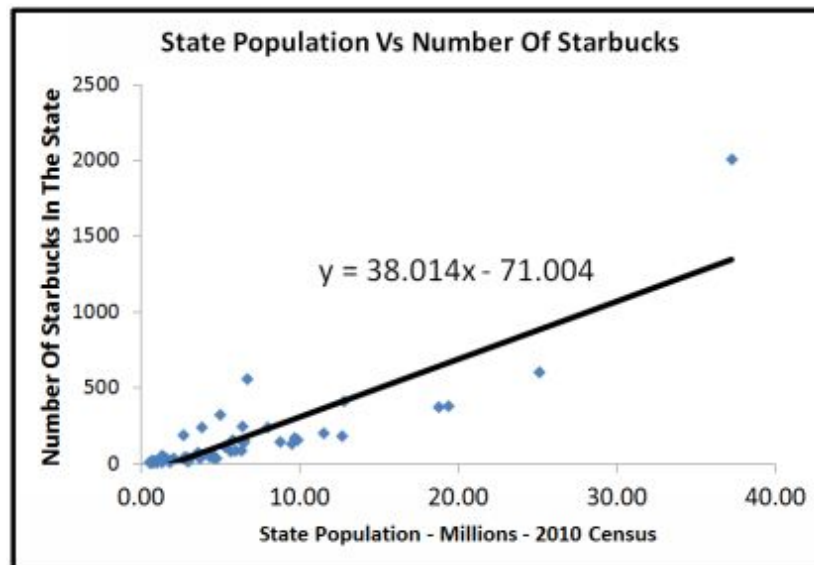
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Regression and Correlation Overview

This book covers linear regression. In doing so, this book covers several other necessary topics in order to understand linear regression. Those topics include correlation, as well as the most common regression metric, R^2 .

Linear regression is a way of predicting an unknown variable using results that you do know. If you have a set of x and y values, you can use a regression equation to make a straight line relating the x and y . The reason you might want to do this is if you know some information, and want to estimate other information. For instance, you might have measured the fuel economy in your car when you were driving 30 miles per hour, when you were driving 40 miles per hour, and when you were driving 75 miles per hour. Now you are planning a cross country road trip and plan to average 60 miles per hour, and want to estimate what fuel economy you will have so that you can budget how much money you will need for gas.

The chart below shows an example of linear regression using real world data. It shows the relationship between the population of states within the United States, and the number of Starbucks (a coffee chain restaurant) within that state.



Likely this is information that is useful to no one. However, the result of the regression equation is that I can predict the number of Starbucks within a state by taking the population (in millions), multiplying it by 38.014, and

subtracting 71.004. So if I had a state with 10 million people, I would predict it had $(10 * 38.014 - 71.004 = 309.1)$ just over 309 Starbucks within the state.

This is a book on linear regression, which means the result of the regression will be a line when we have two variables, or a plane with 3 variables, or a hyperplane with more variables. The above chart was generated in Excel, which can do linear regression. This book shows how to do the regression analysis manually, and more importantly dives deeply into understanding exactly what is happening when the regression analysis is done.

The final result of the regression process was the equation for a line. The equation for a line has this form

The diagram shows the equation $y = a + bx$. Above the equation, the word "Constants" is written in red, with two red arrows pointing down to the terms a and b . Below the equation, the word "Variables" is written in red, with two red arrows pointing up to the terms y and x .

$$y = a + bx$$

Multiple linear regression is linear regression when you have more than one independent variable. If you have a single independent variable, the regression equation forms a line. With two independent variables, it forms a plane. With more than two independent variables, the regression equation forms a hyperplane. The resulting equation for multiple regression is

$$y = a + b_1x_1 + b_2x_2 + b_3x_3 + \dots$$

One metric for measuring how good of a prediction was made is R^2 . This metric measures how much error remains in your prediction after you did the regression vs. how much error you had if you did no regression.

Correlation is nearly the same as linear regression. Correlation is a measure of how much two variables change together. A high value of correlation (high magnitude) will result in a regression line that is a good prediction of

the data. A low correlation value (near zero) will result in a poor linear regression line.

This book covers all of the above topics in detail. The order that they are covered in is

- R^2
- Correlation
- Linear Regression
- Multiple Linear Regression

The reason they are covered in that order, instead of skipping straight to linear regression, is that R^2 builds up some information that is useful to know for correlation. And then correlation is 80% of what you need to know to understand simple linear regression.

Initially, it appears that multiple linear regression is a more challenging topic. However, as it turns out, you can solve multiple regression problems just by doing simple linear regression multiple times. This method for multiple regression isn't the best in terms of number of steps you need to take for very large problems, but the process of repeated simple linear regression is great for understanding how multiple regression works, and that is what is covered in this book.

Get The Data

There are a number of examples shown in this book. All of the examples were generated in Excel. If you want to get the data used in these examples, or if you want to see the equations themselves in action, you can download the Excel file with all the examples for free here

<http://www.fairlynerdy.com/linear-regression-examples/>

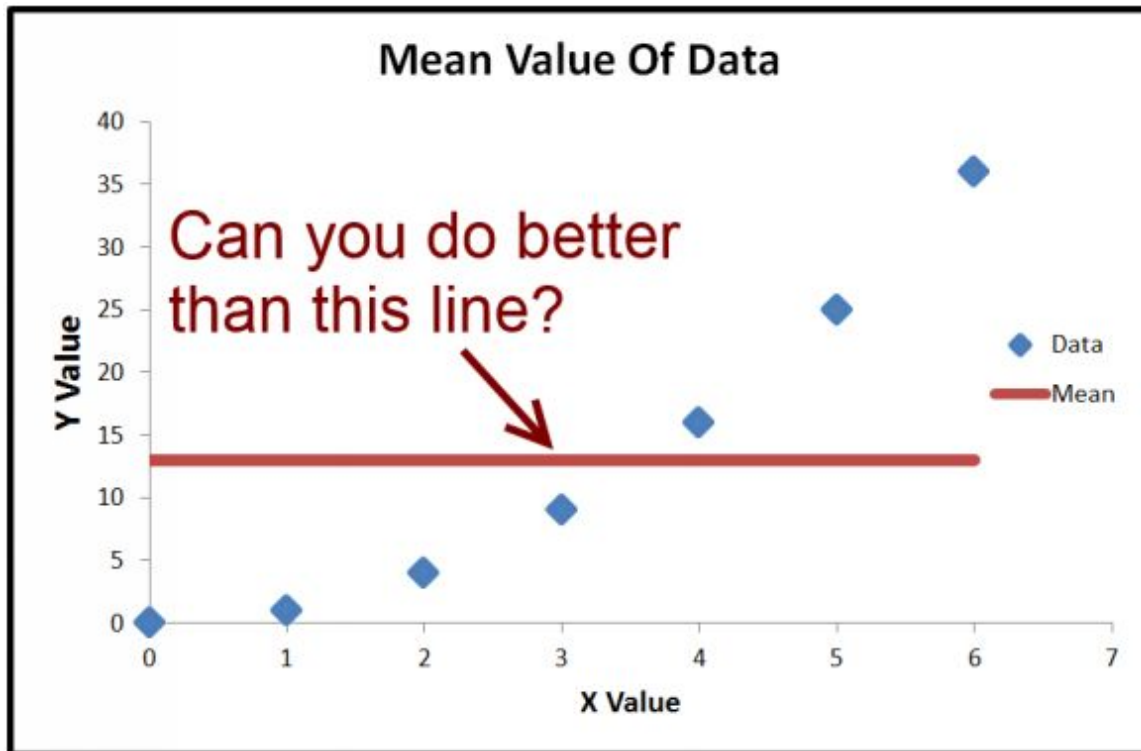
R-Squared – A Way Of Evaluating Regression

Regression is a way of fitting a function to a set of data. For instance, maybe you have been using satellites to [count the number of cars in the parking lot of Walmart](#) stores for the past couple of years. You also know the quarterly sales that Walmart had during that time frame from their earnings report. You want to find a function that relates the two so that you can use your satellites to count the number of cars and predict Walmart's quarterly earnings. (In order to get an advantage in the stock market) In order to generate that function, you can use regression analysis. But after you generate the car to profit relationship function, how can you tell if the quality of the model is good or bad? After all, if you are using that model to try to predict the stock market, you will be betting real money on it. You need to know, is your model a good fit? A bad fit? Mediocre? One commonly used metric for determining the goodness of fit is R^2 .

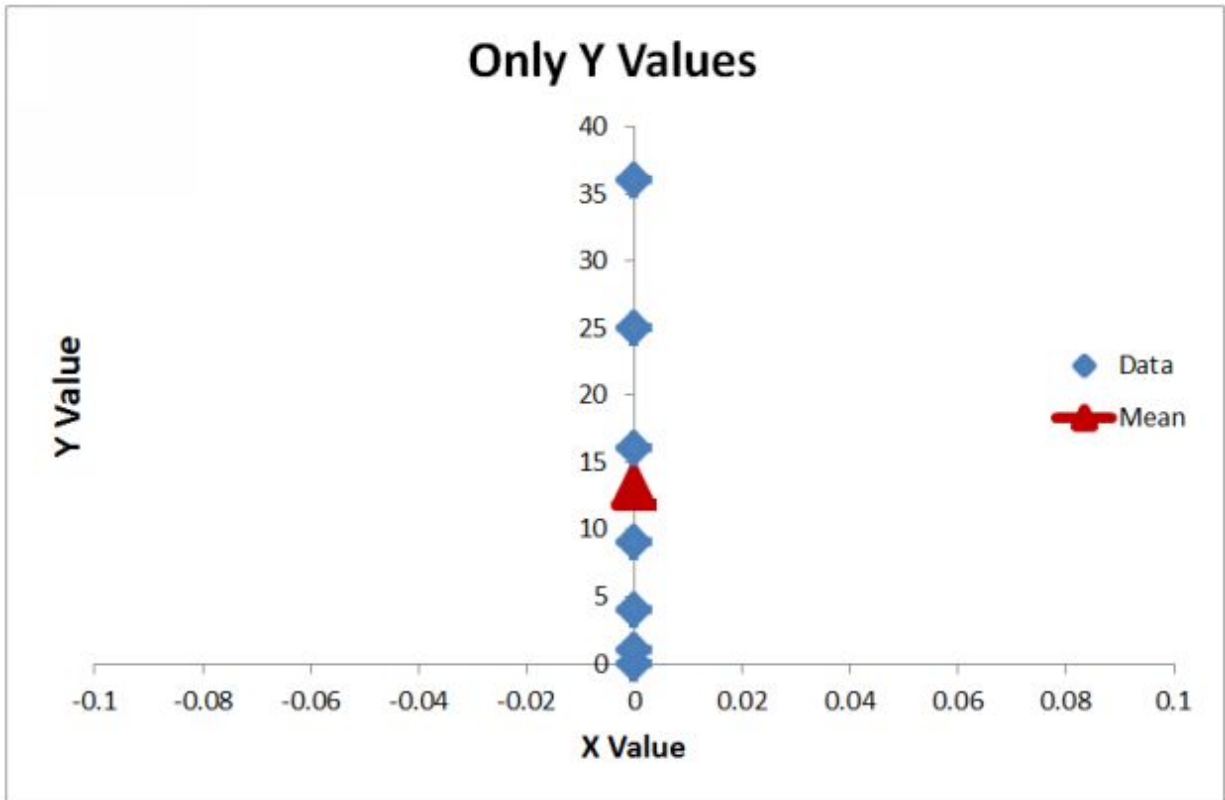
This section goes over R^2 , and by the end, you will understand what it is and how to calculate it, but unfortunately, you won't have a good rule of thumb for what R^2 value is good enough for your analysis because it is entirely problem dependent.

What Is R Squared?

We will get into the equation for R^2 in a little bit, but first what is R^2 ? Simply put, it is how much better your regression line is than a simple horizontal line through the mean of the data. In the plot below the blue dots are the data that we are trying to generate a regression on and the horizontal red line is the average of that data.



The red line, located at the average of all the data points, is the value that gives the lowest summed squared error to the blue data points, assuming you had no other information about the blue data points other than their y value. This is shown in the plot below. In that chart, only the y values of the data points are available. You don't know anything else about those values.



If you want to select a value that gives you the lowest summed squared error, the value that you would select is the mean value, shown as the red triangle. A different way to think about that assertion is this: if I took all 7 of the y points (0, 1, 4, 9, 16, 25, and 36) and randomly selected one of those points from the set (with replacement) and made you repeatedly guess a value for what I drew, what strategy would give you the minimum sum squared error? That strategy is to guess the mean value for all the points.

With regression, the question is now that you have more information (the X values in this case) can you make a better approximation than just guessing the mean value? And the R^2 value answers the question, how much better did you do?

That is actually a pretty intuitive understanding. First calculate how much error you would have if you don't even try to do regression, and instead just guess the mean of all the values. That is the total error. It could be low if all the data is clustered together, or it could be high if the data is spread out.

The next step is to calculate your sum squared error after you do the regression. It will likely be the case that not all of the data points lay exactly

on the regression line, so there will be some residual error. Square the error for each data point, sum them, and that is the regression error.

The less regression error there is remaining relative to the initial total error, the higher the resulting R^2 will be.

The equation for R^2 is shown below.

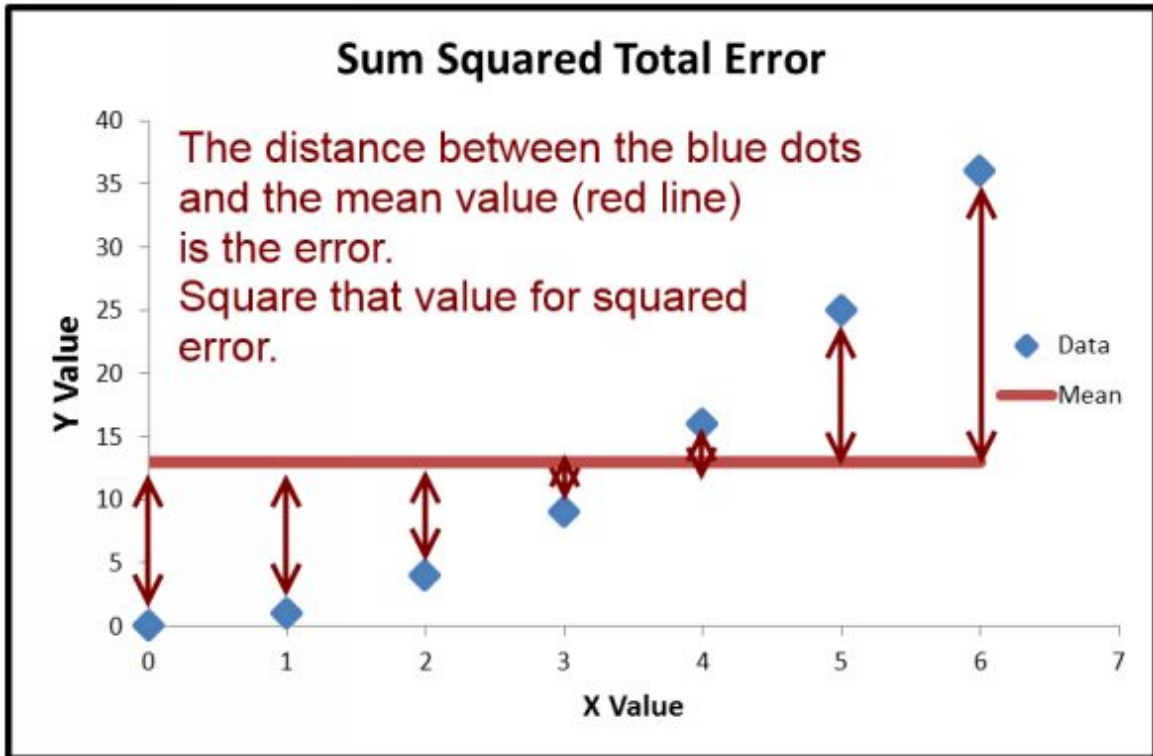
$$R^2 = 1 - \frac{SS_{Regression}}{SS_{Total}}$$

Sum Squared Regression Error

Sum Squared Total Error

SS stands for summed squared error, which is how the error is calculated. To get the total sum squared error you

- Start with the mean value
- For every data point subtract that mean value from the data point value
- Square that difference
- Add up all of the squares. This results in summed squared error



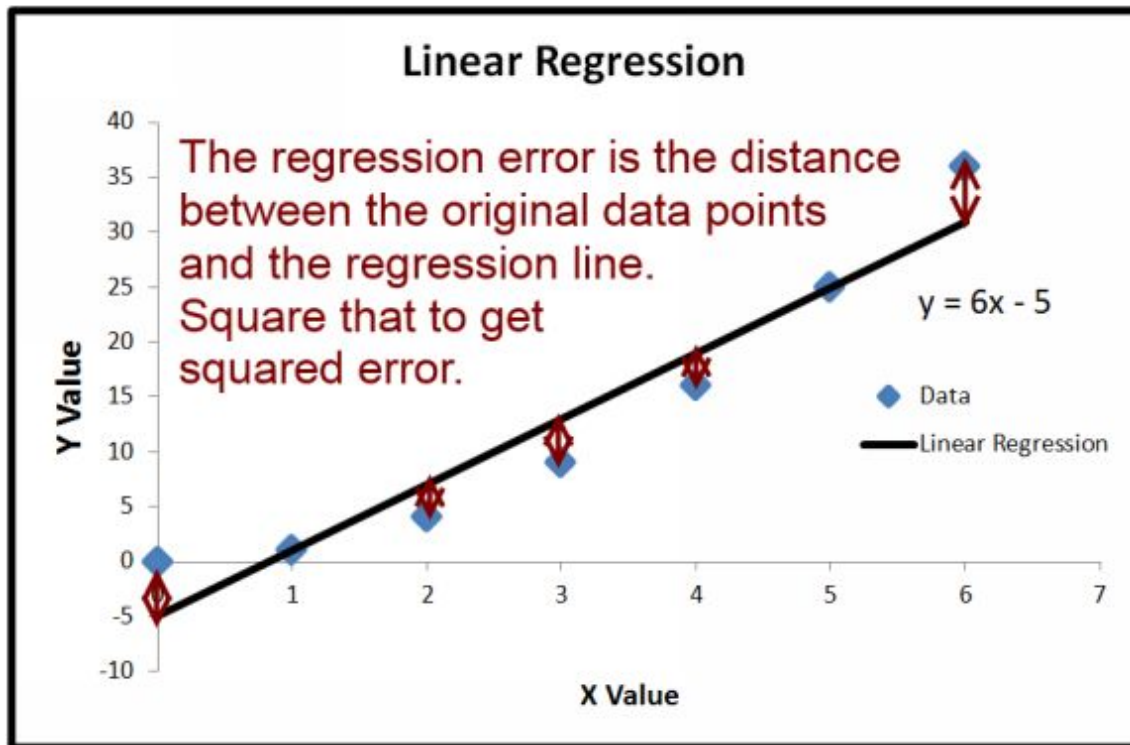
As an equation, the sum squared total error is

$$SS_{Total} = \sum (y_i - \bar{y})^2$$

Annotations for the equation:

- SS_{Total} : Sum Squared Total Error
- $\sum$: Sum Over All The Data Points
- y_i : Each Data Point
- $\bar{y}$: Mean Value
- 2 : Square The Result

Calculate The Regression Error Next, calculate the error in your regression values against the true values. This is your regression error. Ideally, the regression error is very low, near zero.



For the sum squared regression error, the equation is the same except you use the regression prediction instead of the mean value

Sum Over All The Data Points

Square The Result

$$SS_{Regression} = \sum (y_i - y_{Regression})^2$$

Sum Squared Regression Error

Each Data Point

Regression Value

The ratio of the regression error against the total error tells you how much of the total error remains in your regression model. Subtracting that ratio from 1.0 gives how much error you removed using the regression analysis. That is R^2

What is a Good R-Squared Value?

In most statistics books, you will see that an R^2 value is always between 0 and 1, and that the best value is 1.0. That is only partially true. The lower the error in your regression analysis relative to total error, the higher the R^2 value will be. The best R^2 value is 1.0. To get that value, you have to have zero error in your regression analysis.

$$R^2 = 1 - \frac{\overset{\text{Zero Regression Error}}{\rightarrow 0}}{SS_{Total}} \rightarrow 1.0$$

However, R^2 is not truly limited to a lower bound of zero.

For practical purposes, the lowest R^2 you can get is zero, but only because the assumption is that if your regression line is not better than using the mean, then you will just use the mean value.

Theoretically, however, you could use something else. Let's say that you wanted to make a prediction on the population of one of the states in the United States. I am not giving you any information other than the population of all 50 states, [based on the 2010 census](#). I.e. I am not telling you the name of the state you are trying to make the prediction on, you just have to guess the population (in millions) of all the states in a random order. The best you could do here is to take the mean value. Your total squared error would be 2298.2 (The calculation for this error can be found in this free Excel file <http://www.fairlynerdy.com/linear-regression-examples/>) The best you could do would be the mean value. However, you could make a different choice and do worse. For instance, if you used the median value instead of the mean, the summed squared error would be 2447.2

Which when converted into R^2 is

$$R^2 = 1 - \frac{\text{Squared Error Using Median}}{\text{Squared Error Using Mean}} = -.0648$$

And you get a negative R^2 number. The assertion that the R^2 value has to be greater than or equal to zero is based on the assumption that if you get a negative R^2 value, you will discard whatever regression calculation you are using and just go with the mean value.

The takeaway for R^2 is

- An R^2 of 1.0 is the best. It means you have zero error in your regression.
- An R^2 of 0 means your regression is no better than taking the mean value, i.e. you are not using any information from the other variables
- A negative R^2 means you are doing worse than the mean value. However, maybe summed squared error isn't the metric that matters most to you and this is OK. (for instance, maybe you care most about mean absolute error instead)

As for what is a good R^2 value, it is too problem dependent to say. A useful regression analysis is one that explains information that you didn't know before. That could be a very low R^2 for regression on social or personal economic data or a high R^2 for highly controlled engineering data.

R Squared Example

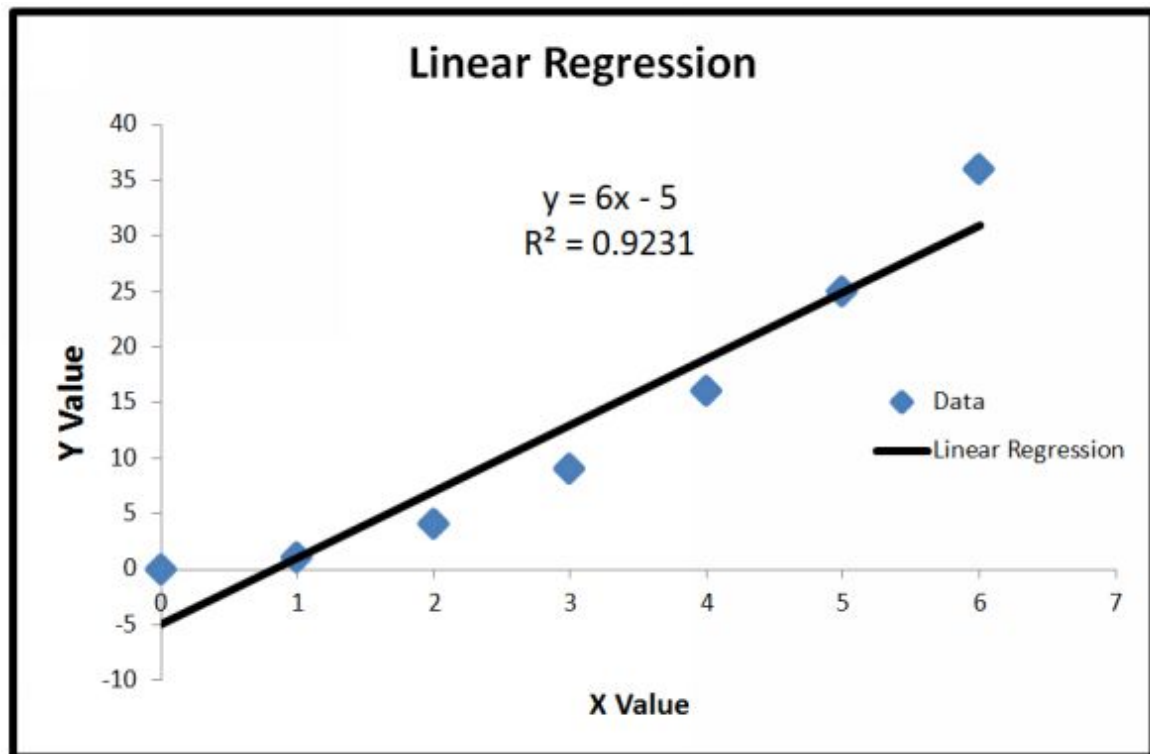
As an example of how to calculate R^2 , let's look at this data

X Value	Y Value = X^2
0	0
1	1
2	4
3	9
4	16
5	25
6	36

This data is just the numbers 0 through 6, with the y value being the square of those numbers. The linear regression equation for this data is

$$y = 6x - 5$$

and is plotted on the graph below



Excel has calculated the R^2 of this equation to be .9231. How can we duplicate that manually?

Well the equation is

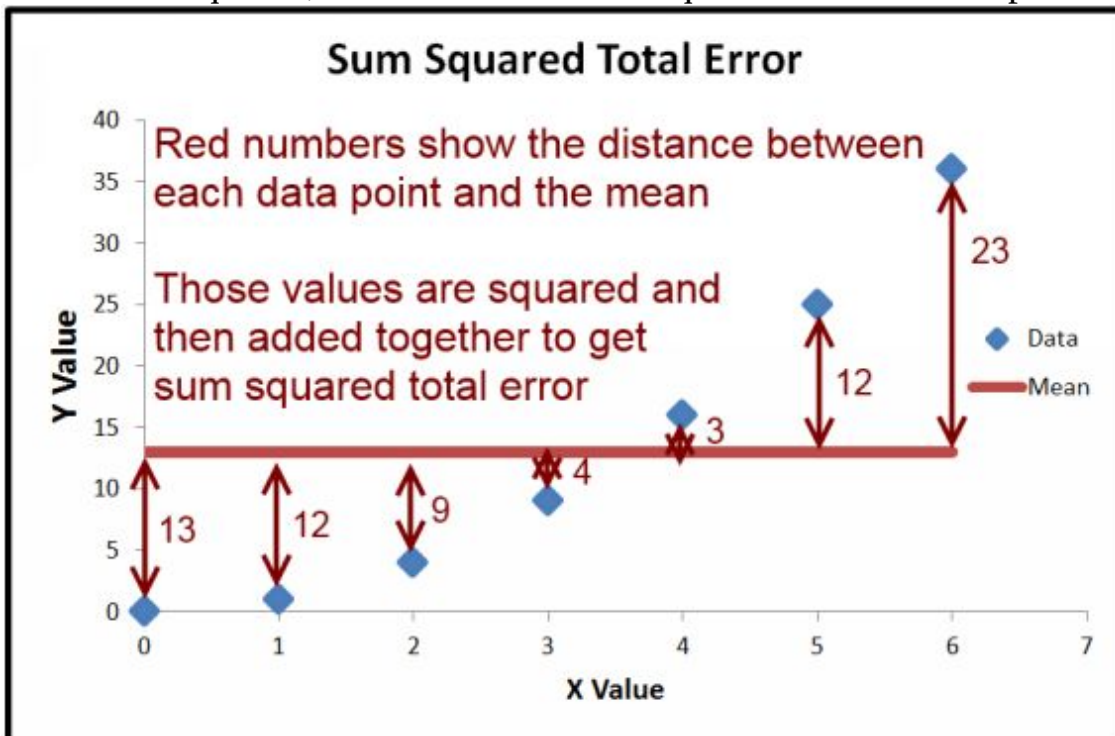
$$R^2 = 1 - \frac{SS_{Regression}}{SS_{Total}}$$

So we need to find the total summed squared error (based on the mean) and the summed squared error based on the regression line.

The mean value of the y values of the data (0, 1, 4, 9, 16, 25, and 36) is 13

X Value	Y Value = X^2
0	0
1	1
2	4
3	9
4	16
5	25
6	36
Mean	13

To find the total summed square error, we will subtract 13 from each of the y values, square that result, and add up all of the squares. Graphically, this is shown below. At every data point, the distance between the red line and the blue line is squared, and then all of those squares are summed up



The total sum squared error is 1092, with most of the error coming from the edges of the chart where the mean is the farthest way from the true value

X Value	Y Value = X^2	Sum Squared Total (Y - Mean) ²
0	0	169
1	1	144
2	4	81
3	9	16
4	16	9
5	25	144
6	36	529
Mean	13	
Total		1092

Now we need to find the values that our regression line of $y = 6x - 5$ predicts, and get the summed squared error of that. For the sum squared value, we will subtract each y regression value from the true value, take the square, and sum up all of the squares

X Value	Y Value = X^2	Sum Squared Total (Y - Mean) ²		Regression Prediction $Y = 6x - 5$	Regression Error Squared
0	0	169		-5	25
1	1	144		1	0
2	4	81		7	9
3	9	16		13	16
4	16	9		19	9
5	25	144		25	0
6	36	529		31	25
Mean	13				
Total		1092			84.0

So the total summed squared error of the linear regression is 84, and the total summed squared error is 1092 based on the mean value.

Plugging these numbers into the R^2 equation, we get

$$R^2 = 1 - \frac{84}{1092} = .9231$$

This is the same value that Excel calculated.

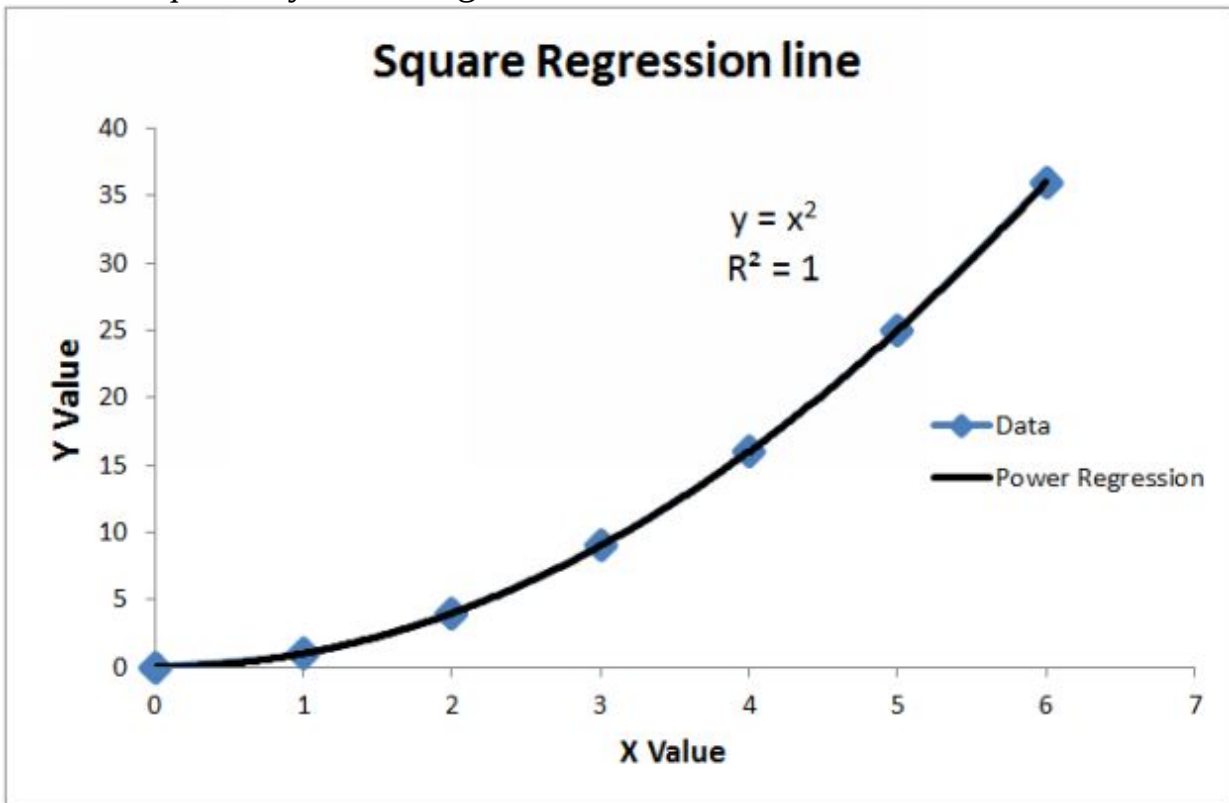
A different way to think about the same result would be that we have $84/1092 = 7.69\%$ of the total error remaining. Basically, if someone had just given us the y values, and then told us that they were randomly ordering those y values and we had to guess what they all were, the best guess we could have made was the mean for each one. But if now they give us the x value, and tell us to try to guess the Y value, we can use the linear regression line and remove 92.31% of the error from our guess.

But Wait, Can't We Do Better?

We just showed a linear regression line that produced an R^2 value of .9231 and said that that was the best linear fit we could make based on the summed squared error metric. But couldn't we do better with a different regression fit?

Well, the answer is yes, of course, we could. We used a linear regression, i.e. a straight line, on this data. However, the data itself wasn't linear. Y is the square of x with this data. So if we used a square regression, and in fact just

used the equation $y = x^2$, we get a much better fit which is shown below

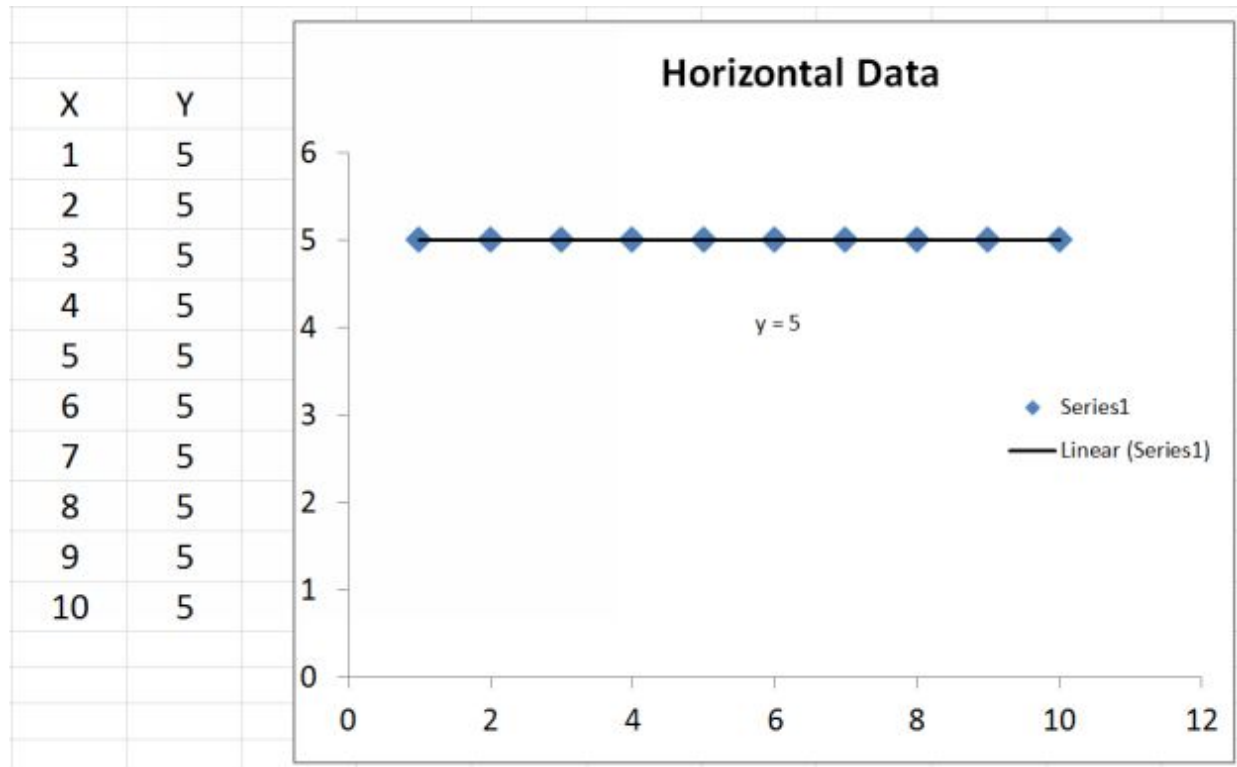


Here we have an R^2 of 1.0, because the regression line exactly matches the data, and there is no remaining error. However, the fact that we were able to do this is somewhat beside the point for this R^2 explanation. We were able to find an exact match for this data only because it is a toy data set. The Y values were built as the square of the X values, so it is no surprise that making a regression that utilized that fact gave a good match.

For most data sets, an exact match will not be able to be generated because the data will be noisy and not a simple equation. For instance, an economist might be doing a study to determine what attributes of a person correlate to their adult profession and income. Some of those attributes could be height, childhood interests, parent's incomes, school grades, SAT scores, etc. It is unlikely that any of those will have a perfect R^2 value; in fact, the R^2 of some of them might be quite low. But there are times that even a low R^2 value could be of interest. Any R^2 value above 0.0 indicates that there could be some correlation between the variable and the result, although very low values are likely just random noise.

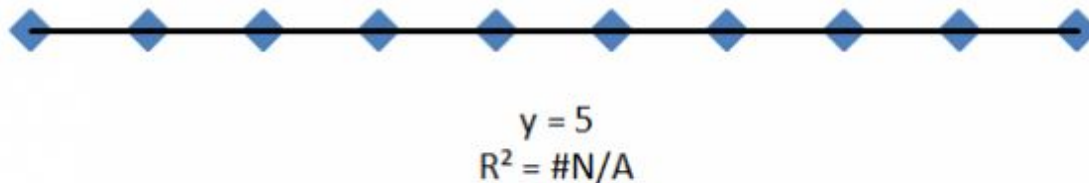
An Odd Special Case For R^2

Just for fun, what do you think the R^2 of this linear regression line for this data is?



Here we have a purely horizontal line; all the data is 5.0. The regression line perfectly fits the data, which means we should have an R^2 of 1.0, right? However, as it turns out, the R^2 value of a linear regression on this data is undefined. Excel will display the value as N/A

Horizontal Data



What has happened in this example is that the total summed squared error is equal to zero. All the data values exactly equal the mean value. So there is zero error if you just estimate the mean value.

Of course, there is also zero error for the regression line. You end up with zero divided by zero terms in the R^2 equation, which is undefined.

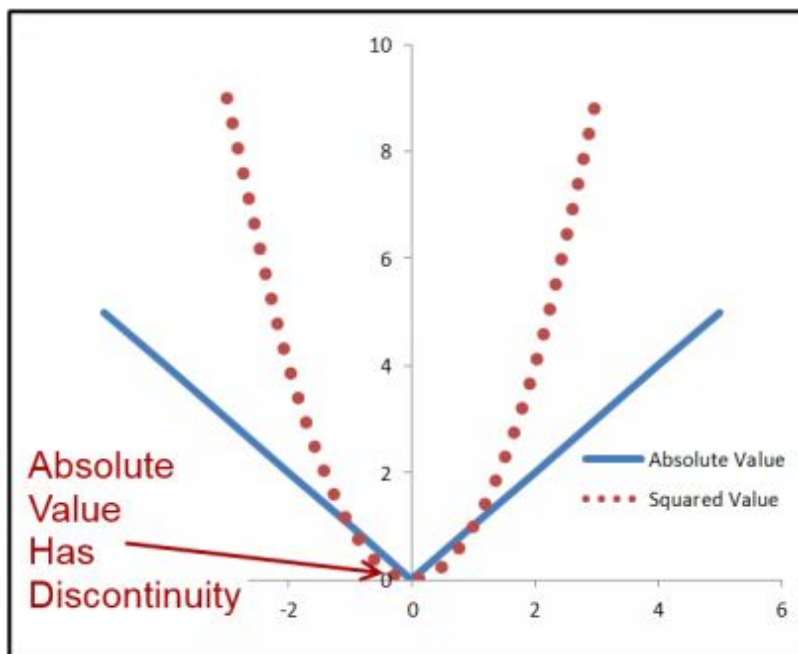
$$R^2 = 1 - \frac{0}{0} = \textit{undefined}$$

More On Summed Squared Error

R^2 is based on summed squared error. Summed squared error is also crucial to understanding linear regression since the objective of the regression function is to find a straight line through the data points which caused the minimum summed squared error.

One reason that sum squared error is so widely used, instead of using other potential metrics such as summed error (no square) or summed absolute error (absolute value instead of square) is that the square has convenient mathematical properties. The properties include being differentiable, and always additive.

For instance, using just summed error would not always be additive. An error of +5 and -5 would cancel out, as opposed to $(+5)^2 + (-5)^2$ which would sum. The absolute value would be additive, but it is not differentiable since there would be a discontinuity.



Summed Squared error addresses both of those issues, which is why it has found its way into many different equations.

Summed Squared Error In Real Life In addition to the useful mathematical properties of summed squared error, there are also a few places where an equivalent of it shows up in real life. One of those is when calculating the center of gravity of an object. An object's center of gravity is the point at which it will balance.

This bird toy has a center of gravity that is located below the tip of its beak, which allows it to balance on its beak surprisingly well



The location of the center of gravity is calculated in the same way as if you were trying to find the location that would give you the minimum summed squared error to every single individual atom of the bird.

What Is Correlation?

Correlation is a measure of how closely two variables move together. Pearson's correlation coefficient is a common measure of correlation, and it ranges from +1 for two variables that are perfectly in sync with each other, to 0 when they have no correlation, to -1 when the two variables are moving opposite to each other.

For linear regression, one way of calculating the slope of the regression line uses Pearson's correlation, so it is worth understanding what correlation is.

The equation for a line is

$$y = a + bx$$

Intercept Slope

One part of the equation for the slope of a regression line is Pearson's correlation. That equation is

$$b = r * \frac{S_y}{S_x}$$

Slope Sample Standard Deviation of y
Pearson's Correlation Sample Standard Deviation of x

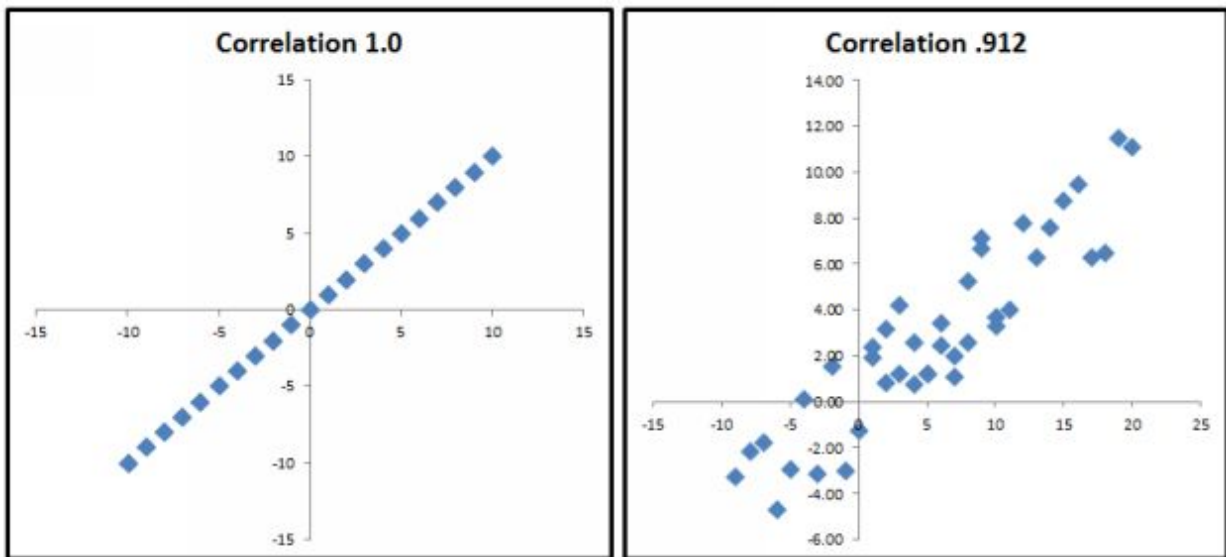
Where

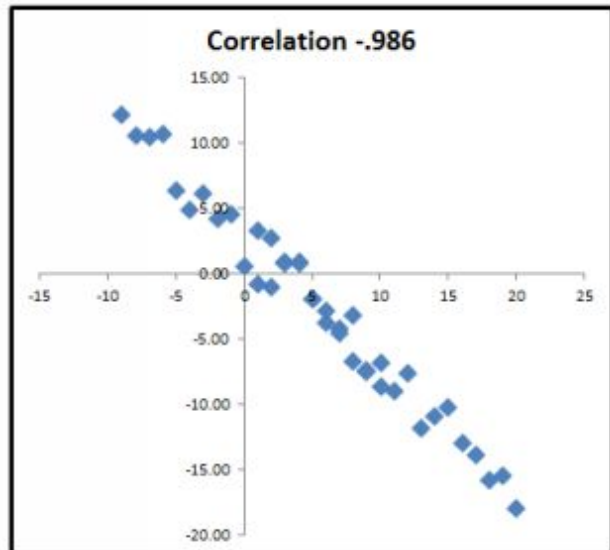
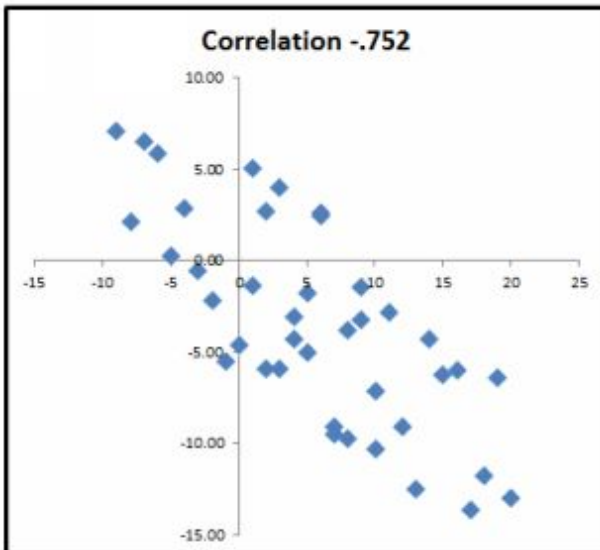
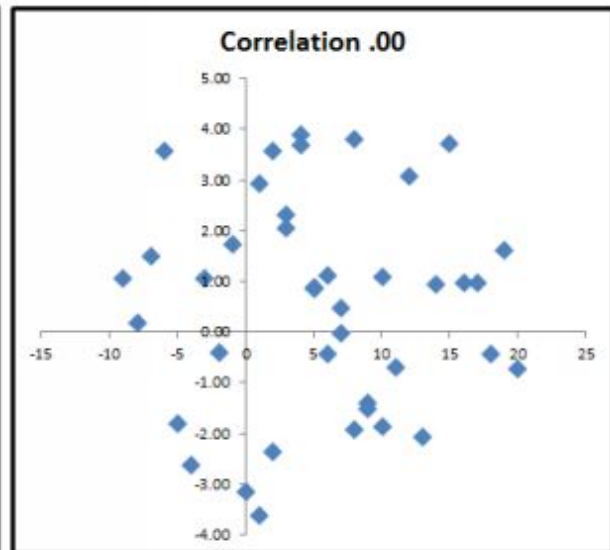
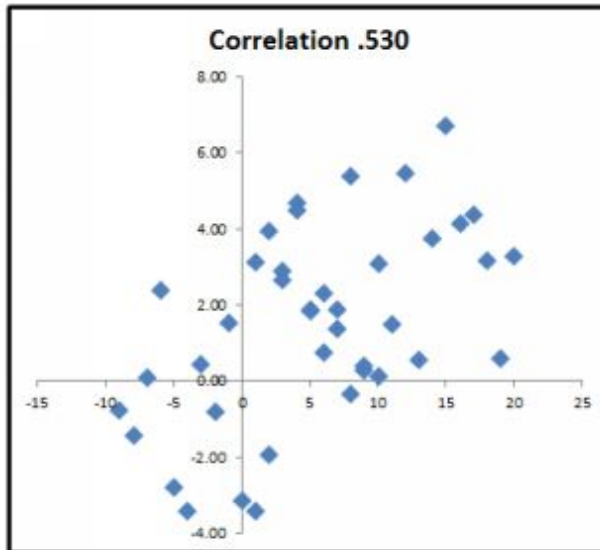
- r = Pearson's correlation coefficient
- s_y = sample standard deviation of y
- s_x = sample standard deviation of x . (Note that these are sample standard deviations, not population standard deviation)

One thing this equation suggests is that if the correlation between x and y is zero, then the slope of the linear regression line is zero, i.e. the regression line will just be the mean value of the y values (the ' a ' in the $y=a + bx$ equation) As an obligatory side note, we should mention that correlation does not imply causation. However, correlation does sort of surreptitiously point a finger and give a discrete wink.

Correlation is one of two terms that gets multiplied to generate the slope of the regression line. The other term is the ratio of the standard deviation of x and y . The correlation value controls the sign of the sign of the regression slope, and influences the magnitude of the slope.

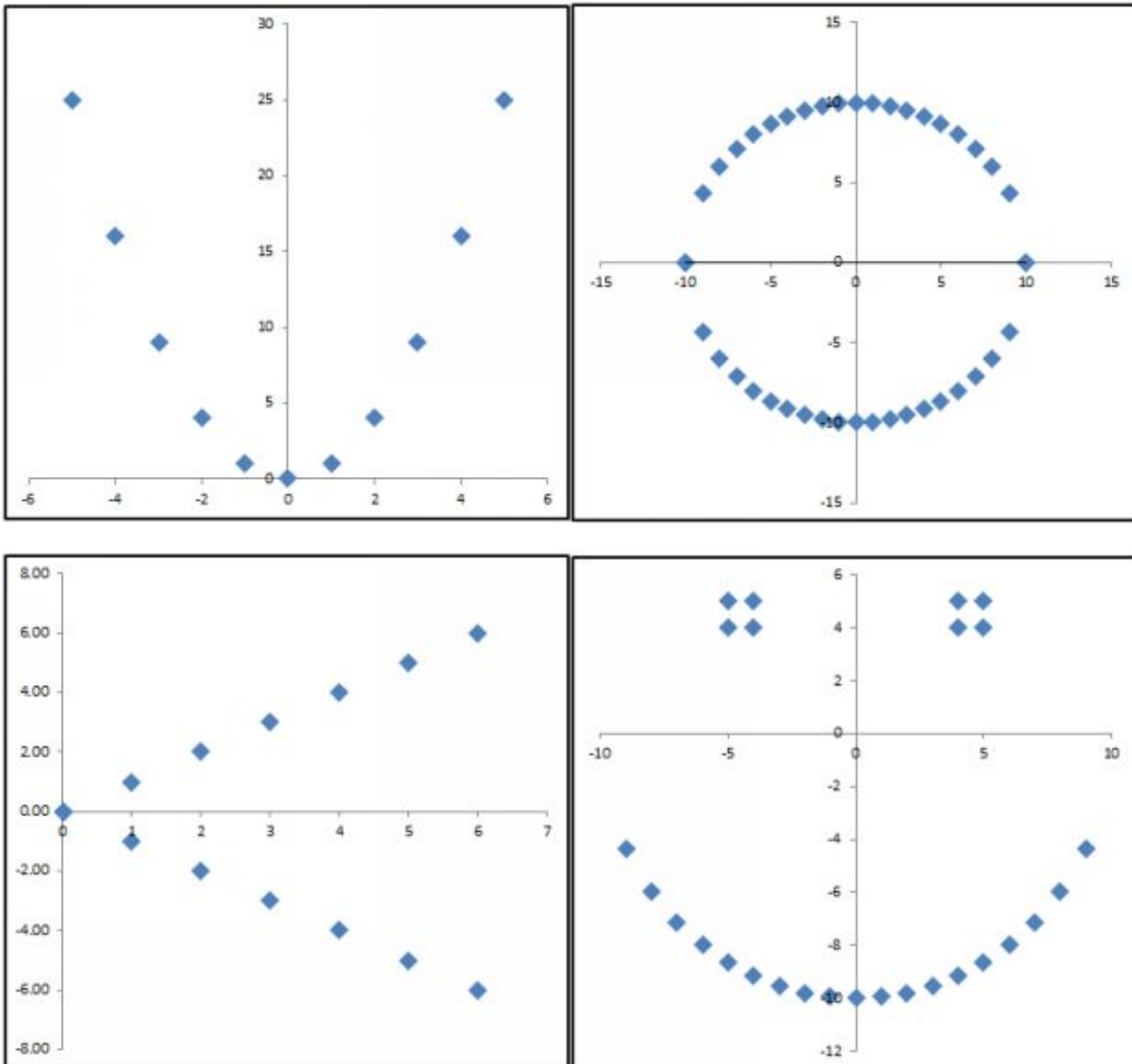
Here are some scatter plots with different correlation values, ranging from highly correlated to zero correlation to negative correlation.





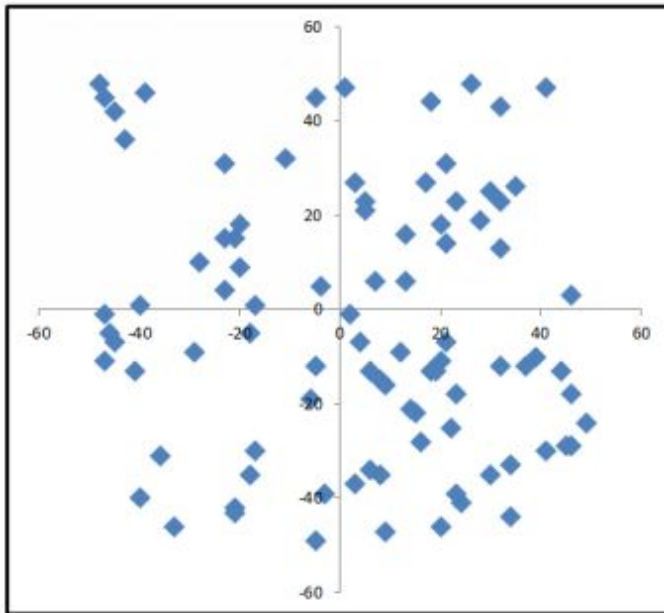
Interestingly, zero correlation does not mean having no pattern. Here are some plots that all have zero correlation even though there is an apparent

pattern



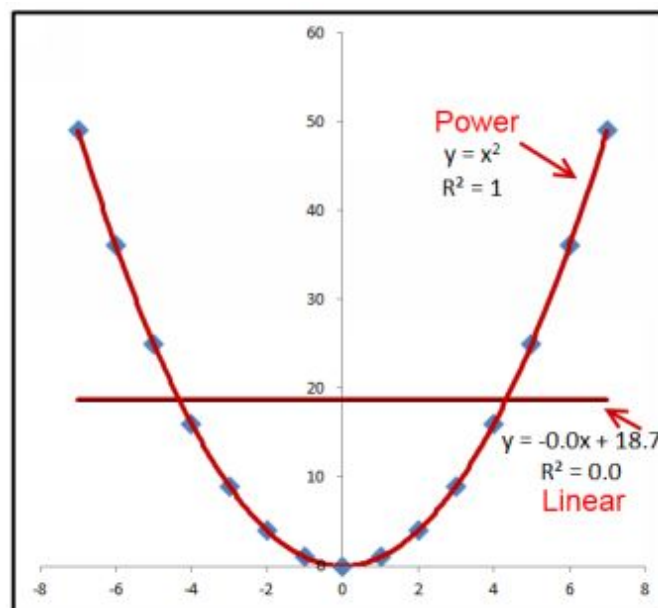
Essentially, zero correlation is the same as saying the R^2 of the linear regression will be zero, i.e. it can't do better than the mean value for linear

regression. This could mean that no regression will be useful, like this scatter



plot

Or it could mean that a different regression would work, like this squared plot below. In this plot, if we used $y = x^2$ we could get a perfect regression. None-the-less, we can't do better than $y = \text{mean value of all } y\text{'s}$ for this set of data with a linear regression.



Correlation Equation

Here is the equation for Pearson's correlation

The diagram shows the Pearson correlation equation with red annotations and arrows pointing to specific parts:

$$r = \frac{\sum ((x - \bar{x}) * (y - \bar{y}))}{(n - 1) * s_x * s_y}$$

Annotations:

- Sum Over All Data Points** points to the summation symbol $\sum$.
- x & y values of each point minus x & y mean values** points to the terms $(x - \bar{x})$ and $(y - \bar{y})$.
- Pearson's Correlation** points to the variable r .
- # of Data Points** points to the variable n .
- Standard Deviation of x & y** points to the variables s_x and s_y .

Where

- r is the correlation value
- n = number of data points
- s_x = sample standard deviation of x
- s_y = sample standard deviation of y
- x, y are each individual data point
- $\bar{x}, \bar{y}$ are the mean values of x and y

There are a couple of different ways that equation can be rearranged, but I like this version the best because it uses pieces we already understand, such as the standard deviation (s_x, s_y) Let's take a look at this equation, and remember that pretty much everything we demonstrate for the correlation

value r also applies to the slope of the regression line since

$$b = r * \frac{s_y}{s_x}$$

Slope $\rightarrow b$

r $\uparrow$ Pearson's Correlation

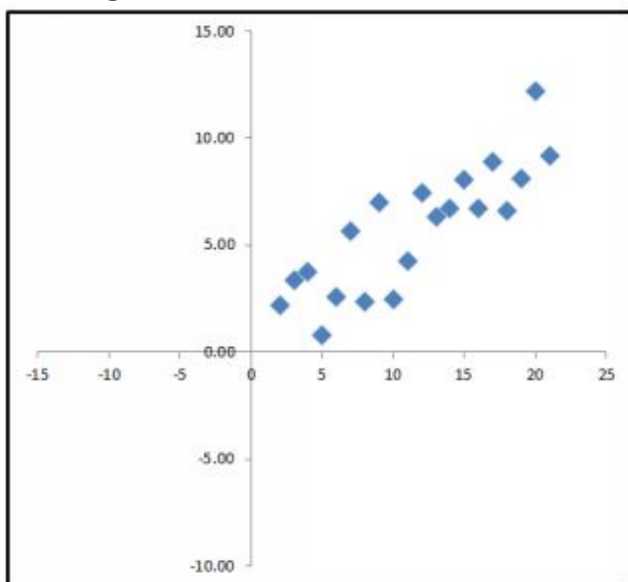
s_y $\nwarrow$ Sample Standard Deviation of y & x

s_x $\swarrow$

In the correlation equation, the number of data points, n , and the standard deviation values are always positive. That means the denominator of the fraction is always positive. The numerator, however, can be positive or negative, so it controls the sign. The numerator of the correlation value is shown below.

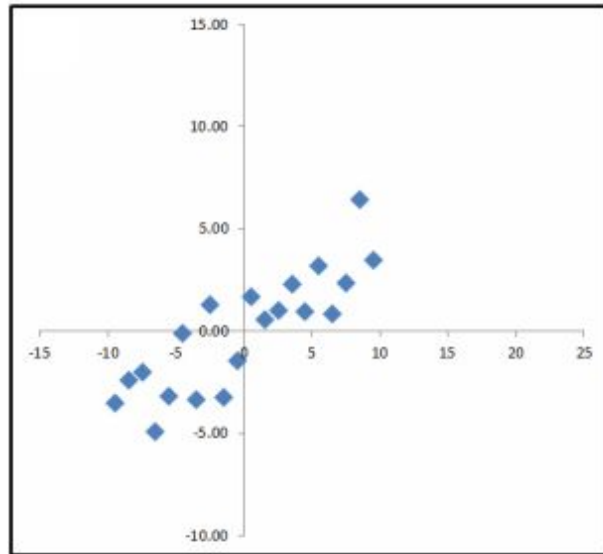
$$\sum ((x - \bar{x}) * (y - \bar{y}))$$

$\bar{x}$ and $\bar{y}$ are the mean values, so subtracting them from x and y is effectively normalizing the chart around the mean. So whether your data is offset like



this

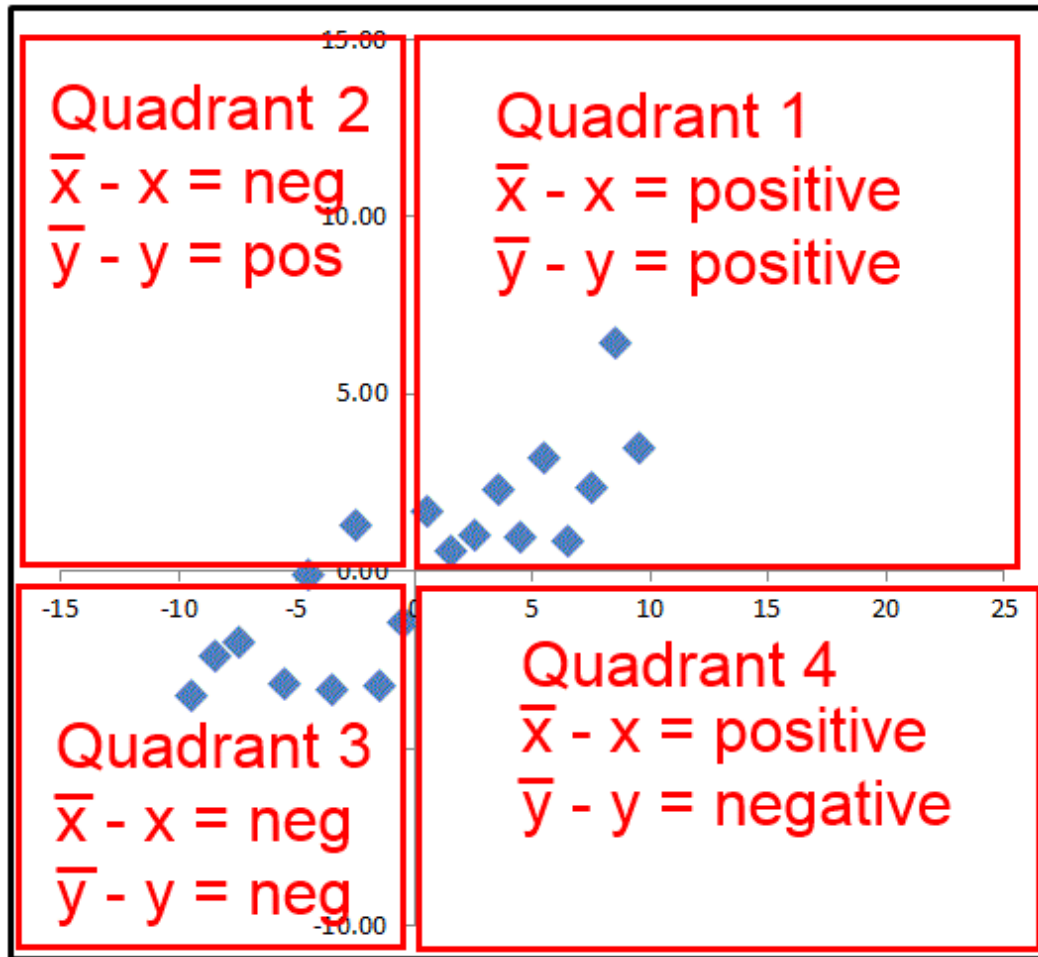
Or centered like this



The

$$\sum ((x - \bar{x}) * (y - \bar{y}))$$

part of the equation will give the same results The value coming from this portion of the equation is positive when x and y have mostly the same sign relative to their mean. For instance, in the chart above, where $(\bar{x}, \bar{y})$ is the origin. We get a negative result when the sign of $(x - \bar{x})$ and $(y - \bar{y})$ don't match. For instance, when x is greater than x average and y is less than x average in Quadrants 2 and 4. There is a positive result in Quadrant 1 and 3 where the sign of $(x - \bar{x})$ and $(y - \bar{y})$ do match. This is shown in the chart below.

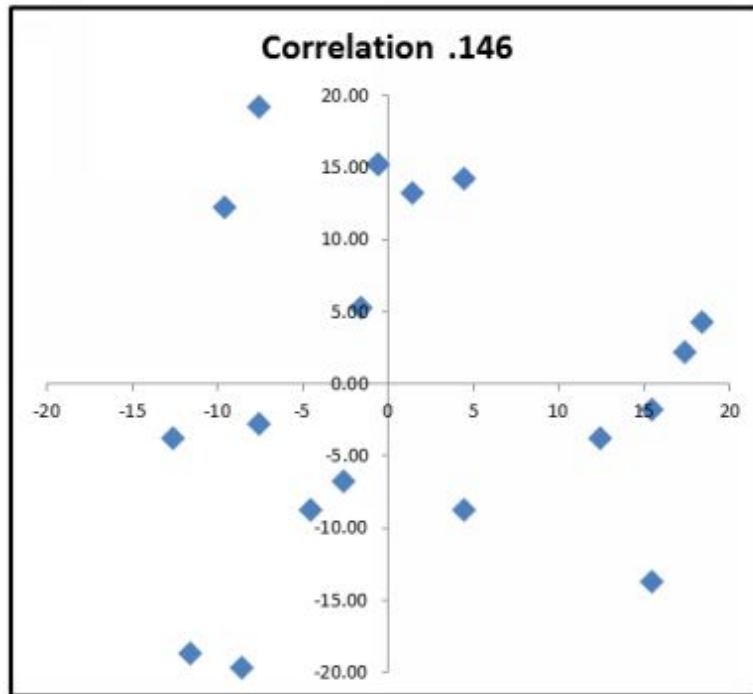


For the blue points in the chart above, most of the points are in Quadrant 1, and Quadrant 3, which means $(x - \bar{x})$ is positive when $(y - \bar{y})$ is positive, most of the time, and $(x - \bar{x})$ is negative when $(y - \bar{y})$ is negative, most of the time.

Since the product of two positive or two negative numbers are positive, data in quadrants 1 or 3 relative to the mean value results in a positive value, hence positive correlation and positive result of the linear regression line slope.

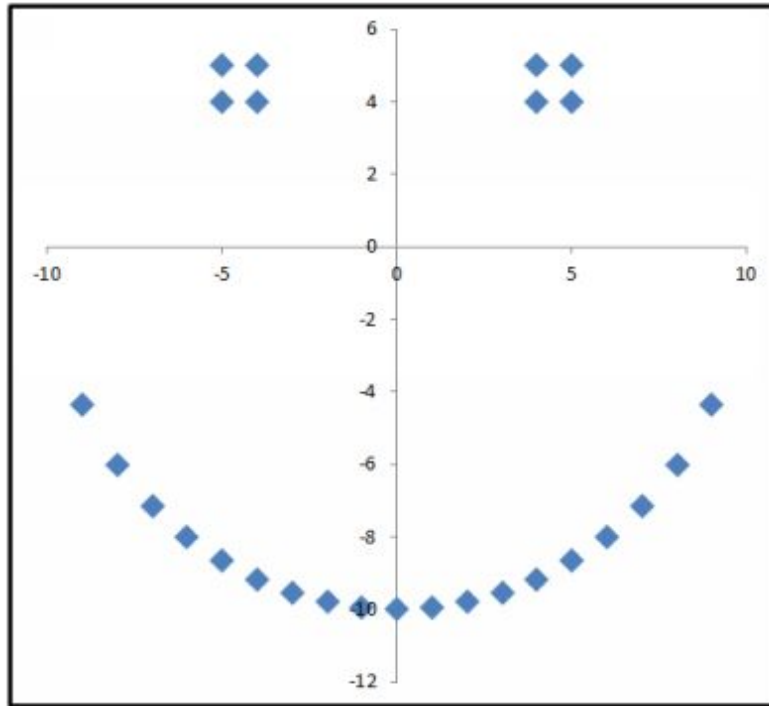
Data in quadrants 2 or 4 would result in negative correlation. Basically, if you center the scatter plot on the mean value, any point in quadrant 1 or 3 would contribute to positive correlation, and any point in quadrant 2 or 4 would contribute to a negative correlation.

If we have positive and negative results, those can cancel out. We would then get either a very low correlation value or in rare cases zero.



The chart above is centered on the mean. It has low correlation because there is fairly even scatter in all quadrants about the mean.

In fact, one easy way to get zero correlation (not the only way) is to have symmetry around either $\bar{x}$, $\bar{y}$, or both. That makes the numbers exactly cancel out. That is what we see in the image below, which has zero correlation.



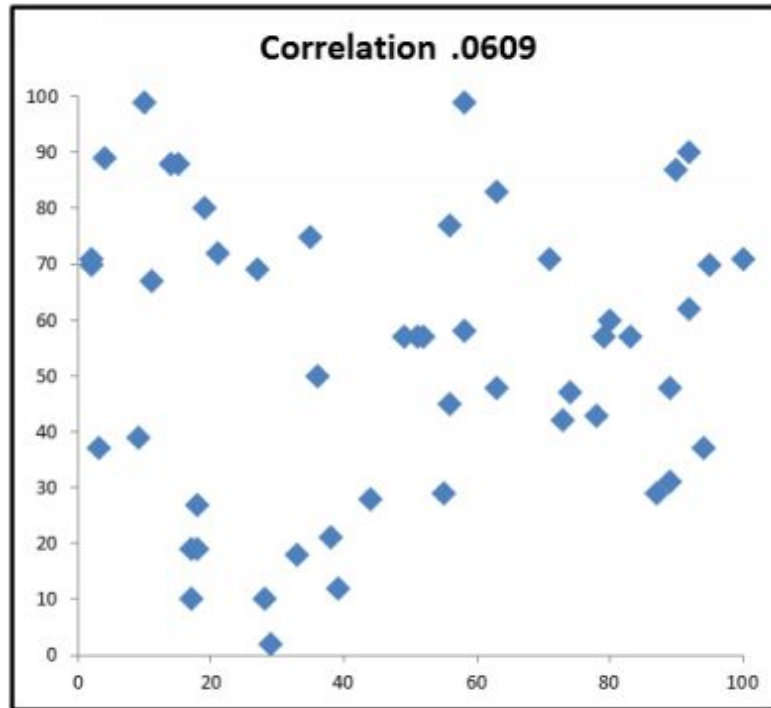
The image above has symmetry around $\bar{x}$, this means that for every data point, there will be a matching point that has the same $(y - \bar{y})$ and an opposite sign but same magnitude on $(x - \bar{x})$. Those two values cancel each other out. You could also have symmetry about $\bar{y}$.

Of course, symmetry isn't required to get

$$\sum ((x - \bar{x}) * (y - \bar{y}))$$

to sum to zero and get zero correlation. All you need is for the magnitude of all the positive points to cancel with all the negative points.

Realistically though, nearly any real world data set will end up with some correlation. Here is a scatter of 50 points generated by 50 pairs of Random numbers between 1 and 100.



Which shouldn't have much correlation because both the x and y values were randomly generated, but the correlation is non-zero. We expect that two streams of random numbers will have zero correlation given a large enough sample size. However, we see that for these 50 points, the correlation isn't that high, but it is non-zero. (Since we have been discussing quadrants, we should note here that the quadrants refer to location relative to the average x and average y values, in this case, that would be an x, y of approximately 50, 50)

Denominator of Pearson's Correlation We've focused so far on the numerator of Pearson's correlation equation but what about the denominator?

The denominator of the equation is

$$(n - 1) * s_x * s_y$$

Where s_x and s_y are the sample standard deviations of the data. (As opposed to the population standard deviation).

The equation for sample standard deviation is

$$s_x = \sqrt{\frac{\sum (x - \bar{x})^2}{(n - 1)}}$$

Note The Summed Square Error

Standard deviation is a way of measuring how spread out your data is. Data that is tightly clustered together will have a low standard deviation. Data that is spread out will have a high standard deviation. Since there is a squared term in the equation, the most outlying data points will have the largest impact on the standard deviation value.

Note the denominator here is (n-1), instead of n which it would have been if we were using the population standard deviation. The same equation would hold true for the sample standard deviation of y, except with y terms instead of x terms.

There are other ways to rearrange this equation. If we are just looking at the denominator of Pearson's correlation equation, that is shown below.

$$= \frac{\sum (x - \bar{x})(y - \bar{y})}{(n - 1) * s_x * s_y}$$

We could cancel out the (n-1) with the two square roots of (n-1) that are part of the standard deviations of x and y to rearrange the denominator to be

$$= \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2} * \sqrt{\sum (y - \bar{y})^2}}$$

If you prefer. Personally, I like keeping the equation in terms of the standard deviations, but it is the same equation either way.

These values have the effect of normalizing the results of the correlation against the numerator. I.e. the denominator will end up with the same units of measurement as the numerator. If we assume or adjust the values such that

$\bar{x}$ and $\bar{y}$ are zero, the numerator ends up being $\sum xy$

And the denominator ends up being

$$= \frac{\sum xy}{\sqrt{\sum x^2} * \sqrt{\sum y^2}}$$

For that special case where $\bar{x}$ and $\bar{y}$ are zero (note don't use these modified equations for general numbers). Notice that both the numerator and denominator end up having units of xy . When they are divided, the result is a unit less value. Which basically means a correlation calculation where you are comparing your truck's payload vs fuel economy will have the same result whether the units are pounds and miles per gallon, or the units are kilograms and kilometers per liter.

The correlation value, r , will be the same for either set of units. Note however that the slope of the regression line won't be the same since

$$b = r * \frac{S_y}{S_x}$$

And the standard deviation parts of the equation still have units baked into them.

Correlation Takeaways

We did a lot of looking at equations in this section. What are the key takeaways?

The key takeaway is that correlation is an understandable equation that relates the amount of change in x and y. If the two variables have consistent change, there will be a high correlation; otherwise, there will have a lower correlation.

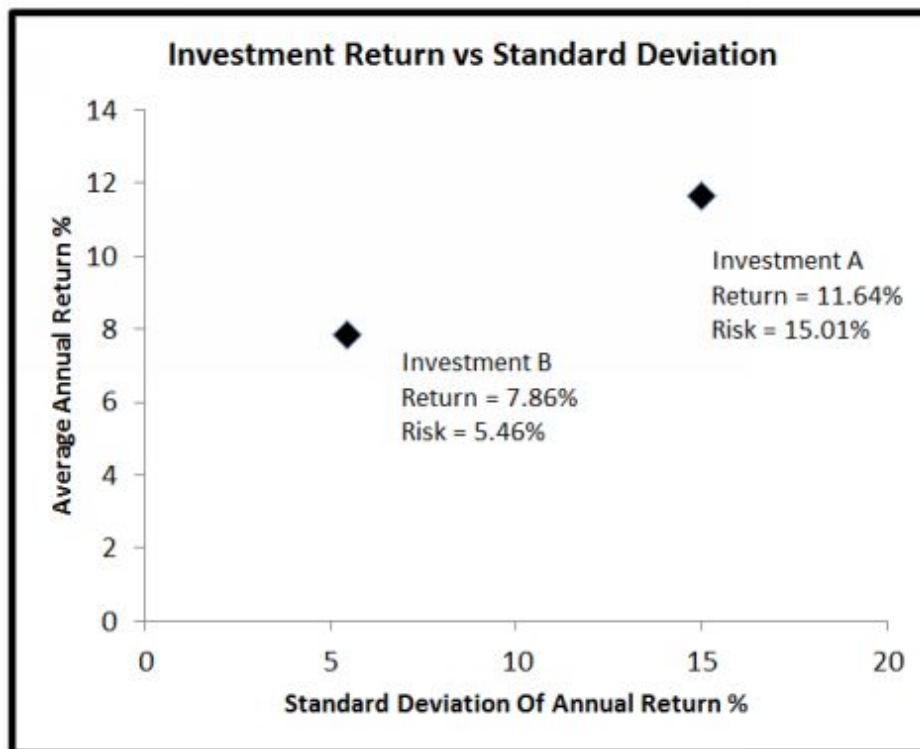
Uses For Correlation

Although we will be using correlation as part of the linear regression equations, correlation has other interesting applications independent of regression that are worth knowing about. One common use for correlation analysis is in investment portfolio management.

Let's say that you have two investments, stocks for instance. If you have their price histories, you can calculate the correlation between those two investments over time. If you do, what can you do with that result?

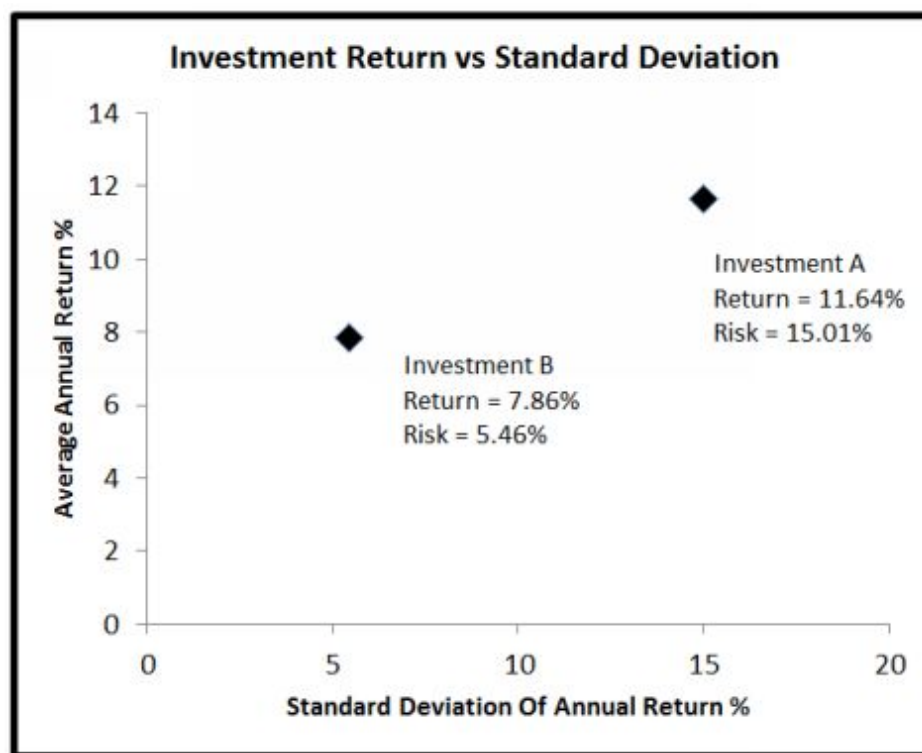
Well, if you are a hedge fund on Wall Street with access to high-frequency trading, you might be able to observe the price movement of one stock and predict the direction of movement for another. That type of analysis isn't useful for the everyday small investor. But the correlation is still useful for long term investment.

Here is a chart that shows two risk and return profiles for investments A and B



The y-axis shows the average annual return as a percentage, and the x-axis shows the standard deviation of that return. The best investment would be as high as possible (high return) and as far left as possible (low risk) (note returns can be negative, but standard deviation is always greater than or equal to zero.) The ideal investment would have absolutely zero variance in return. For instance, if you average a 12% return in a year, you would prefer that it paid out 1% every single month, compared to one that was +5%, -10%, +4%, +6%, -4%, etc., even if the more volatile investment had the same 12% total return. The benefit of a higher return is obvious. The benefit of smaller volatility is that it allows you to invest more money with less held back as a safety net, it reduces your risk of going broke due to a string of bad returns, or of making a bad choice and selling at the wrong time.

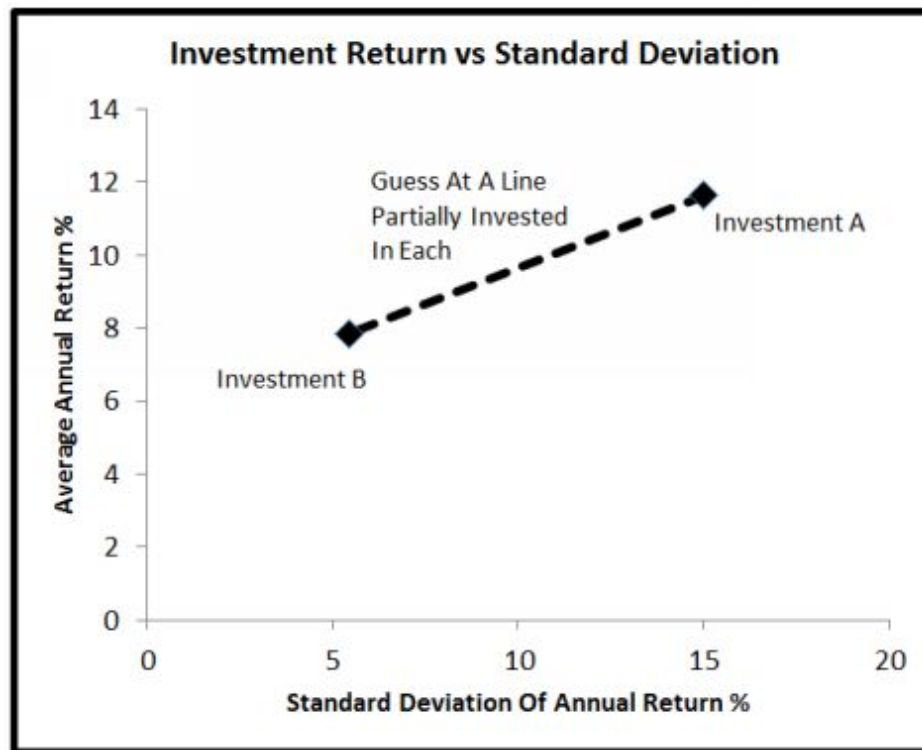
So knowing that you prefer high return and low risk, which of these two investments is better?



The answer is, you can't tell. It varies based on what your objective is. One person might be able to take on more risk for more return. A different person might prefer less variation in their results. So you might have person 1 who prefers investment A, and person 2 who prefers investment B.

Now suppose that you have person 3 who has a little bit of both qualities. They are willing to accept some more risk, for some greater return, so they split their money between investments A & B. What does their risk vs. return profile look like?

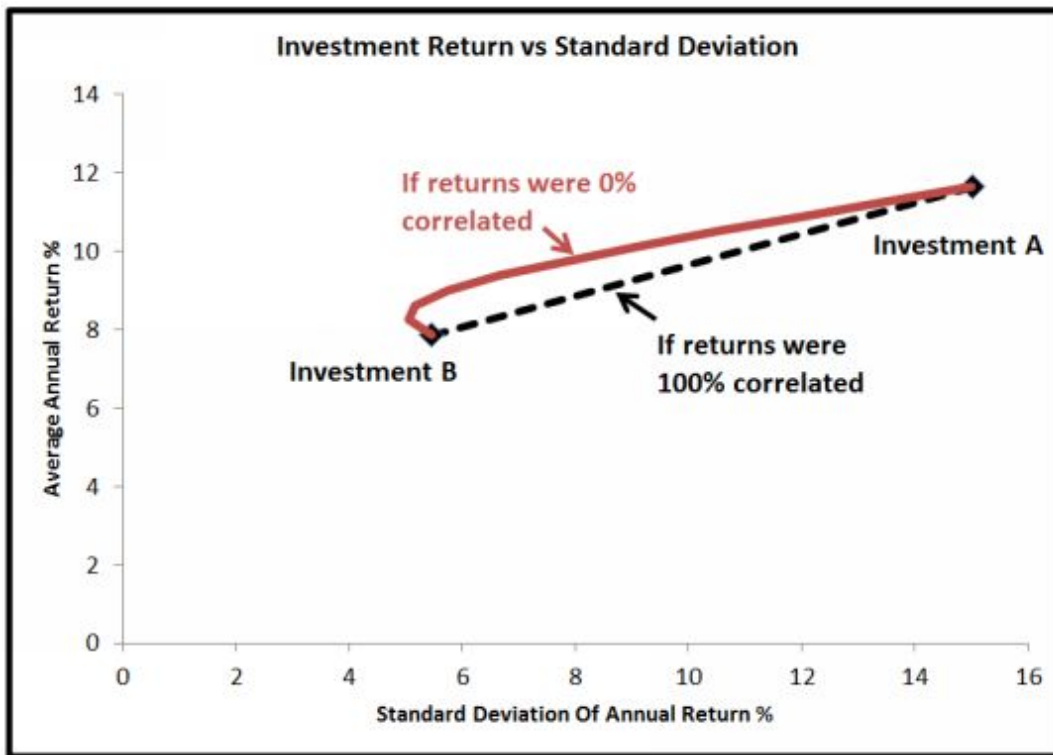
The first assumption is that they end up somewhere along a line that falls between A & B



And if they invest 50% in A and 50% in B, they will fall halfway between the A and B results. If they invested in A & B in different ratios, they would fall elsewhere on that line.

But that result is true only if A and B are perfectly correlated. I.e. have a correlation of 1.0. If they are not perfectly correlated, you can do better.

With investments that have less than 1.0 correlation, the result looks like



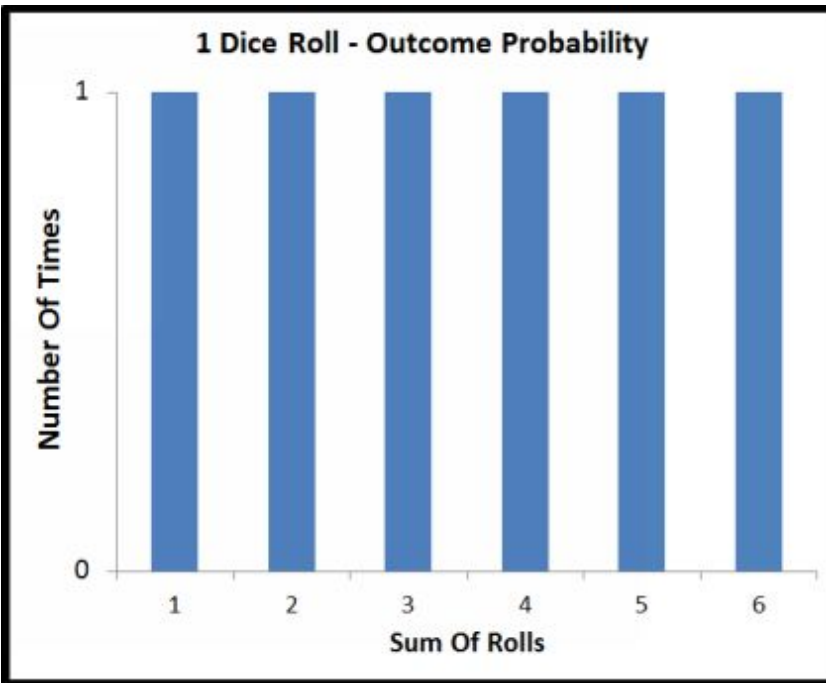
By finding investments with low correlation, Person 3 now has an unequivocal benefit. For the same level of risk, they have a higher return. They have an area where they can get more money without additional risk.

How is this possible?

Remember that we are measuring risk as the total standard deviation of results. That standard deviation is lower for the sum of independent events than it is for a single event because the highs on one investment will cancel out the lows on another investment.

One intuitive way to think about this is with dice. Imagine you have an investment that has an equal likelihood of returning 1, 2, 3, 4, 5, or 6% in a year. You can simulate that with the roll of a die, and your probability

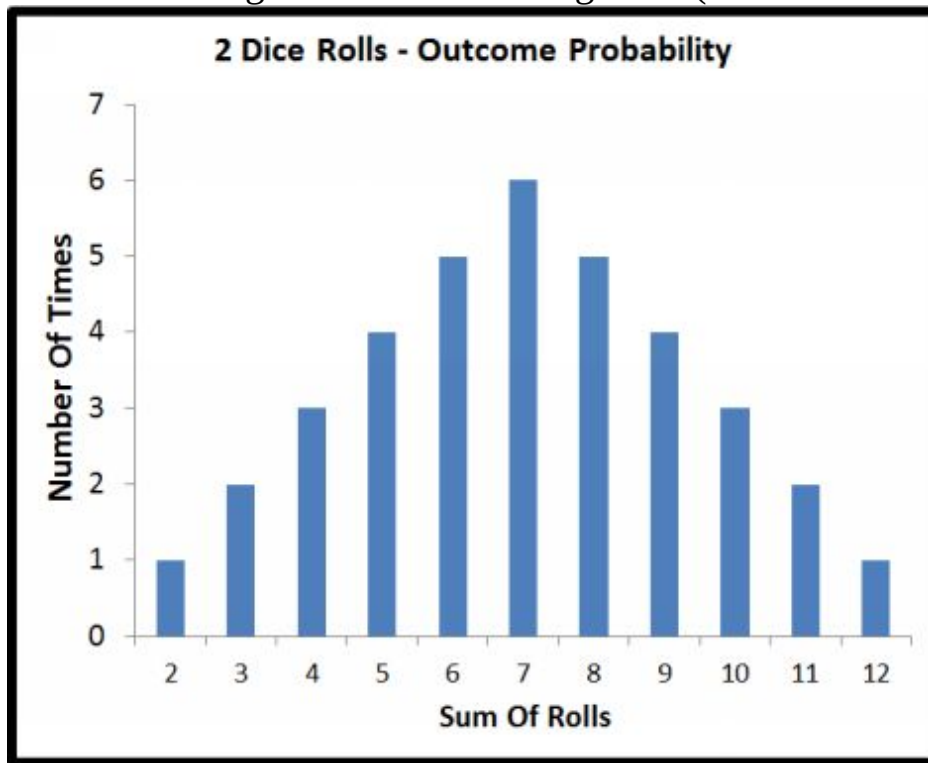
distribution looks like this



Your average return is 3.5%, and the standard deviation of results is the population standard deviation of (1, 2, 3, 4, 5, 6) which is 1.708

Now you take half of your money and move it to a different investment in a different industry. The correlation of the two investments is zero, so we simulate it with a second die. To get your total results, you roll both dice and add them according to their weightings.

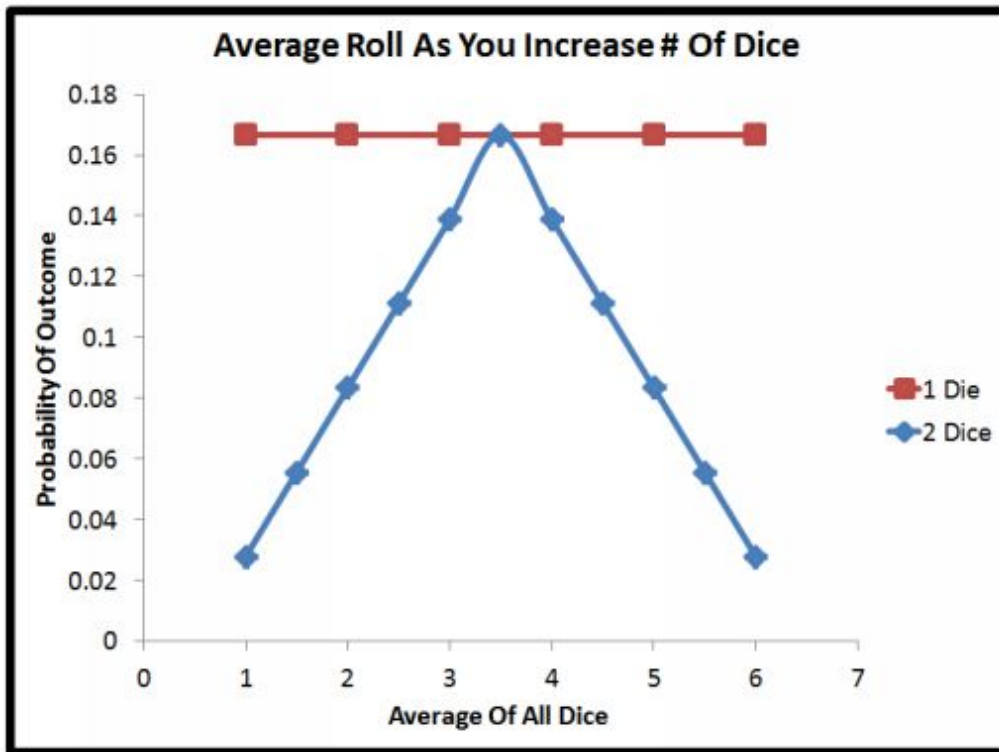
The standard results for rolling 2 dice and summing them (without



weightings) is

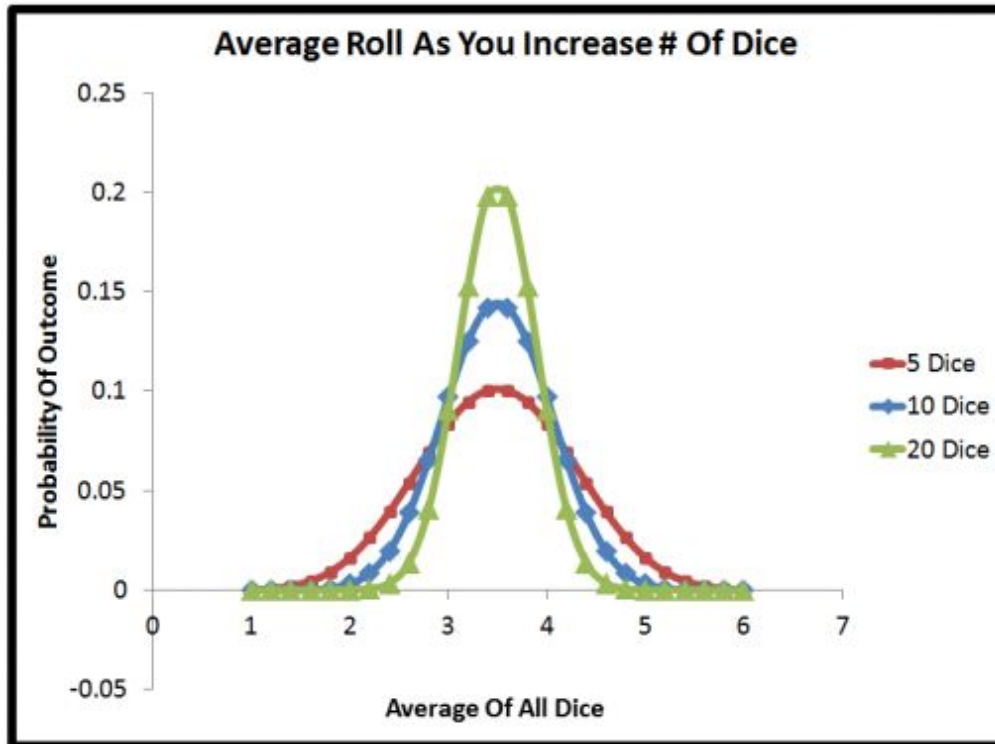
Since we have 50% weightings on both dice, we can divide the sum by 2 to get the average roll. When that average roll is plotted against the average roll

for 1 die, the results are



Rolling two dice still has an average return of 3.5%, but it has a standard deviation of 1.207. This is lower than the standard deviation of a single die, which is 1.7078. So we have essentially gotten the same return with less risk.

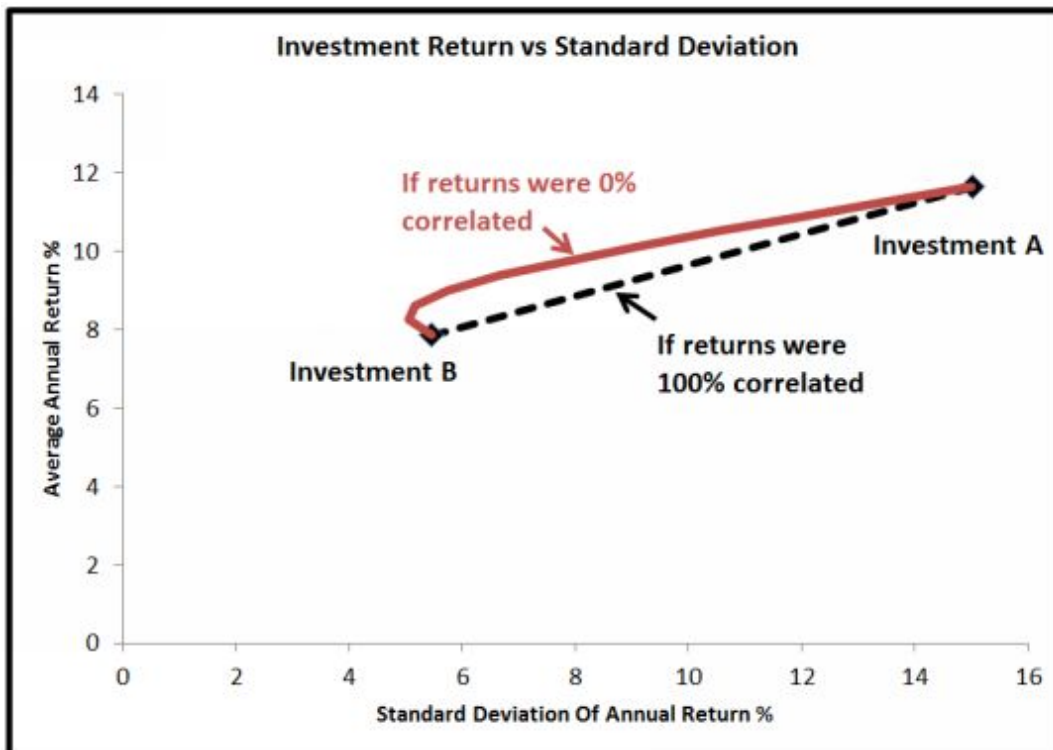
If you had a way to keep adding identical but uncorrelated investments, you could continue to make these results more narrowly spread around the mean



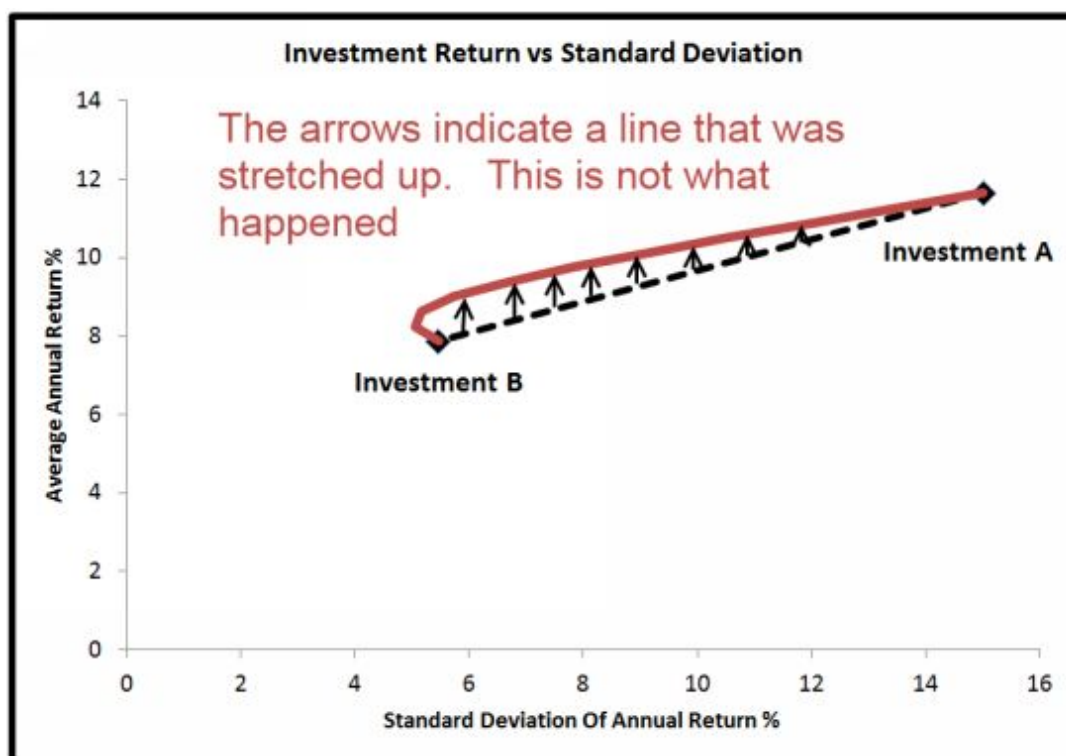
The chart above shows the results of increasing the number of completely uncorrelated events that you are sampling from. What we see is that return is a weighted average of the events, but that standard deviation decreases with the number of events. Although the above chart was made with dice, it could have been the result of 0% correlated stock returns. Of course, in real life, you are faced with this problem.

- Finite number of potential investments
- They don't have the same return or standard deviation
- The investments are not completely uncorrelated

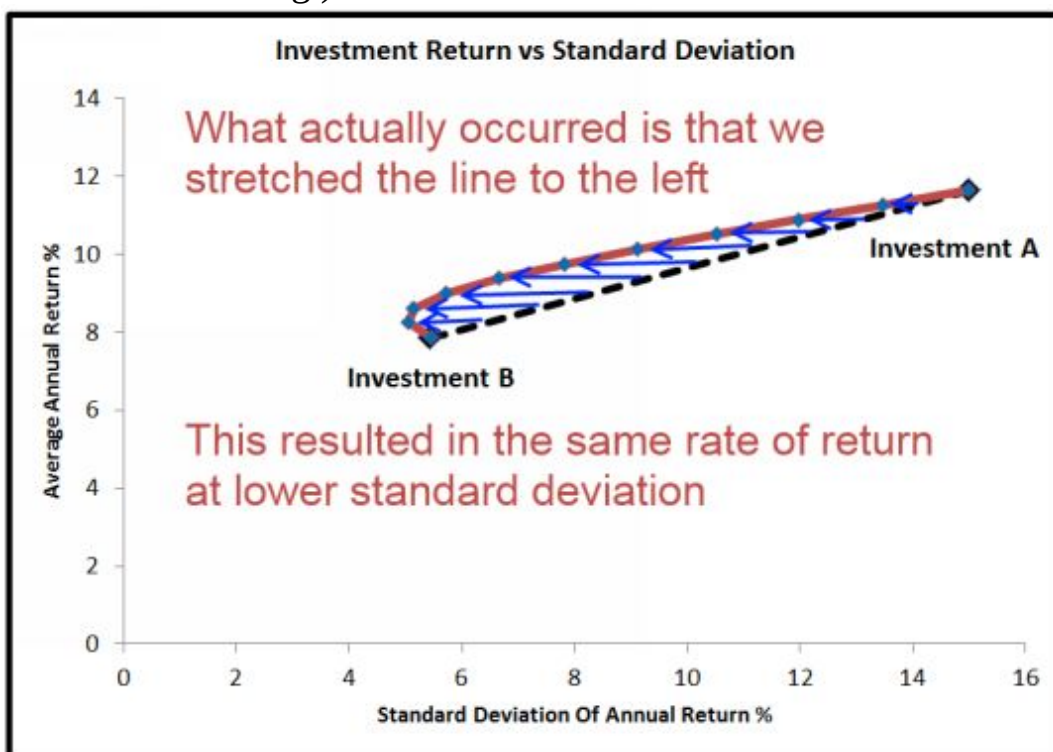
One important thing to realize is that we didn't raise the rate of return at all. What we really did was reduce the risk. For instance in this chart



We did not stretch this line upwards



What we really did was pull it to the left, i.e. reduce risk. So for a given weighting of investment A and investment B, there was the same rate of return as if you had done a linear interpolation between the two, but that rate of return is achieved for less risk. (I should note here that this section is focused on the math behind correlation, and is not investment advice. As a result, I'm completely ignoring some things that could impact your return, such as rebalancing.)

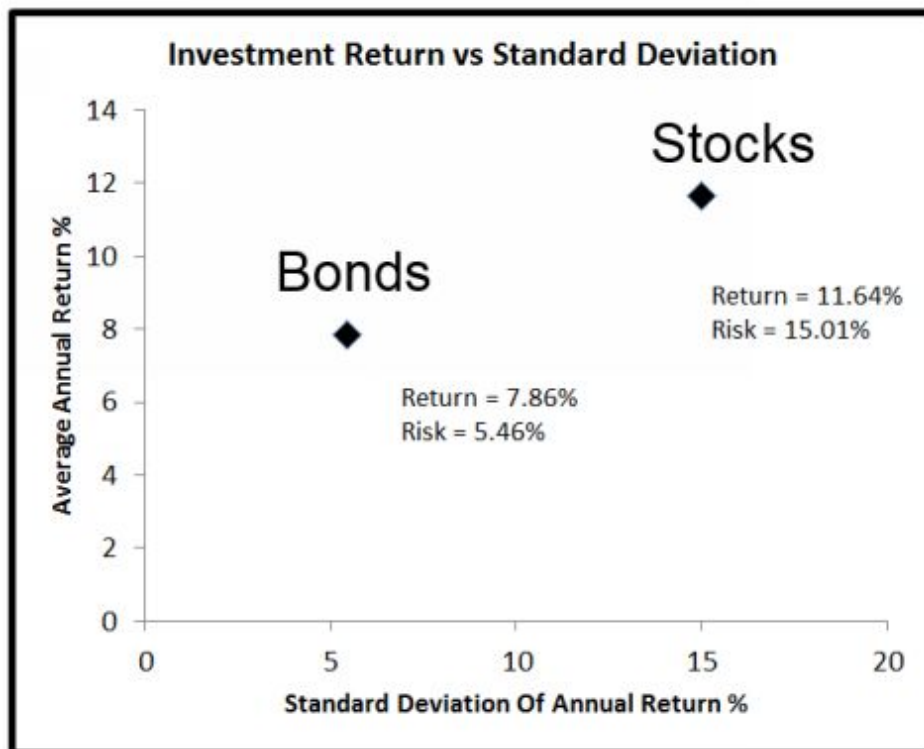


The maximum rate of return is still bounded by the return rate of the highest investment. We can't go higher than the 11.64% that we see for investment A. In fact, our total rate of return is still just the weighted average of all the investments.

Are You Diversified?

This page is why people ask, "Are you diversified?" Being diversified is a big benefit of investing in index funds over individual stocks. By owning the whole market, the investor is getting the same average return as if they owned a handful of stocks, but they have reduced the variance of that return. Looking at that same statement another way, owning a handful of stocks

instead of the whole market means you are taking on additional risk, and not getting compensated for it. (Assuming, of course, that you are an average investor. If you are a stock picker that actually can beat the market, that statement doesn't apply) The chart that we have been looking at is the average risk/reward of stocks and bonds from 1976 to 2015. The stocks are the S&P 500, and the bonds are the Barclays Aggregate Bond Index



I should note that the real life results don't have a zero correlation between stocks and bonds. That was a simplification for these charts. The real efficient frontier of investing would be different than the dashed line previously shown. (And would be different again if you consider things like international investments, real estate, etc.)

Correlation Of The Stock Market

Let's calculate the correlation of 2 stocks. The stocks I chose are Chevron (Ticker CVX) and Exxon Mobil (Ticker XOM). I downloaded the daily closing price in 2016 from Google finance. Since they are both major oil companies, we expect them to be highly correlated. Presumably, their profits are driven by the price of oil and how good the technology is that allows them to extract that oil inexpensively.

The price of oil and state of technology is the same for both companies. There are other factors that are different between the two companies, like how well they are managed or the situations at their local wells. These differences mean that the two companies won't get exactly the same results over time, and hence won't be completely correlated.

To start the correlation, we need to decide exactly what we want to correlate on. We have a year's worth of data, approximately 252 trading days. We need to choose the time scale that we want to correlate. Should it be day to day, week to week, month to month? This matters because two items can be uncorrelated over one scale, for instance how the stocks trade minute to minute, but still be highly correlated over another scale, say their total returns over a quarter.

In the interest of long-term investing, and of having few enough data points to fit on a page of this book, let's look at the monthly correlation. This is the stock price on the first trading day of every month in 2016, plus the last day of 2016

Date	Exxon Mobil	Chevron
4-Jan-16	77.46	88.85
1-Feb-16	76.29	85.29
1-Mar-16	81.28	86.13
1-Apr-16	82.96	94.26
2-May-16	89.13	103.31
1-Jun-16	89.24	101.13
1-Jul-16	93.84	104.15
1-Aug-16	85.86	99.11
1-Sep-16	86.84	100.21
3-Oct-16	87.05	102.45
1-Nov-16	83.65	106.48
1-Dec-16	87.24	113.29
30-Dec-16	90.26	117.70

Note that we are looking at the price here, which is not necessarily the same as total return for these dividend paying stocks. A different analysis, one which was actually focused on the stock results as opposed to demonstrating how correlation works, might include things like reinvested dividends into the stock price.

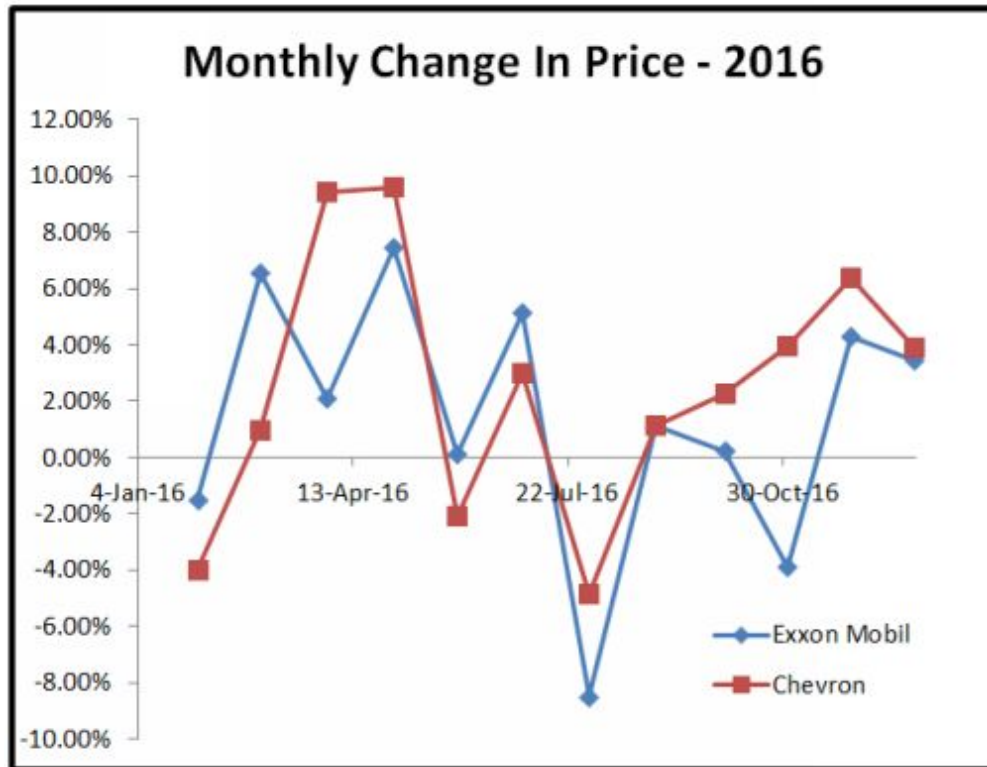
We could do the correlation analysis on this price data as it is. However, I'm going to make one additional modification to the data to make it be a monthly change in price as a percentage.

Monthly Change In Price

Date	Exxon Mobil	Chevron
4-Jan-16		
1-Feb-16	-1.51%	-4.01%
1-Mar-16	6.54%	0.98%
1-Apr-16	2.07%	9.44%
2-May-16	7.44%	9.60%
1-Jun-16	0.12%	-2.11%
1-Jul-16	5.15%	2.99%
1-Aug-16	-8.50%	-4.84%
1-Sep-16	1.14%	1.11%
3-Oct-16	0.24%	2.24%
1-Nov-16	-3.91%	3.93%
1-Dec-16	4.29%	6.40%
30-Dec-16	3.46%	3.89%

The effect of price vs. change in price is small for this data, but for times when there are a couple of months in the middle that have a large change, it can affect the correlation value.

If we plot those results, what we see is



There certainly seems to be some correlation between those results. To get the actual value for correlation, we will use Pearson's correlation equation again, and go through all the steps to get mean, standard deviation, the sum of xy , etc.

Correlation		0.22
Monthly % Change In Price		
Date	Chevron	Coke
4-Jan-16		
1-Feb-16	-4.01	-0.82
1-Mar-16	0.98	2.64
1-Apr-16	9.44	-8.70
2-May-16	9.60	1.03
1-Jun-16	-2.11	-24.00
1-Jul-16	2.99	15.93
1-Aug-16	-4.84	-0.21
1-Sep-16	1.11	5.63
3-Oct-16	2.24	-4.93
1-Nov-16	3.93	-2.56
1-Dec-16	6.40	12.45
30-Dec-16	3.89	12.48

The result is much lower, but there is still some correlation. In fact, most equities will show at least some correlation to each other, which is why in broad market swings many stocks will gain or lose value at the same time.

One good way to show the correlation among multiple items is a correlation matrix

Correlation Matrix of Montly Return Of Stocks In 2016

	Exxon Mobil	Chevron	Conoco Phillips	BP	Caterpillar	Cummins	Deere	Coke	Pepsi	Kellogg
Exxon Mobil	1.00									
Chevron	0.63	1.00								
ConocoPhillips	0.44	0.91	1.00							
BP	0.59	0.53	0.50	1.00						
Caterpillar	0.29	0.37	0.39	0.09	1.00					
Cummins	0.22	0.26	0.26	-0.11	0.83	1.00				
Deere	0.45	0.35	0.33	0.25	0.26	0.44	1.00			
Coke	0.35	0.22	0.12	0.55	0.32	0.15	0.38	1.00		
Pepsi	-0.09	-0.02	-0.05	0.15	-0.11	-0.38	-0.80	0.08	1.00	
Kellogg	0.17	0.03	-0.16	0.32	-0.06	-0.21	-0.57	0.35	0.77	1.00

A value in any given square is the correlation between its row item and column item. Here, for instance, we can see that all the oil stocks are fairly highly correlated

Correlation Matrix of Oil Stocks

	Exxon Mobil	Chevron	Conoco Phillips	BP
Exxon Mobil	1.00			
Chevron	0.63	1.00		
ConocoPhillips	0.44	0.91	1.00	
BP	0.59	0.53	0.50	1.00

Those energy stocks are less correlated to heavy equipment stocks like Caterpillar and Deere.

	Exxon Mobil	Chevron	Conoco Phillips	BP	Caterpillar	Cummins	Deere
Exxon Mobil	1.00						
Chevron	0.63	1.00					
ConocoPhillips	0.44	0.91	1.00				
BP	0.59	0.53	0.50	1.00			
Caterpillar	0.29	0.37	0.39	0.09	1.00		
Cummins	0.22	0.26	0.26	-0.11	0.83	1.00	
Deere	0.45	0.35	0.33	0.25	0.26	0.44	1.00

Correlation between oil companies and heavy manufacturers

And less correlated again to consumer stocks like Coke, Pepsi, and Kellogg's.

Correlation Matrix of Monthly Return Of Stocks In 2016

	Exxon Mobil	Chevron	Conoco Phillips	BP	Caterpillar	Cummins	Deere	Coke	Pepsi	Kellogg
Exxon Mobil	1.00									
Chevron	0.63	1.00								
ConocoPhillips	0.44	0.91	1.00							
BP	0.59	0.53	0.50	1.00						
Caterpillar	0.29	0.37	0.39	0.09	1.00					
Cummins	0.22	0.26	0.26	-0.11	0.83	1.00				
Deere	0.45	0.35	0.33	0.25	0.26	0.44	1.00			
Coke	0.35	0.22	0.12	0.55	0.32	0.15	0.38	1.00		
Pepsi	-0.09	-0.02	-0.05	0.15	-0.11	-0.38	-0.80	0.08	1.00	
Kellogg	0.17	0.03	-0.16	0.32	-0.06	-0.21	-0.57	0.35	0.77	1.00

One interesting thing is how little Coke and Pepsi are correlated. (Only .08)
 One would expect that since they are in the same sector, they might have a similar level of correlation between them as you see in the oil companies (between .5-.9), but the actual correlation between Pepsi and Coke is fairly low. That could be because they are more direct competitors, and one company's gain is another's loss, or it could be for some completely different reason.

Getting Started With Regression

Up until now, we've looked at correlation. Let's now look at regression. With correlation, we determined how much two sets of numbers changed together. With regression, we want to use one set of numbers to make a prediction on the value in the other set. Correlation is part of what we need for regression. But we also need to know how much each set of numbers change individually, via the standard deviation, and where we should put the line, i.e. the intercept.

The regression that we are calculating is very similar to correlation. So one might ask, why do we have both regression and correlation? It turns out that regression and correlation give related but distinct information.

Correlation gives you a measurement that can be interpreted independently of the scale of the two variables. Correlation is always bounded by ± 1 . The closer the correlation is to ± 1 the closer the two variables are to a perfectly linear relationship. The regression slope by itself does not tell you that.

The regression slope tells you the expected change in the dependent variable y when the independent variable x changes one unit. That information cannot be calculated from the correlation alone.

A fallout of those two points is that correlation is a unit-less value, while the slope of the regression line has units. If for instance, you owned a large business and were doing an analysis on the amount of revenue in each region compared to the number of salespeople in that region, you would get a unit-less result with correlation, and with regression, you would get a result that was the amount of money per person.

The Regression Equations

With linear regression, we are trying to solve for the equation of a line, which is shown below.

$$y = a + bx$$

The values that we need to solve for are b, the slope of the line, and a, the intercept of the line. The hardest part of calculating the slope, b, is finding the correlation between x and y, which we have already done. The only modification that needs to be made to that correlation is multiplying it by the ratio of the standard deviations of x and y, which we also already calculated when finding the correlation. The equation for slope is shown below

$$b = r * \frac{S_y}{S_x}$$

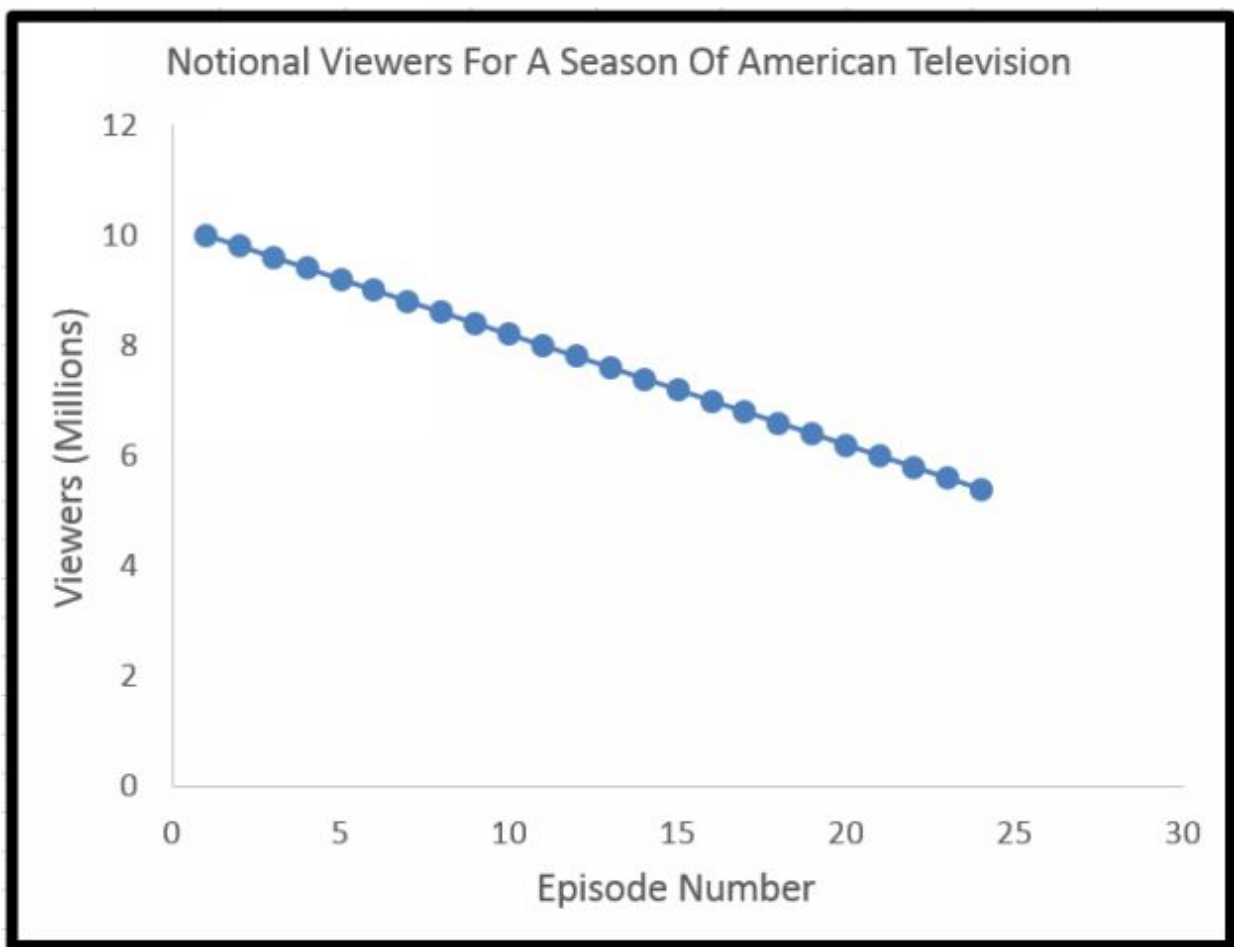
Once we have the slope, getting the intercept is easy. Assuming that you are using the standard equations for correlation and standard deviation, which go through the average of x and y ($\bar{x}, \bar{y}$), the equation for intercept is

$$a = \bar{y} - b\bar{x}$$

A later section in the book shows how to modify those equations when you don't want your regression line to go through ($\bar{x}, \bar{y}$). An example of how to use these regression equations is shown in the next section.

A Regression Example For A Television Show

Modern Family is a fairly popular American sitcom that airs on ABC. As of the time of this writing, there have been 7 seasons that have aired, and it is in the middle of season 8. American television shows typically have 20-24 episodes in them. (Side note, I wanted to do this example with a British television show, but, sadly, couldn't find any that had more than 5 episodes in a season.) Modern Family, along with many shows, experiences a trend where the number of viewers starts high at the beginning of the season and then drops as the season progresses.

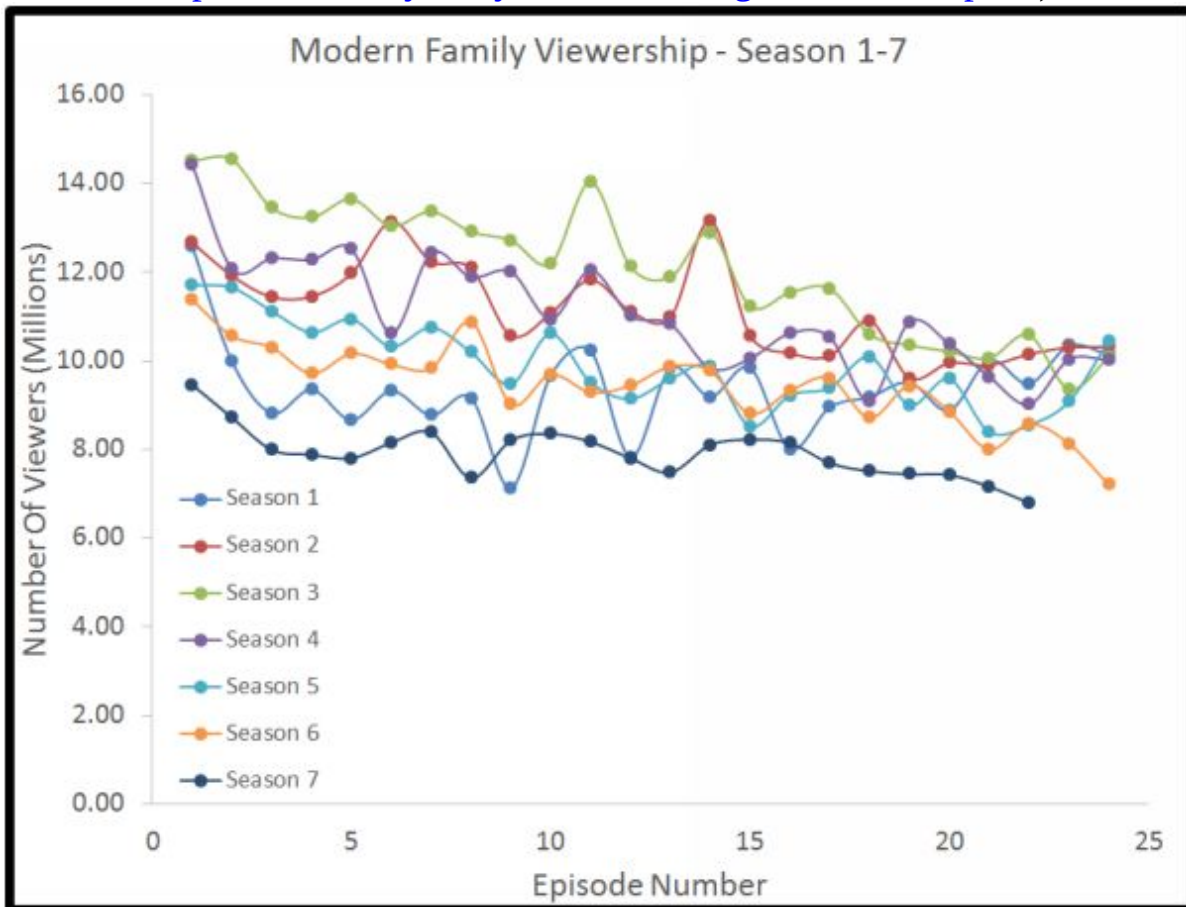


Let's pretend that you are an advertising executive about to make an ad purchase with ABC. The premiere of "Modern Family" season 8 has just been shown, and you are deciding whether to buy ads for the rest of the season or more importantly, how much you are willing to pay for them.

All you care about is getting your product in front of as many people as possible as cheaply as possible. And if an episode of “Modern Family” will only deliver 6 million viewers, you won’t pay as much for an ad as if it had 10 million viewers.

You could just believe the television company when they tell you their expected viewership for the season, or you could do a regression analysis and make your own prediction.

This is a chart of the data you have for viewership of the first 7 seasons of Modern Family. (Pulled from Wikipedia here https://en.wikipedia.org/wiki/List_of_Modern_Family_episodes , along with the other examples in this book compiled into a spreadsheet you can get for free here <http://www.fairlynerdy.com/linear-regression-examples/>)



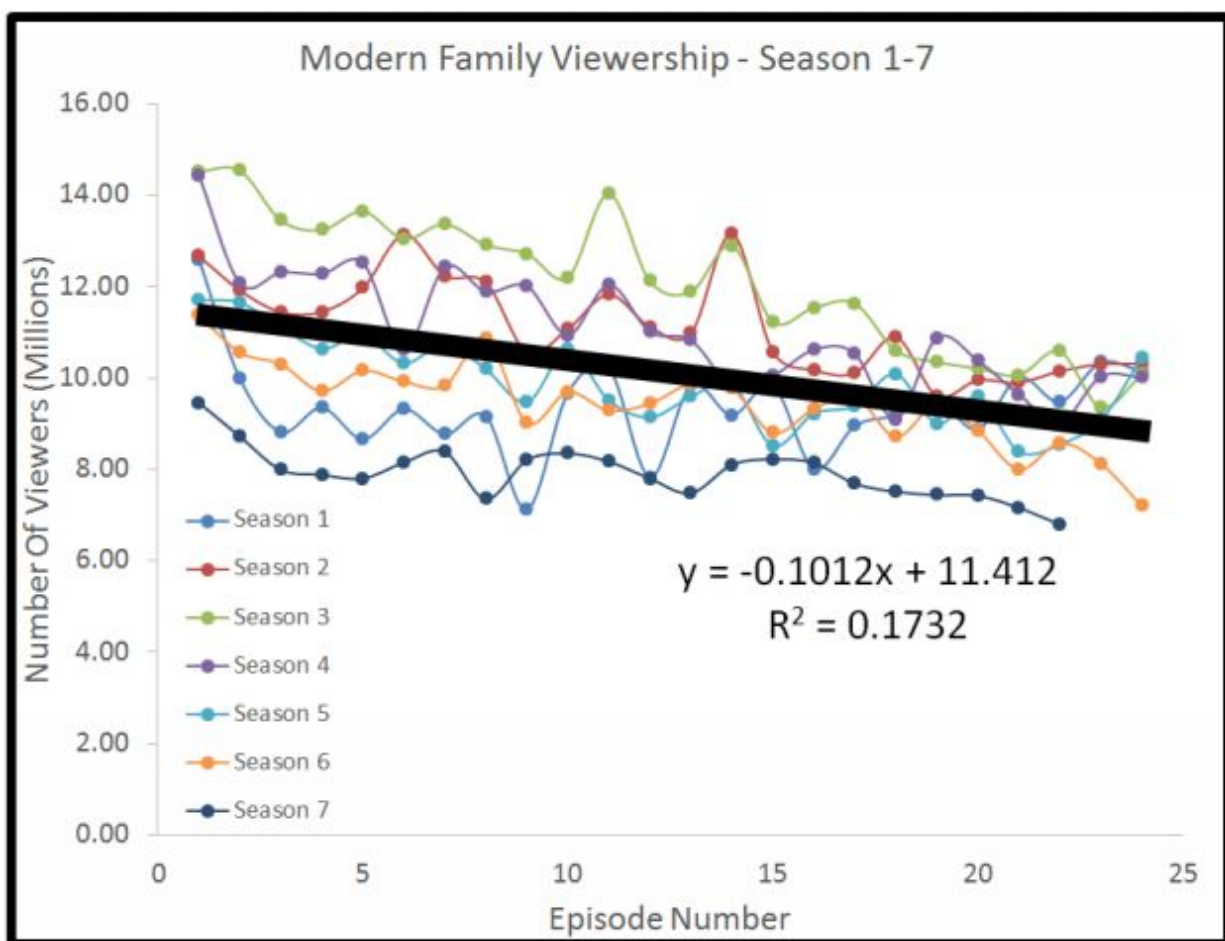
Each line represents a distinct season. The x-axis is the episode number in a given season, and the y-axis is the number of viewers in millions. As you suspected, there is a clear drop-off in number of viewers as the weeks

progress. But there is also quite a bit of scatter in the data, particularly between seasons.

Below is the same data in a table.

Number of Viewers Of Modern Family - Unmodified Data								
Season	1	2	3	4	5	6	7	8
Episode	Viewers (Millions)							
1	12.60	12.67	14.52	14.44	11.71	11.38	9.46	8.24
2	9.99	11.92	14.54	12.08	11.65	10.56	8.72	7.41
3	8.82	11.44	13.45	12.31	11.12	10.30	7.99	6.97
4	9.35	11.45	13.24	12.28	10.64	9.71	7.88	7.50
5	8.66	11.97	13.65	12.52	10.94	10.16	7.80	7.38
6	9.32	13.14	13.04	10.62	10.32	9.92	8.15	7.23
7	8.77	12.24	13.37	12.43	10.75	9.83	8.38	7.33
8	9.16	12.09	12.91	11.89	10.19	10.88	7.35	6.81
9	7.12	10.57	12.72	12.01	9.47	9.02	8.20	6.81
10	9.67	11.07	12.20	10.94	10.61	9.69	8.35	7.57
11	10.22	11.83	14.03	12.04	9.51	9.29	8.17	7.59
12	7.83	11.12	12.12	11.01	9.14	9.44	7.80	
13	9.83	10.98	11.89	10.83	9.59	9.87	7.48	
14	9.19	13.16	12.90	9.83	9.87	9.77	8.10	
15	9.84	10.57	11.23	10.05	8.52	8.80	8.22	
16	8.01	10.17	11.54	10.62	9.22	9.32	8.14	
17	8.95	10.11	11.63	10.53	9.39	9.61	7.69	
18	9.18	10.90	10.60	9.09	10.09	8.71	7.51	
19	9.51	9.61	10.35	10.88	9.00	9.43	7.44	
20	8.88	9.95	10.21	10.38	9.59	8.85	7.42	
21	10.01	9.90	10.06	9.64	8.39	8.00	7.16	
22	9.48	10.15	10.58	9.02	8.55	8.57	6.79	
23	10.34	10.30	9.36	10.03	9.08	8.13		
24	10.14	10.31	10.07	10.01	10.45	7.20		

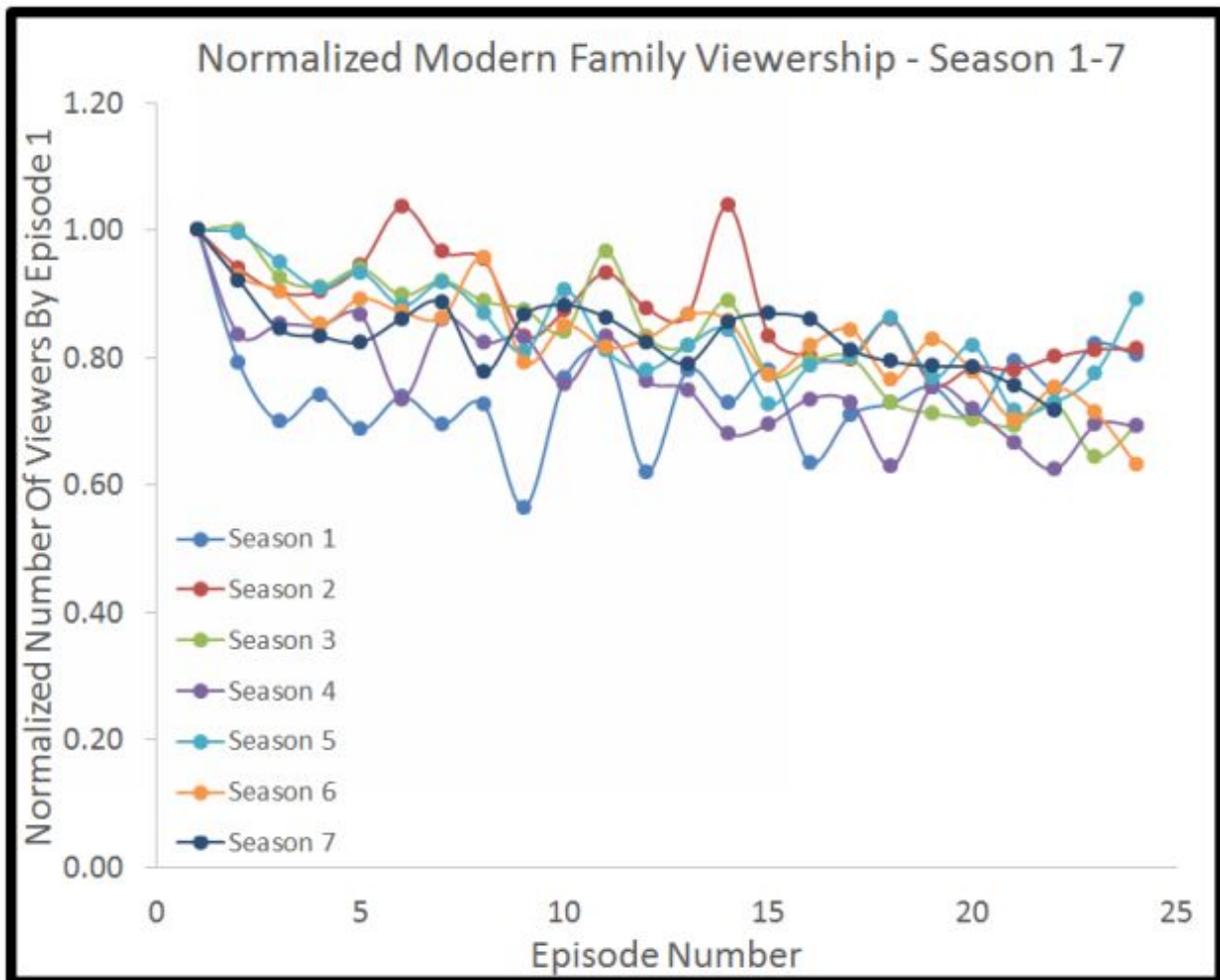
In order to scope out the problem, before diving into the equations for how to calculate a regression line, let's just see a regression line generated by Excel. A linear regression line of all the data is shown in the thick black line in the chart below.



The regression line doesn't appear to match the data particularly well, and the intercept of the regression line is quite a bit away from the number of viewers for the season 8 premiere. If you make your prediction based on this line, you'll probably end up with fewer viewers than you expected.

One solution is to normalize the input data based on the number of viewers in the premiere of each season. That is, divide the viewers from every episode by the number of viewers in episode 1 of its respective season. This ends up with much tighter data clustering. You've essentially removed all the season to season variation, and just have a single variable plotted to show the

change within a season



We did the previous regression line in Excel, but we will do this one manually in order to demonstrate the math.

Even though this is 7 seasons, it is really one data set. Instead of each season getting its own column in the data table, it is easier to put all 166 data points in one column.

Episode Number		
	In A Season	Normalized Viewers
	(X)	(Y)
Season 1	1	1.00
	2	0.79
	3	0.70
	4	0.74
	5	0.69
	6	0.74
	7	0.70
	8	0.73
	9	0.57
	10	0.77
	11	0.81
	12	0.62
	13	0.78
	14	0.73
	15	0.78
	16	0.64
	17	0.71
	18	0.73
	19	0.75
	20	0.70
	21	0.79
	22	0.75
	23	0.82
	24	0.80
Season 2	1	1.00
	2	0.94
	3	0.90
	4	0.90
	5	0.94
	6	1.04
	7	0.97
	8	0.95

Note, truncated data

To generate a regression line, we have to solve for the ‘a’ and ‘b’ coefficients in the equation $y = bx + a$, where ‘a’ is the intercept and ‘b’ is the slope of the

regression line. We'll start by finding the slope of the regression line, then the intercept. We have already seen the equation for the slope, b, it is

$$b = r * \frac{s_y}{s_x}$$

To refresh, the equation for Pearson's correlation, r, is

$$r = \frac{\sum ((x - \bar{x}) * (y - \bar{y}))}{(n - 1) * s_x * s_y}$$

Where s_y is the sample standard deviation of y, and s_x is the sample standard deviation of x.

First, let's calculate $\bar{x}$ and $\bar{y}$, because these are simple. They are just the average of the 166 data points.

Average	12.367	0.819
Count	166	

Episode Number In A Season (X)	Normalized Viewers (Y)
1	1.00
2	0.79
3	0.70
4	0.74
5	0.69
6	0.74
7	0.70
8	0.73
9	0.57
10	0.77

Note, truncated data

The averages that we get are 12.367 for episode number, and .819 for normalized number of viewers. The episode number is an average of 12.367 because we start with episode 1, and end with episode 24 for most seasons. This makes the average episode number 12.5 for those seasons, even though half of the number of episodes would be 12. (Note, one season only had 22 episodes, so the average episode number for all the seasons ended up 12.367 not 12.5). The .819 for the average number of viewers means that any given episode in a season got, on average, 81.9% of the number of viewers received for that season premiere.

Next, we will make another column and put the result of each x minus $\bar{x}$ in that column. This is $(x - \bar{x})$. We will do the same for y to get $(y - \bar{y})$

Season 1	Episode Number In A Season (X)	Normalized Viewers (Y)	(x - $\bar{x}$)	(y - $\bar{y}$)
	1	1.00	-11.37	0.18
	2	0.79	-10.37	-0.03
	3	0.70	-9.37	-0.12
	4	0.74	-8.37	-0.08
	5	0.69	-7.37	-0.13
	6	0.74	-6.37	-0.08
	7	0.70	-5.37	-0.12
	8	0.73	-4.37	-0.09
	9	0.57	-3.37	-0.25
	10	0.77	-2.37	-0.05
	11	0.81	-1.37	-0.01
	12	0.62	-0.37	-0.20
	13	0.78	0.63	-0.04
	14	0.73	1.63	-0.09
	15	0.78	2.63	-0.04
	16	0.64	3.63	-0.18

All Modern Family tables of this type have truncated data

We can multiply each pair of cells in those two columns to get $(x - \bar{x}) * (y - \bar{y})$.
And summing that column gives us

$$\sum ((x - \bar{x}) * (y - \bar{y}))$$

The $(x - \bar{x})$ and $(y - \bar{y})$ columns can be squared to get $(x - \bar{x})^2$ and $(y - \bar{y})^2$

respectively. Those can be summed for

$$\sum (x - \bar{x})^2$$

And

$$\sum (y - \bar{y})^2$$

Putting those equations into the table results in

Average		12.367	0.819	Column			
Count		166		Sum	-66.00	7804.58	1.46

Season 1	Episode Number In A Season (X)	Normalized Viewers (Y)	$(x - \bar{x})$	$(y - \bar{y})$	$(x - \bar{x}) * (y - \bar{y})$	$(x - \bar{x})^2$	$(y - \bar{y})^2$
	1	1.00	-11.37	0.18	-2.06	129.22	0.03
	2	0.79	-10.37	-0.03	0.27	107.48	0.00
	3	0.70	-9.37	-0.12	1.12	87.75	0.01
	4	0.74	-8.37	-0.08	0.64	70.01	0.01
	5	0.69	-7.37	-0.13	0.97	54.28	0.02
	6	0.74	-6.37	-0.08	0.51	40.54	0.01
	7	0.70	-5.37	-0.12	0.66	28.81	0.02
	8	0.73	-4.37	-0.09	0.40	19.07	0.01
	9	0.57	-3.37	-0.25	0.86	11.34	0.06
	10	0.77	-2.37	-0.05	0.12	5.60	0.00
	11	0.81	-1.37	-0.01	0.01	1.87	0.00
	12	0.62	-0.37	-0.20	0.07	0.14	0.04
	13	0.78	0.63	-0.04	-0.02	0.40	0.00
	14	0.73	1.63	-0.09	-0.15	2.67	0.01
	15	0.78	2.63	-0.04	-0.10	6.93	0.00
	16	0.64	3.63	-0.18	-0.67	13.20	0.03

If we divide the $(x - \bar{x})^2$ and $(y - \bar{y})^2$ sums each by $(n-1)$ i.e. $(166-1)$ and take the square root, we get the sample standard deviation of the x and y values of our data points.

The results we get are that the standard deviation in episode number is 6.878 and the standard deviation in normalized number of viewers is .094. If we didn't want to calculate the standard deviation using this method, we could have gotten the same result with STDEV.S() in Excel.

Std Deviation	6.878	0.094	Column			
Average	12.367	0.819	Sum	-66.00	7804.58	1.46
Count	166					

	Episode Number In A Season (X)	Normalized Viewers (Y)	(x - $\bar{x}$)	(y - $\bar{y}$)	(x - $\bar{x}$) * (y - $\bar{y}$)	(x - $\bar{x}$) ²	(y - $\bar{y}$) ²
Season 1	1	1.00	-11.37	0.18	-2.06	129.22	0.03
	2	0.79	-10.37	-0.03	0.27	107.48	0.00
	3	0.70	-9.37	-0.12	1.12	87.75	0.01
	4	0.74	-8.37	-0.08	0.64	70.01	0.01
	5	0.69	-7.37	-0.13	0.97	54.28	0.02
	6	0.74	-6.37	-0.08	0.51	40.54	0.01
	7	0.70	-5.37	-0.12	0.66	28.81	0.02
	8	0.73	-4.37	-0.09	0.40	19.07	0.01
	9	0.57	-3.37	-0.25	0.86	11.34	0.06
	10	0.77	-2.37	-0.05	0.12	5.60	0.00
	11	0.81	-1.37	-0.01	0.01	1.87	0.00
	12	0.62	-0.37	-0.20	0.07	0.14	0.04
	13	0.78	0.63	-0.04	-0.02	0.40	0.00
	14	0.73	1.63	-0.09	-0.15	2.67	0.01
	15	0.78	2.63	-0.04	-0.10	6.93	0.00
	16	0.64	3.63	-0.18	-0.67	13.20	0.03

Notice that the sum of $(x - \bar{x}) * (y - \bar{y})$ is a negative number, -66.00 in this case. We stated before that this sum controls the slope of the regression line, and the sign of the correlation value. Based on this negative value we know that episode number and number of viewers are negatively correlated, and that the slope of the regression line will be negative. Of course, we already knew that by looking at the scatter plot and seeing that the number of viewers decreases as each season progresses, but this is the mathematical basis of that result.

At this point, we have all the building blocks we need and can use this equation to get correlation

$$r = \frac{\sum ((x - \bar{x}) * (y - \bar{y}))}{(n - 1) * s_x * s_y}$$

$$r = \frac{-66.0}{(166 - 1) * 6.878 * .094} = -.619$$

And this equation to get the slope of the line

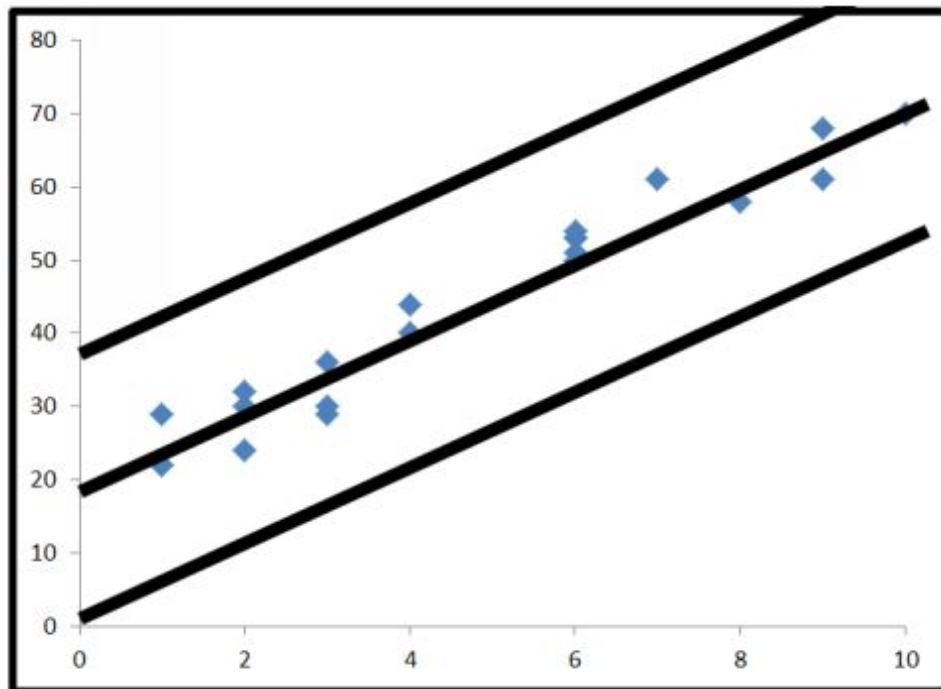
$$b = r * \frac{S_y}{S_x}$$

$$b = -.619 * \frac{.094}{6.878} = -.0085$$

So the slope of the regression line for this data is -.0085. Since this data is based on a percentage of the viewers of the first episode, this result means that each additional episode loses 0.85% of the viewers of the first episode, relative to the previous episode.

Regression Intercept

So far we have solved the slope of the regression line, but that is only half of what we need to fully define a line. (In this case, the slope is the more difficult half). The other piece of information that we need is the intercept of the line. Without an intercept, multiple different lines can have the same slope but be located differently. The chart below shows 3 lines with the same slope, but different intercepts for some sample data.



The way that we are going to calculate the intercept is to take the one point that we know the line passes through, and then use the slope to determine where that would fall on the y-axis.

The Intercept Of The Modern Family Data A line is defined by a slope and an intercept. We've solved the equations to find the slope; now we need to do the same thing for the intercept. The line equation is

$$y = a + bx$$

Rearranging for the intercept, a , we have

$$a = y - bx$$

Our slope equations used $\bar{x}$ and $\bar{y}$. That had the effect of forcing the regression line through $(\bar{x}, \bar{y})$. (More on that later.) Since we know the regression line goes through $(\bar{x}, \bar{y})$ we can substitute in those mean values for

$$\begin{array}{c}
 \text{Intercept} \quad \quad \quad \text{Slope} \\
 \downarrow \quad \quad \quad \downarrow \\
 a = \bar{y} - b\bar{x} \\
 \uparrow \quad \quad \quad \nearrow \\
 \text{Y \& X Average}
 \end{array}$$

x and y and get

So for this example, the intercept is

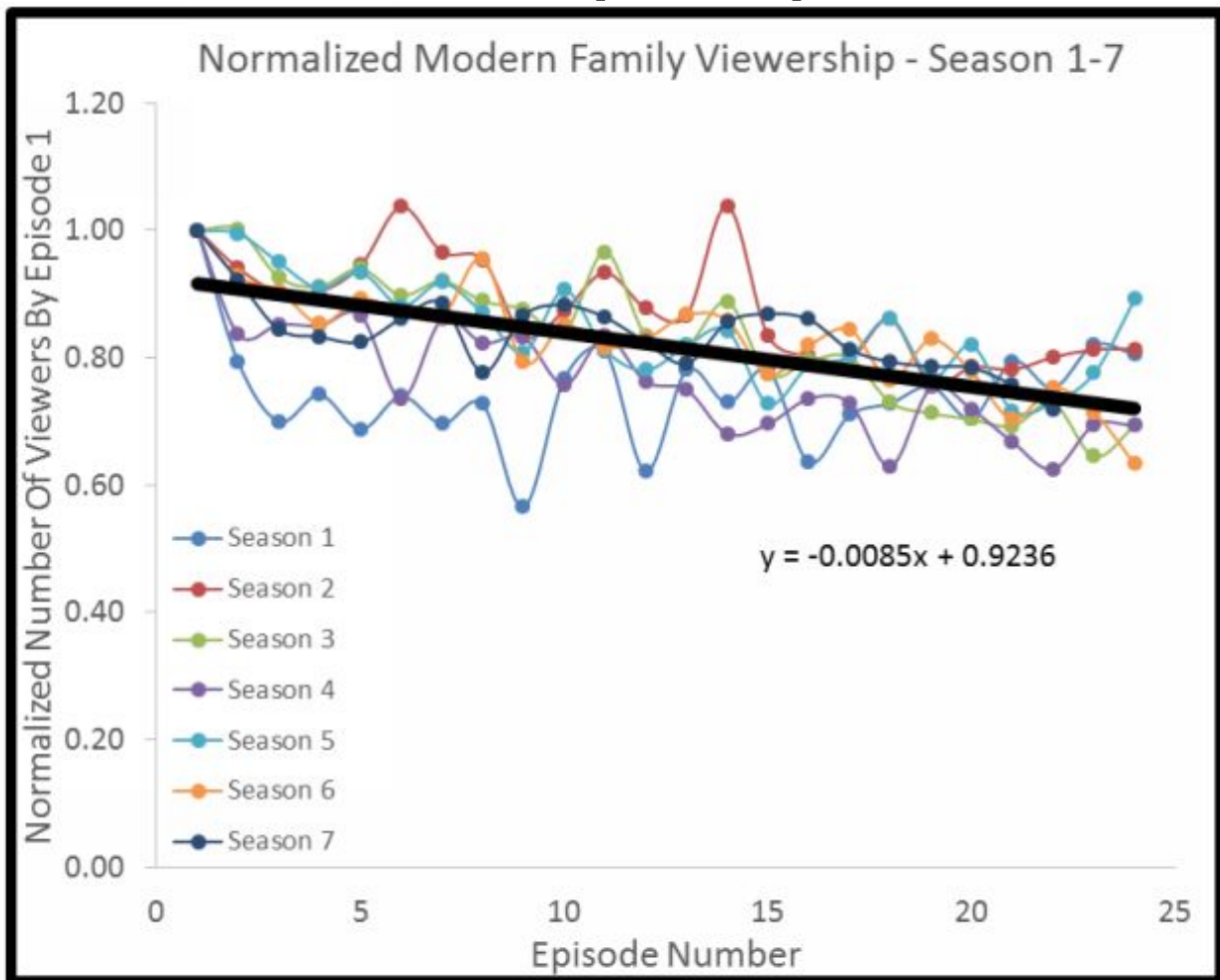
$$a = .819 - (-.0085 * 12.367) = .9236$$

Now that we have solved for the slope and the intercept, our final regression equation is

$$y = .9236 - .0085x$$

We know that the number of viewers for episode 1 of each season of the data should be 1.0 because we forced it to be so by using episode 1 to normalize the data. The intercept value of .9236 (less .0085 for the first episode) shows that we are under predicting the first episode. We can plot this regression

line to see how it looks for the other episodes compared to the actual data



The regression line is clearly capturing the overall trend of the data, and just as clearly is not capturing all of the episode to episode variations.

Calculating R-Squared of the Regression Line

To get a quantitative assessment of how good the linear regression line is, we can calculate the R^2 value.

$$R^2 = 1 - \frac{SS_{Regression}}{SS_{Total}}$$

While calculating the regression line, we already calculated the Summed Squared Total Error. The equation for Sum Squared Total Error is

$$SS_{Total} = \sum (y_i - \bar{y})^2$$

The total sum of $(y - \bar{y})$ squared was one of the columns we calculated in the regression analysis, so we can just reuse that value of 1.46

Std Deviation	6.878	0.094					
Average	12.367	0.819	Sum	-66.00	7804.58	1.46	
Count	166						
This Is Total Error							
Season 1	Episode Number In A Season (X)	Normalized Viewers (Y)	$(x - \bar{x})$	$(y - \bar{y})$	$(x - \bar{x}) * (y - \bar{y})$	$(x - \bar{x})^2$	$(y - \bar{y})^2$
	1	1.00	-11.37	0.18	-2.06	129.22	0.03
	2	0.79	-10.37	-0.03	0.27	107.48	0.00
	3	0.70	-9.37	-0.12	1.12	87.75	0.01
	4	0.74	-8.37	-0.08	0.64	70.01	0.01
	5	0.69	-7.37	-0.13	0.97	54.28	0.02
	6	0.74	-6.37	-0.08	0.51	40.54	0.01
	7	0.70	-5.37	-0.12	0.66	28.81	0.02
	8	0.73	-4.37	-0.09	0.40	19.07	0.01
	9	0.57	-3.37	-0.25	0.86	11.34	0.06

To get the regression squared error, we have to first make the prediction for each data point, using the regression equation.

$$y = .9236 - .0085x$$

We plug in each x to get a regression y for each point. Then for each data point, we can calculate the regression squared error

$$SS_{Regression} = \sum (y_i - y_{Regression})^2$$

We can calculate the regression value and error for each episode

Episode Number In A Season (X)	Normalized Viewers (Y)	Regression Normalized Viewers	Sum		Regression Error
				0.90	
				Regression Error, Squared	
1	1.00	0.915		0.007	-0.085
2	0.79	0.907		0.013	0.114
3	0.70	0.898		0.039	0.198
4	0.74	0.890		0.022	0.148
5	0.69	0.881		0.038	0.194
6	0.74	0.873		0.018	0.133
7	0.70	0.864		0.028	0.168
8	0.73	0.856		0.017	0.129
9	0.57	0.848		0.080	0.282
10	0.77	0.839		0.005	0.072
11	0.81	0.831		0.000	0.019
12	0.62	0.822		0.040	0.201
13	0.78	0.814		0.001	0.034

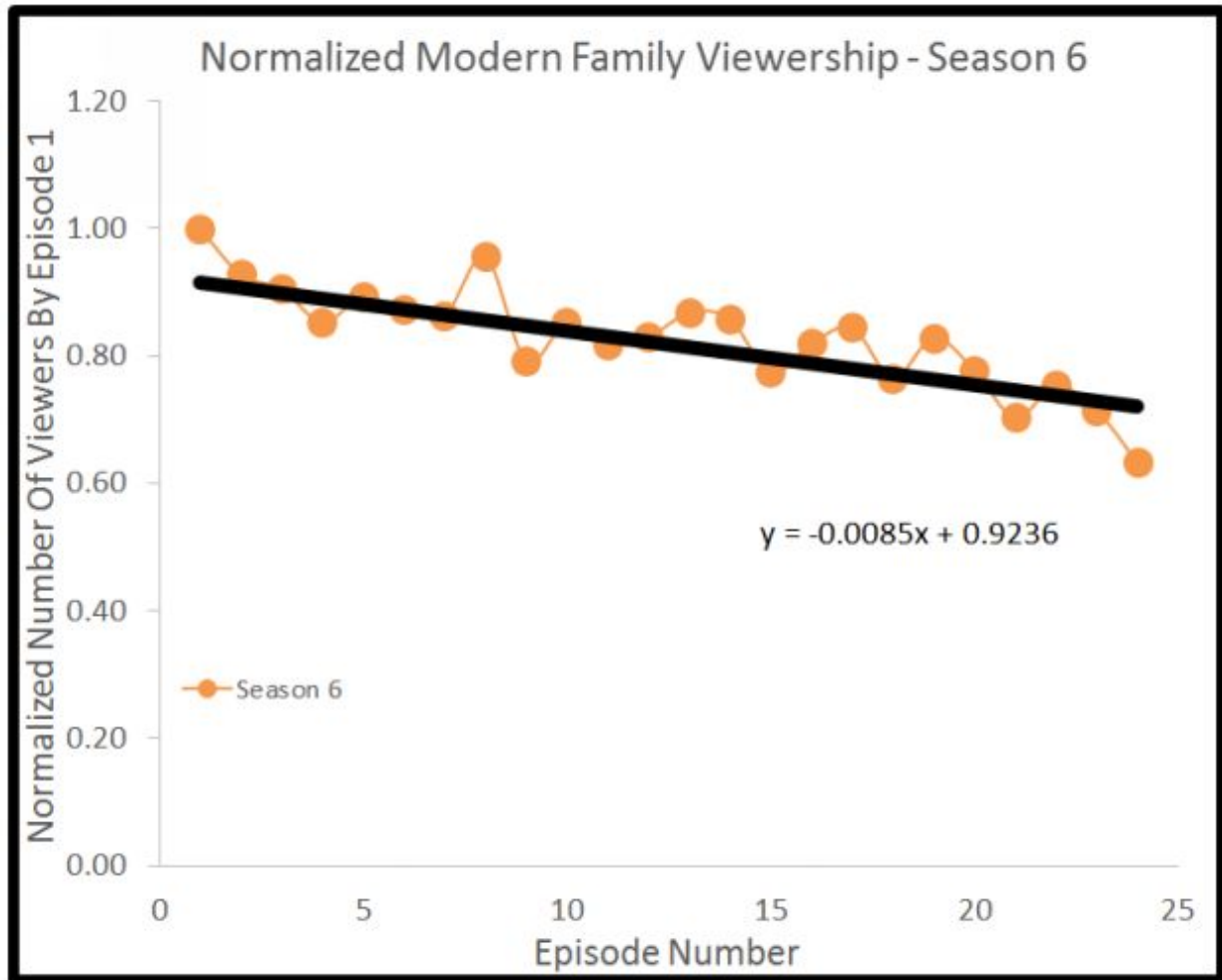
When we sum up all of the squares of the regression error, the value is .9.
The resulting equation for R^2 is

$$R^2 = 1 - \frac{.9}{1.46} = .383$$

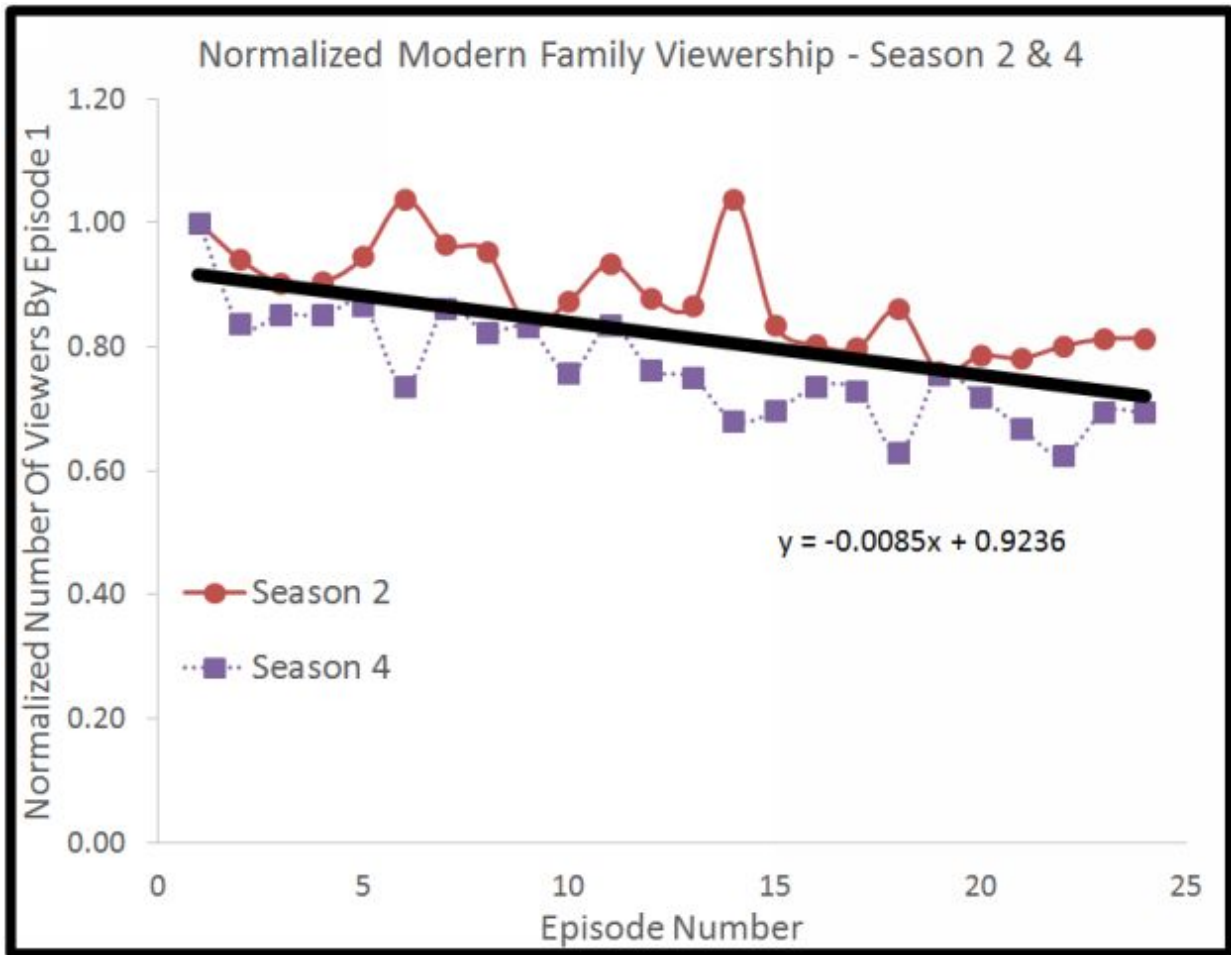
The result in the equation above is an R^2 value of .383. So is that a good value? Well, it is hard to say. If you as the advertising executive have a better model for predicting viewership, then this linear regression analysis won't get used. However are these results better than nothing, or better than just making a guess? Probably.

As a side note, at this point, we should make a comment on R^2 and r (correlation). Despite being the same letter, those values are not necessarily the same. R^2 will be the same as the square of r only if you have no constraints on the regression. If you put restraints on the regression, for instance by enforcing an intercept or some other point the regression line must pass through, then R^2 will not be the same as the correlation square. In this case we did not have any other constraints, so our R^2 value of .383 is the square of the correlation value of -.619.

Let's take a look at how this model would have done on a previous season. This is a view of Season 6

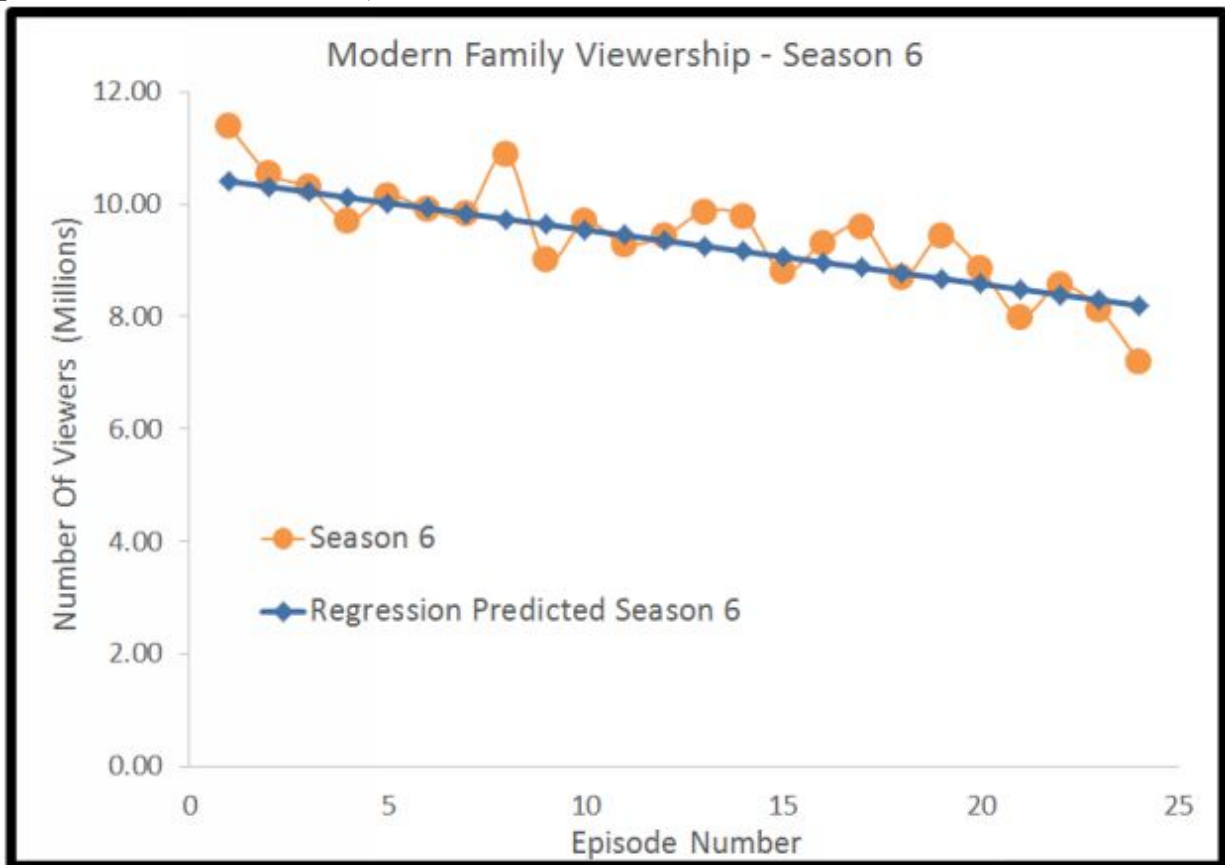


This line looks pretty good. After episode 1, it underpredicts the number of viewers in ~7 episodes, it overpredicts the number of viewers in ~5 episodes and gets the number pretty much exactly right in 9 episodes. Looking at the other seasons, seasons 3, 5, 6, 7 all seem fairly well predicted by this regression line. Season 1 has the largest error, probably because it was the first season and the viewership trends had not solidified yet. Season 2 was consistently under-predicted by this regression line, and season 4 was consistently over-predicted.



With this linear regression, we are predicting the ratio of future episodes to the first episode of the season. We can multiply this regression line by the number of viewers in episode 1 of each season, to get a regression prediction for total number of viewers. When that is plotted for total viewers, for season 6 the result is below. (Note the change in y scale relative to the

previous season 6 chart)



What we see for this season is that some episodes were predicted too high, and some too low, but overall the results aren't that bad. As an advertising executive buying for the entire season, you probably care about the total number of viewers through the season. If we sum the results for the second episode in each season through the last episode in each season (we are ignoring episode 1 because we are assuming that it already happened before

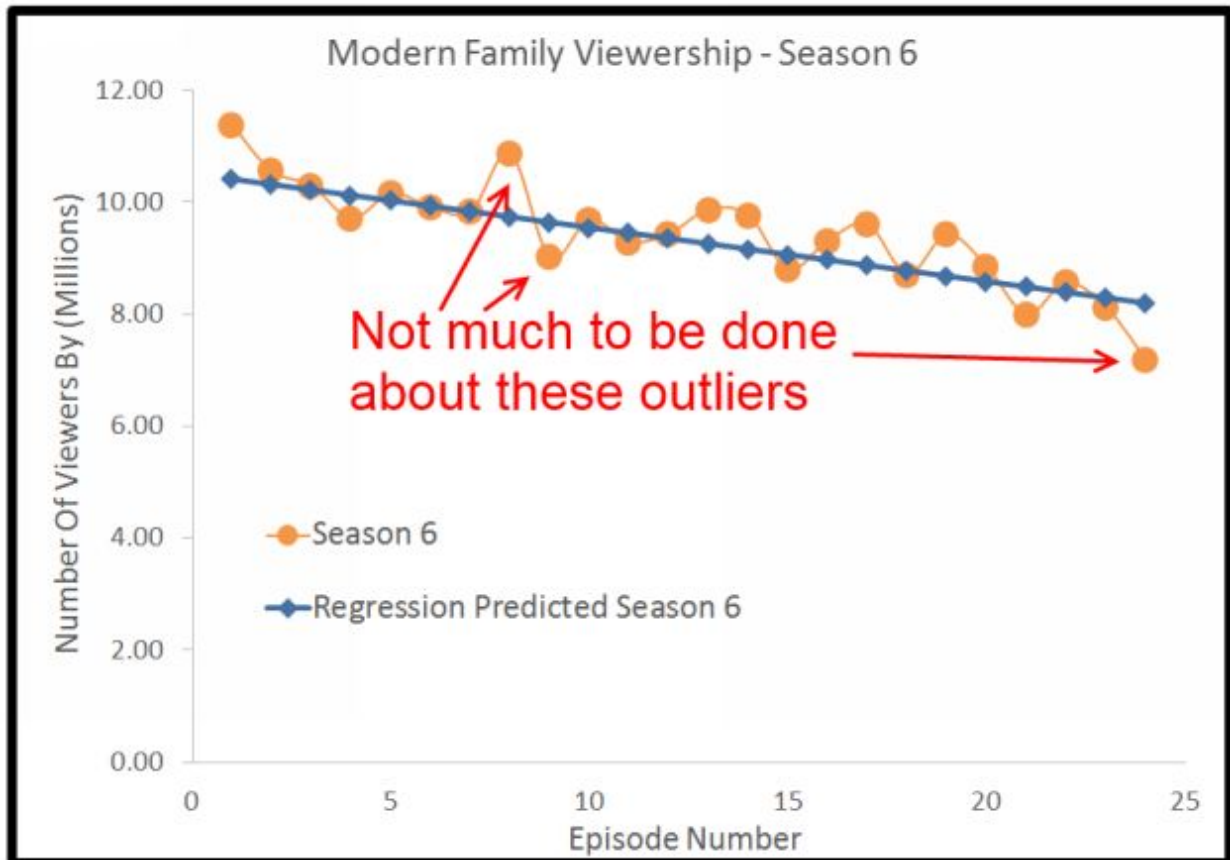
you are buying the ads), these are the results

Total Viewers (Millions) - Episode 2 - Season Finale				
Season	Actual Viewers	Regression Prediction	Error	Error %
1	212.27	235.81	-23.5	-11.1%
2	254.95	237.12	17.8	7.0%
3	275.69	271.74	4.0	1.4%
4	251.04	270.24	-19.2	-7.6%
5	226.08	219.15	6.9	3.1%
6	215.06	212.97	2.1	1.0%
7	164.74	163.33	1.4	0.9%

For the most recent seasons, we are only off by a few percent. Even Season 2 & 4 are only off by 7%. What we see here is that some of the values that were too high canceled out with the values that were too low. There are probably ways to refine this analysis to get a better estimate, but it is probably already better than the results you would get if you were not doing the analysis on your own and just trusting the sales people from the television studio.

Can We Make Better Predictions On An Individual Episode?

With linear regression based only on the episode number in a season, the results we got were pretty much as good as we can do. After all, there is only so much we can do to capture a wavy line with a straight line regression.

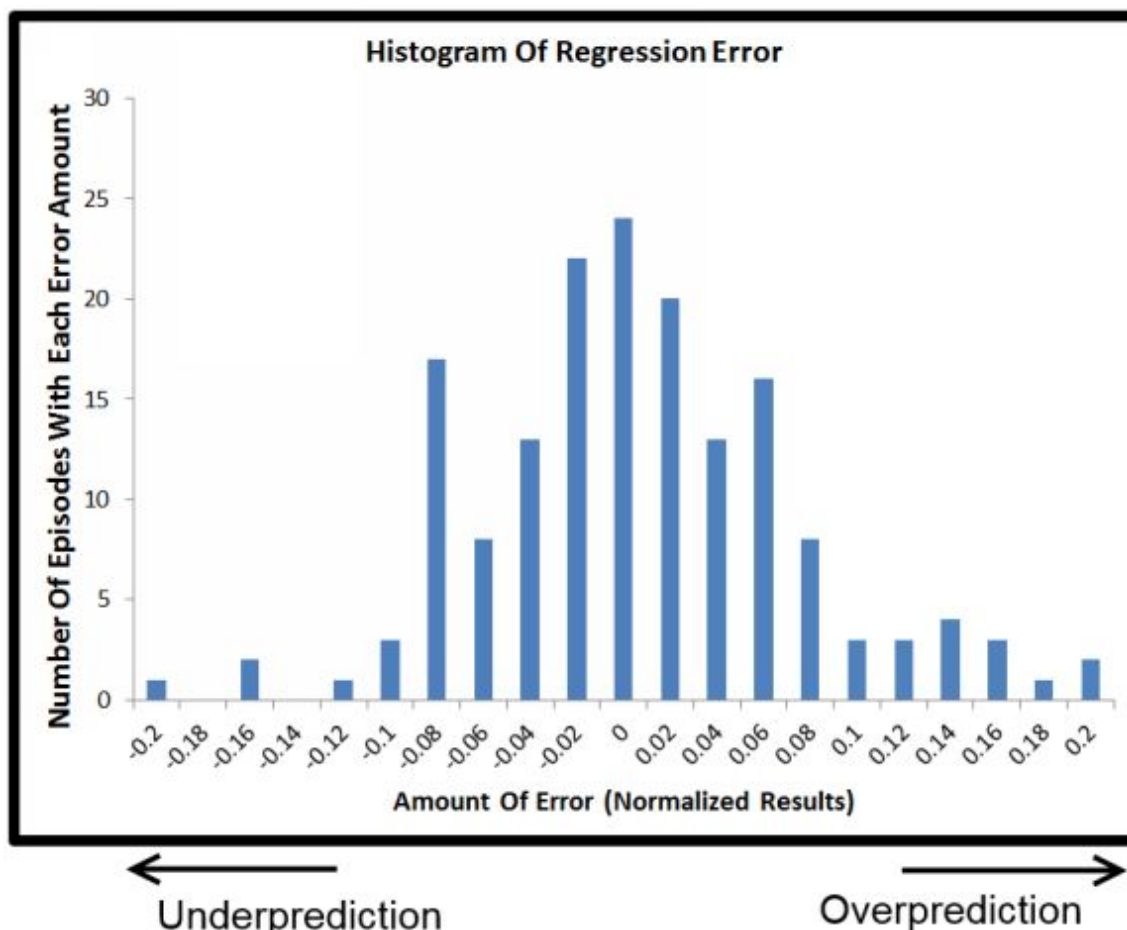


If you had more data, such as which episodes fell on holidays, what the ratings for the lead in show were, etc. and were using a more complicated Machine Learning technique, there very well could be additional patterns that you could extract. The season finale might always do poorly; the Christmas episode might always do well.

But with the data on hand, the above results are about as good as we can get with linear regression. However, even though we don't expect to improve our results, let's see if we can at least quantify how far off we expect the individual episodes to be from the regression line.

To do this, we can make a regression prediction for each episode, and subtract it from the actual value to get our error for each episode. We will do

this for the normalized numbers of viewers. When we calculated R^2 , we squared this value and summed it to get summed squared error. Here we will just take the error, group the error, count the number in each group and make a histogram of the results.

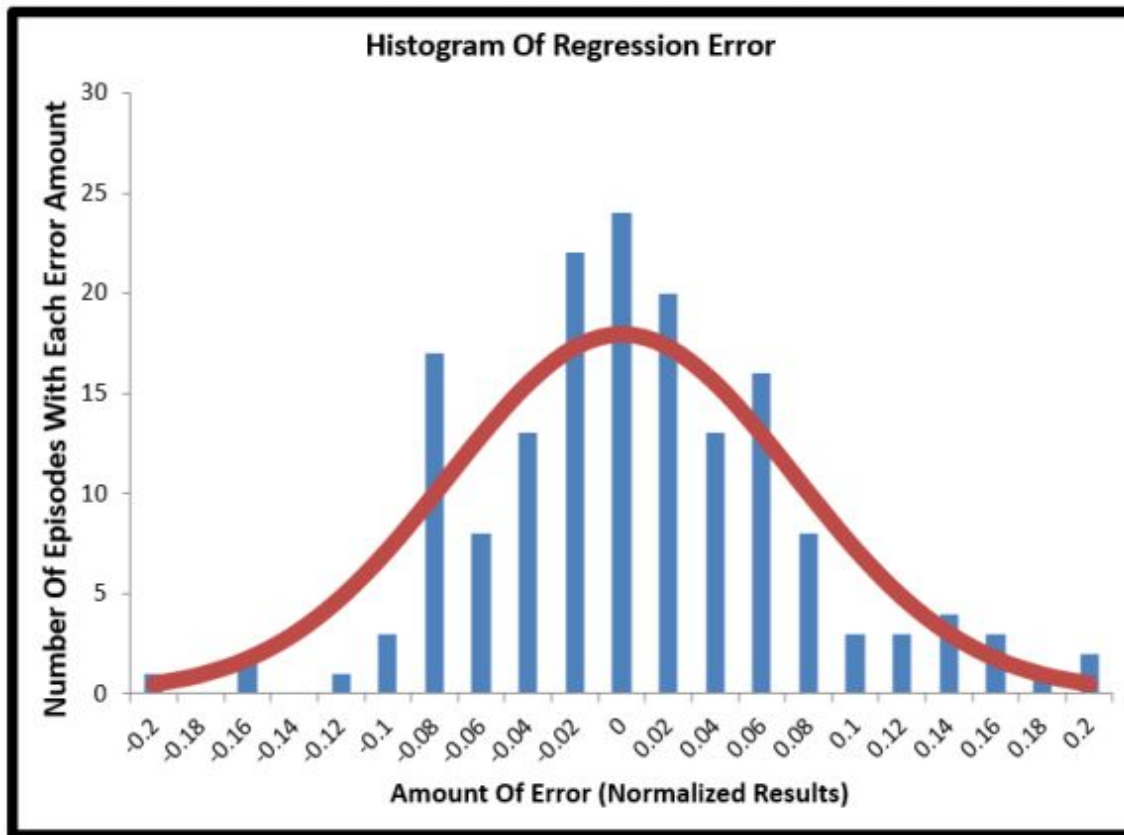


What we see is bell curve shape, centered around zero error. Which leads us to believe we can use typical normal distribution processes. i.e. we can find the standard deviation of the error and estimate that

- 68% are within 1 standard deviation
- 95% are within 2 standard deviations
- 99.7% are within 3 standard deviations

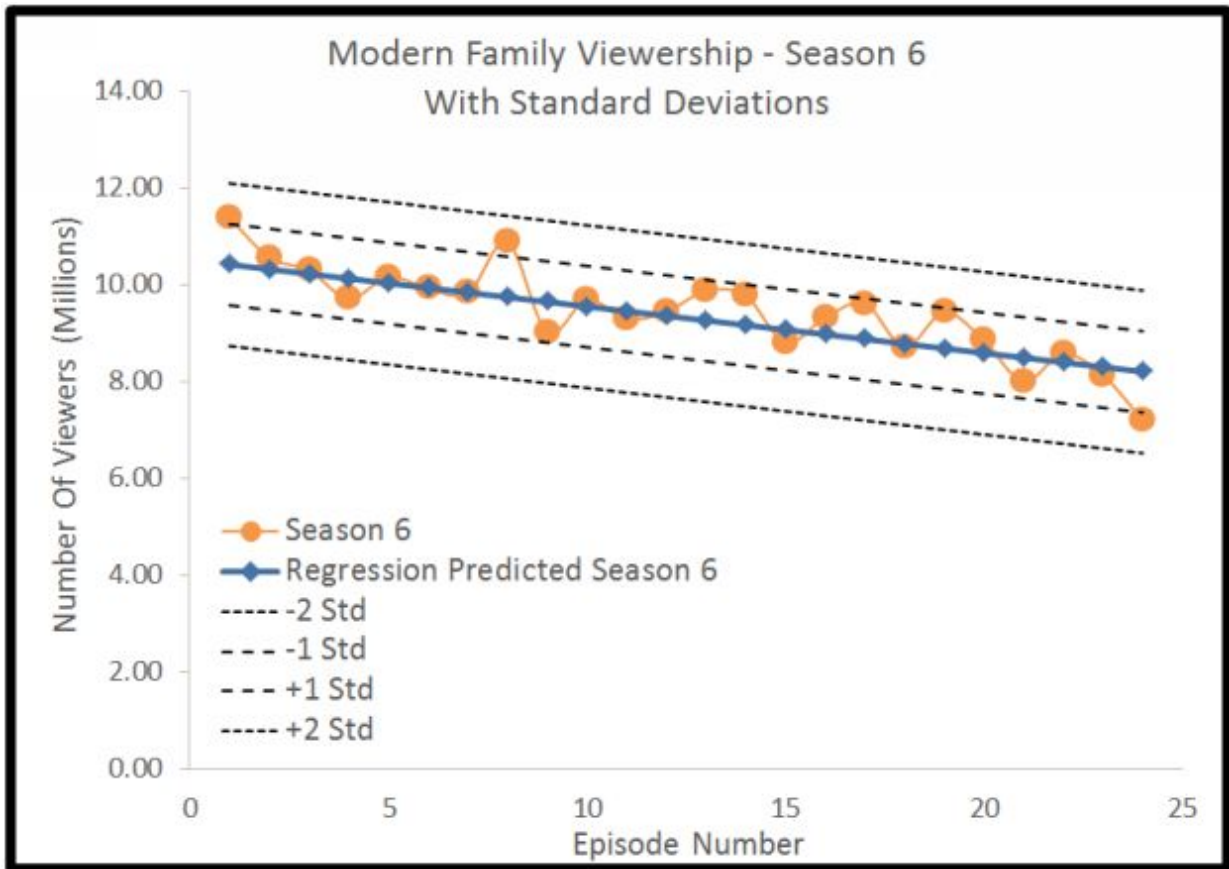
The standard deviation of the error of the regression line against the true data is .074. If we plot a normal curve with a standard deviation of .074 against this data, we see that the data is a reasonable representation of the normal

curve (although far from perfect)



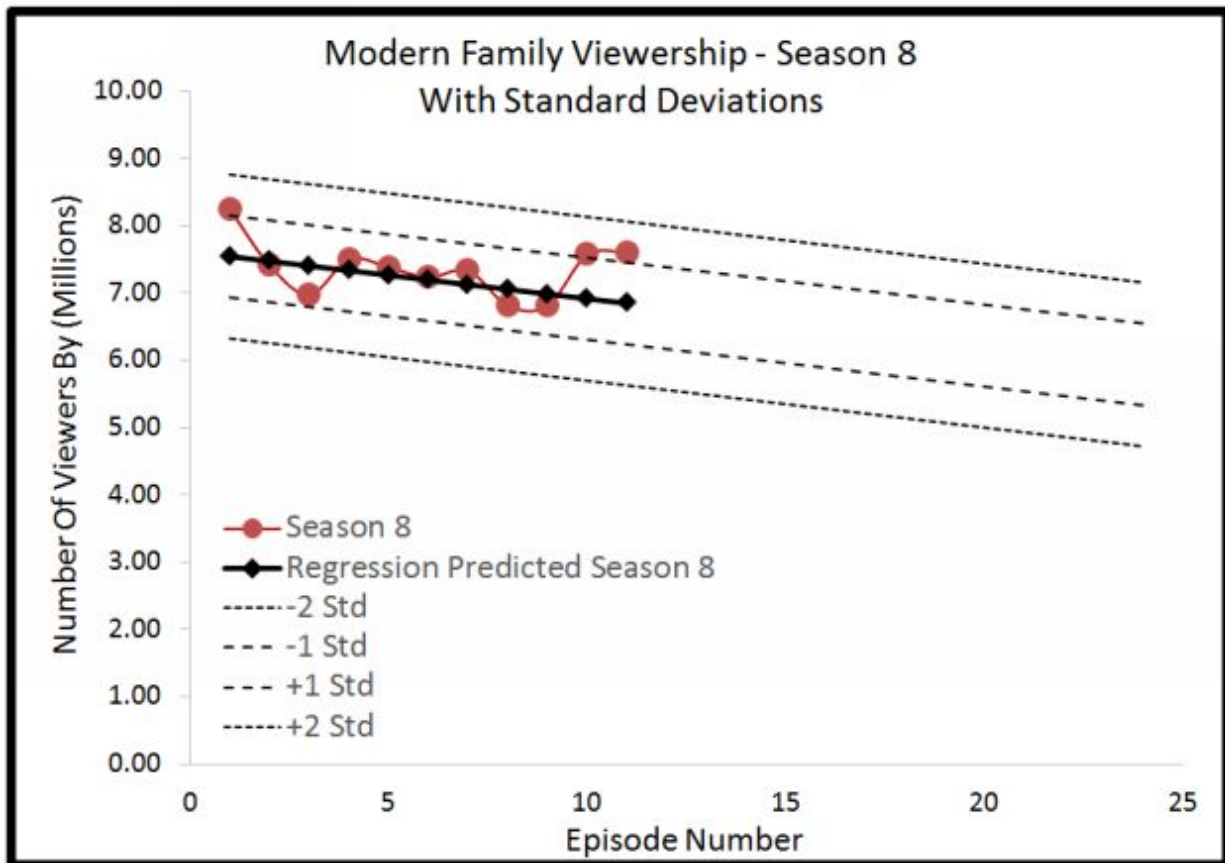
None-the-less, the normal approximation is close enough to make using a normal distribution reasonable for this data.

We can multiply the standard deviation of .074 by the number of viewers in episode 1 to get the standard deviation in number of viewers. If we put the regression line with 1 and 2 standard deviations around it, what we get is for season 6 is shown below.



As expected, most of the data points lie within 1 standard deviation and nearly all lie within 2 standard deviations of the regression line. Even though we can't predict viewership exactly for a given episode, we can use the regression equation to create the best fit and have some estimate of how much error we expect to see.

At the time of this writing, Modern Family is 11 episodes into season 8. Based on the equation above and on the viewership of episode 1 of season 8, here is the regression curve for season 8, with the first 11 episodes and expected error bands plotted.



Viewership results can be found in Wikipedia here https://en.wikipedia.org/wiki/List_of_Modern_Family_episodes if you want to see how well this projection did for future (future to me) results. And this spreadsheet can be downloaded for free here <http://www.fairlynerdy.com/linear-regression-examples/>.

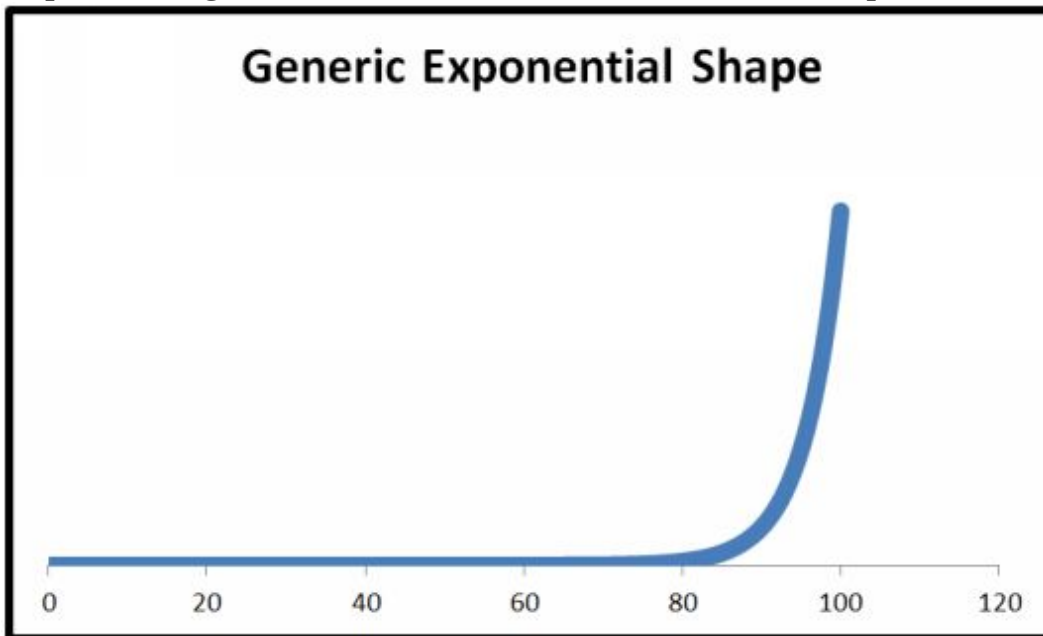
Exponential Regression – A Different Use For Linear Regression

There are some common types of data that a linear regression analysis is ill-suited for. It just doesn't get good results. One of those is when the input data is experiencing exponential growth. This occurs when the current value is a multiple of a previous value. Common occurrences of exponential growth can be found in things like investments or population growth.

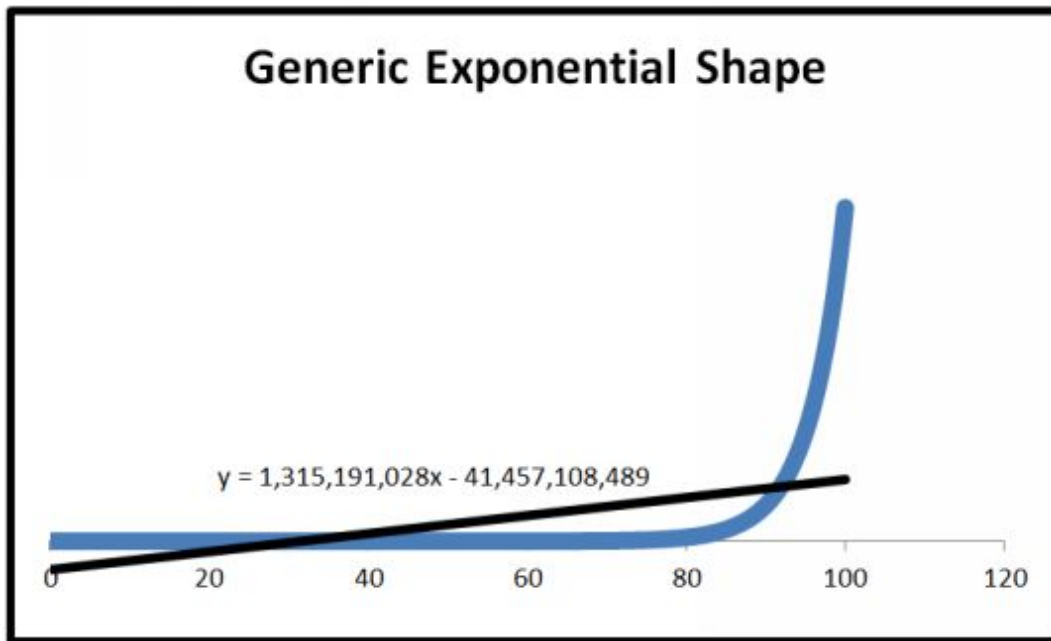
For instance, the amount of money you have in a bank might be the amount of money from last year plus 5% interest. The number of invasive wild rabbits loose in your country might be the number from last year plus 50% annual growth. Those are both examples of exponential functions.

This section shows how to use linear regression to do a regression analysis on an exponential function.

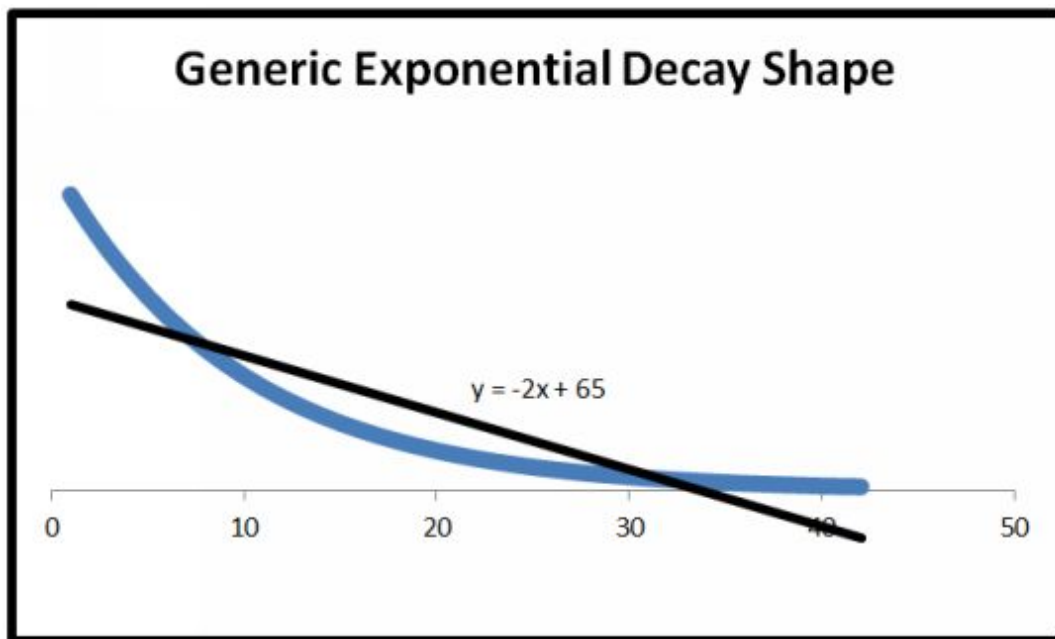
Exponential growth functions have a characteristic shape similar to



Where the amount of change increases each time step. If you attempt to fit a linear regression to this data, the result will be something like this

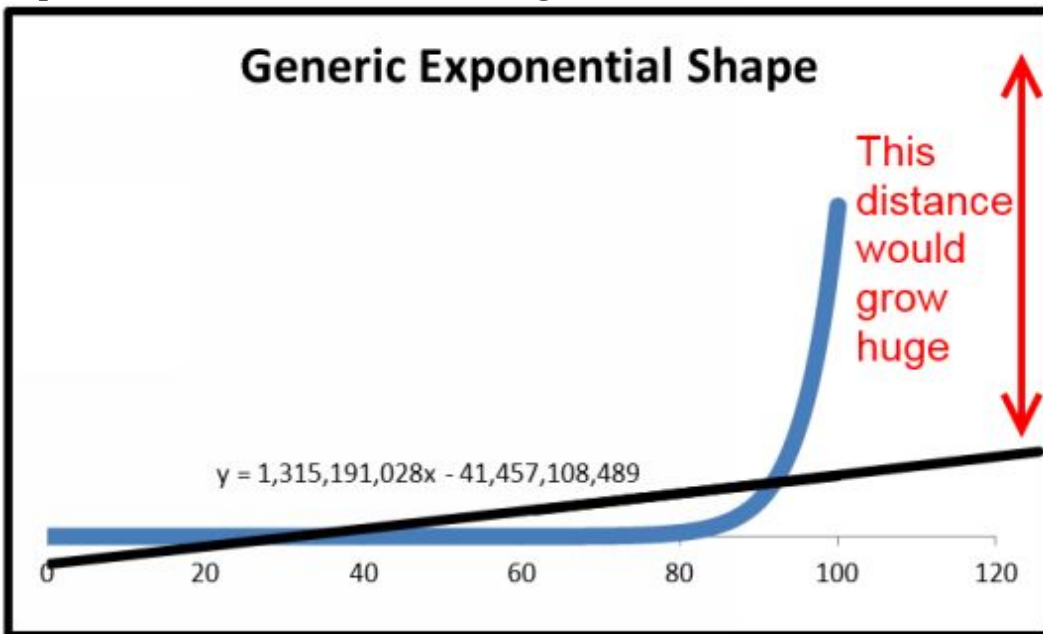


The linear curve will invariably be too low at either end and too high in the middle. This would also be true for exponential decay, as opposed to growth.



Using any regression line for extrapolation can be iffy, but on this exponential data with a linear curve fit, it will be extremely bad, because the

exponential will continue to diverge from the linear line



Fortunately, we can do an exponential curve fit instead of a linear one, and get an accurate regression line. And here is the beautiful part; we don't need a new regression equation. We can use the same linear regression equation that we have been using with one small piece of data manipulation before using the regression equation, and the inverse of that data manipulation after using the regression equation.

The Data Manipulation Trick

Exponential regression functions typically have the form

$$y = e^{a+bx}$$

Where e here is the mathematical constant used as the base of the natural logarithm, approximately equal to 2.71828. However, the process would work the same if you were working in base 10, or base 2, or anything else other than e.

Because of how the exponential works, this is the same as writing

$$y = e^a * e^{bx}$$

And since e^a is a constant, you typically see an exponential regression

$$y = c * e^{bx}$$


 $c = e^a$

function of the form

But whichever form the equation is in, it is just a manipulation of

$$y = e^{a+bx}$$

The $a+bx$ part of the equation is a line, and it should look familiar. If the equation were just $y = a + bx$, we could use linear regression. However, the exponential is getting in the way.

The inverse of e is the natural logarithm ln. If we take the natural log of both sides of the equation we get

$$\ln(y) = \ln(e^{a+bx})$$

$$\ln(y) = a + bx$$

At this point, we have manipulated the right side of the equation to be in the expected form for linear regression. We could do the regression as is, or we could modify the left side of the equation slightly to make it look more like the regression equation that we saw before. We can do that by defining another variable equal to ln(y). We will call that variable y'.

$$y' = \ln(y)$$

$$y' = a + bx$$

Now, this equation is in standard form, and we can do the linear regression as we typically do. The y' will remind us to do the inverse of the natural logarithm after we finish with the regression analysis.

So the steps we will follow are

- Obtain data relating x & y, and recognize it is an exponential function.
- Take the natural log of y
- Find the linear regression equation for x vs. ln(y)
- Solve for y by raising e to the result of that regression equation

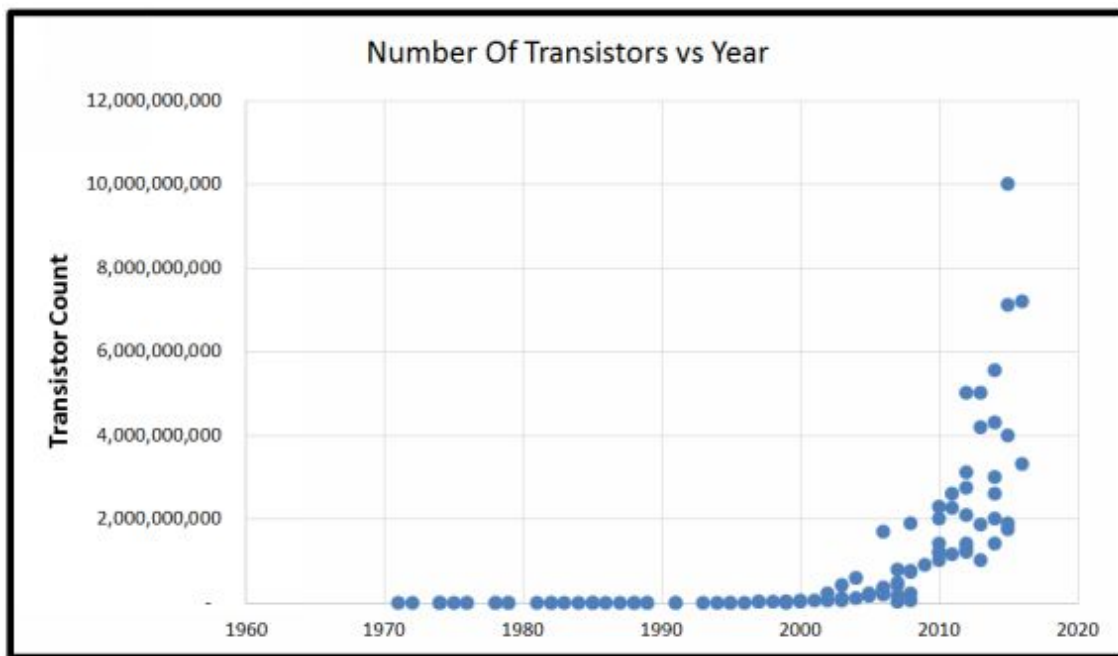
We are showing this process for exponentials, but the same process would work for any function that had an inverse. If you can apply a function to a set of data get a linear result as the output, you can do the linear regression on that result, and then apply the inverse of that regression to find the regression of the original data.

Exponential Regression Example – Replicating Moore’s Law

Here is an example of that process in action. This data is the number of transistors on a microchip, retrieved from this Wikipedia page https://en.wikipedia.org/wiki/Transistor_count January 2017. The Excel file with that data can be downloaded here <http://www.fairlynerdy.com/linear-regression-examples/>.

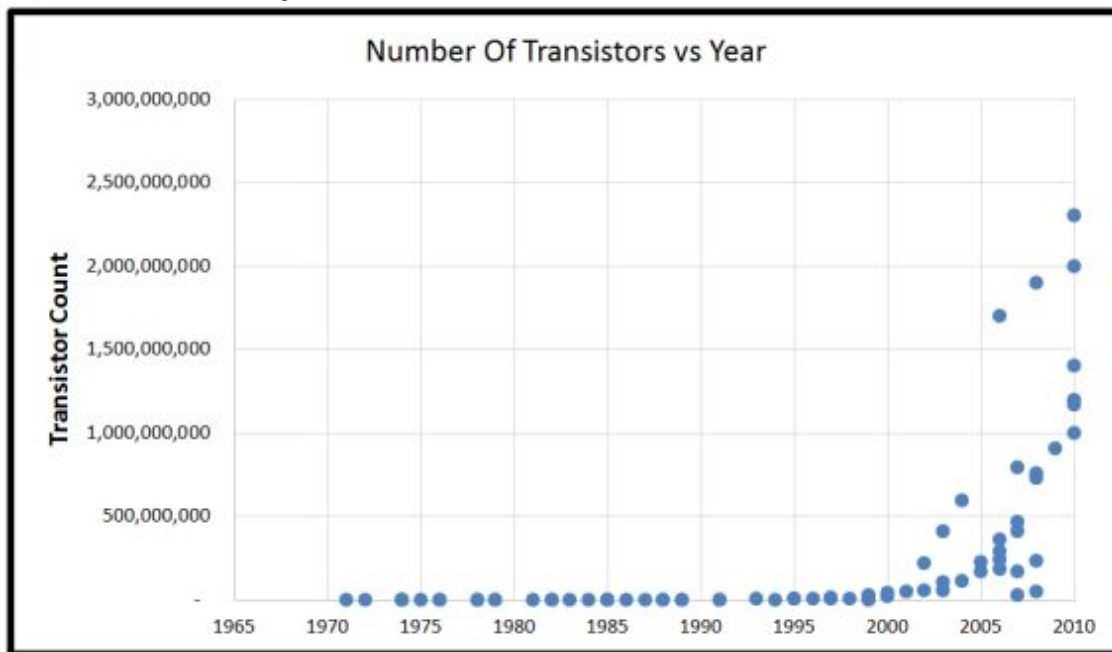
This is the type of data that famously gave rise to Moore’s law stating that the number of transistors on a chip will double every 18 months (later revised to doubling every 2 years). No attempt was made to scrub outliers from this data. So some of these chips could be from computers or phones, be expensive or cheap.

Let’s plot the data and see what we get



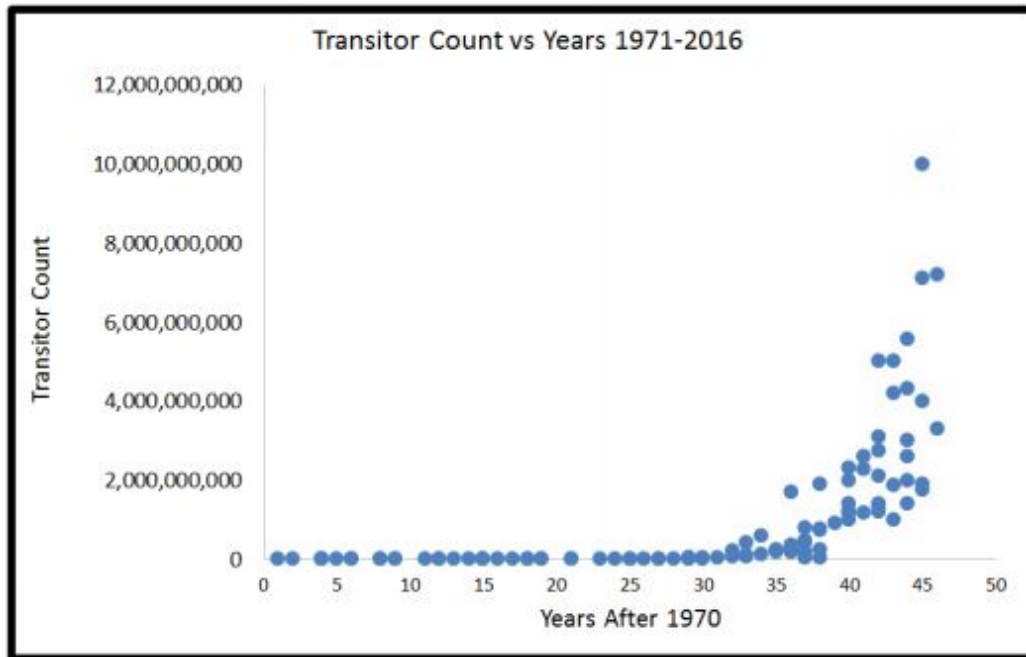
The first thing that we notice is that the data appears relatively flat early on but then gets really big really fast in the past several years.. If we take out

the most recent 5 years



It still looks relatively flat but then gets really big really fast. In fact, for pretty much any time segment it looks like the most recent few years are very large, and the time before that is small. This is characteristic of exponentials. They are always blowing up, with the most recent time period dwarfing what came before. The only difference is the rate that the change occurs at.

Let's make one change to the data. Instead of using the year for the x-axis, let's put the number of years after 1970. We could have left that constant in the data, but removing it by subtracting 1970 from the year will make the final regression equation a bit cleaner.

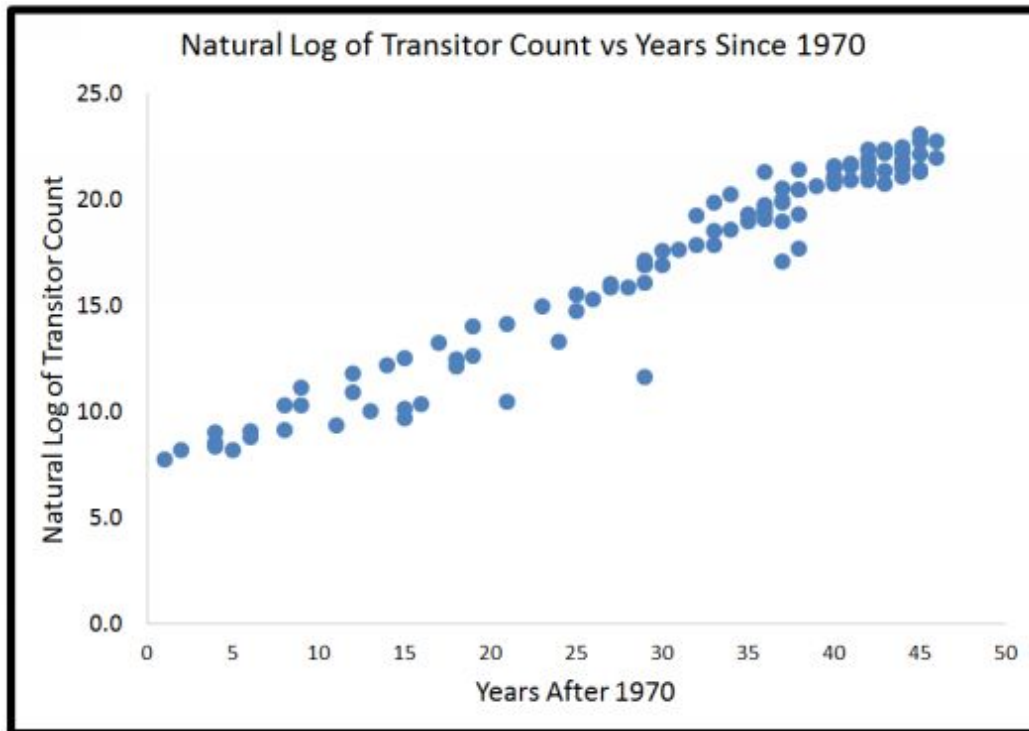


We know that a standard linear regression on this exponential data will be bad. So let's modify the data by taking the natural logarithm.

Year after 1970	Transistor count	ln(transistors)
1	2300	7.741
2	3500	8.161
4	4500	8.412
4	4100	8.319
4	5000	8.517
4	8000	8.987
5	3510	8.163
6	6500	8.780
6	8500	9.048
8	29000	10.275
8	9000	9.105
9	29000	10.275
9	68000	11.127
11	11500	9.350
12	55000	10.915
12	134000	11.806
13	22000	9.999
14	190000	12.155
15	25000	10.127
15	275000	12.525
15	16000	9.680
16	30000	10.309

Truncated Data

When plotted the modified data is



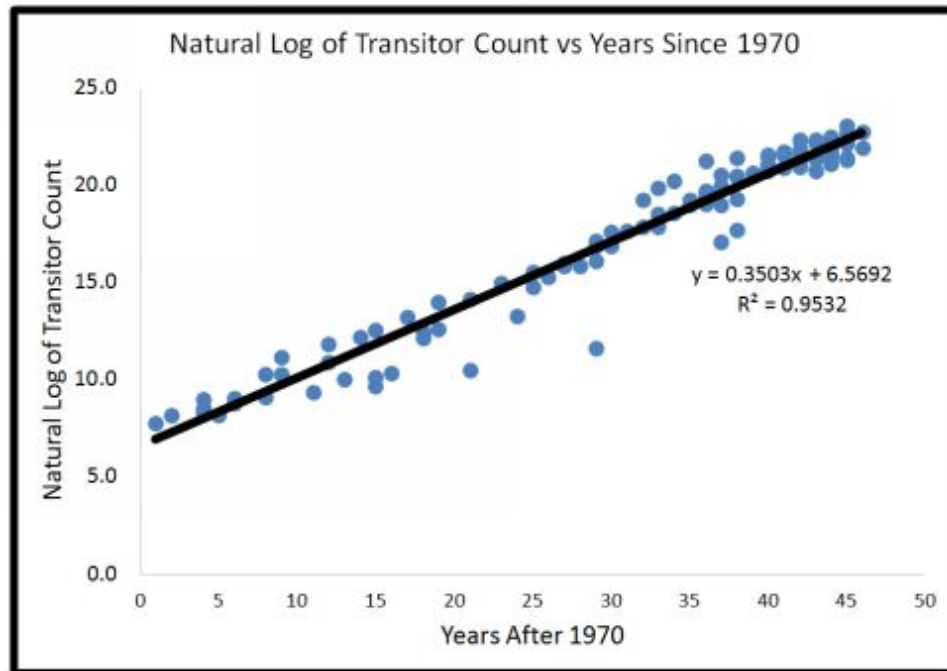
Now instead of blowing up at the end, the data is more or less linear. We can use the standard equations to get the slope and intercept just like in the previous examples. This results in

$$y' = .3503x + 6.5692$$

$$R^2 = .9532$$

With an R^2 of

Plotting that regression line on the data



And it is not bad. There are certainly some outliers, but on the whole, the regression line is capturing the trends of the data.

So the regression line we have is

$$y' = .3503x + 6.5692$$

But remember this y' was really the natural logarithm of our original data.
So our real regression line is

$$\ln(y) = .3503x + 6.5692$$

If we raise e to both sides, we get

$$e^{\ln(y)} = e^{.3503x + 6.5692}$$

The e and natural logarithm cancel out, and we get

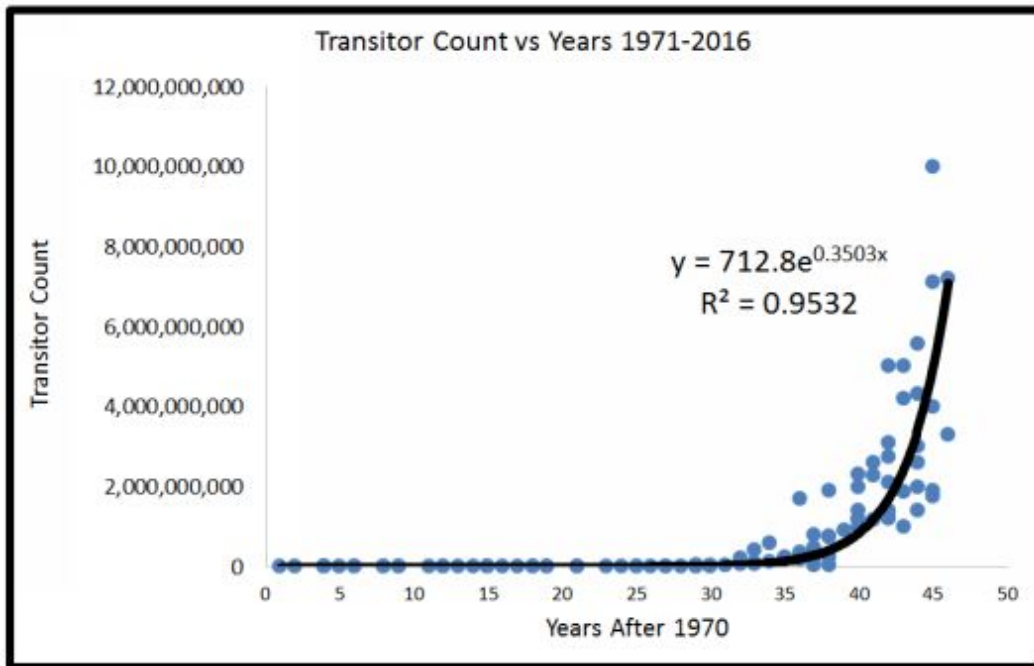
$$y = e^{.3503x+6.5692}$$

And that is the regression line. We can also rearrange it as

$$y = e^{6.5692} * e^{.3503x}$$

$$y = 712.8 * e^{.3503x}$$

This is the same result we would get from an exponential regression in Excel.



Key Points For Exponential Regression

- The c or e^a controls the intercept. When x equals zero

$$E^{bx} = e^b * 0 = e^0 = 1$$

- This mean the intercept is just c or e^a

- The b value controls the “slope.”

In the transistor example, b is .3503. And e^b is 1.419. e^{2b} is 2.015. This means every time x increase by 2, y will increase by a factor of 2.015

This is showing us that every 2 years (x increasing by 2) y goes up by a factor of approximately 2. As you probably know, Moore’s law is that the number of transistors per square inch on a component will double every two years. What we have done with this regression is show that y , the transistor count, increases by a factor of 2 every 2 years, effectively recreating Moore’s law.

Exponential Regression Side Note

One thing to be aware of with exponential regression is that it will not work with negative values or zero values. Your y variable must have strictly positive values. The reason is that it is impossible to raise a positive base, like e , to any value and get a negative number.

If you raise e to a positive value you will get a number greater than one. If you use a negative exponent, it is equivalent to taking 1 divided by e to the positive value of that exponent, which will give a value between zero and one. You can only get a value of zero by having an exponent of negative infinity, which isn't realistic for regression, and you can't raise a positive number to any exponent and get a negative number.

The result of this is that to do an exponential regression; all the Y values must be greater than zero. If you have data that you think would work with an exponential regression, but it has some negative values, you can try offsetting the Y values by adding a positive number to all the results or simply scrubbing those negative values from your dataset.

Linear Regression Through A Specific Point

So far all the regression we have done has had only one goal, generate the regression line which gives the lowest summed squared error, i.e. the line with the best R^2 . However, sometimes you might have an additional objective. You might want the best possible regression line that goes through a certain point. Often, the point specified is the y-intercept, and frequently that intercept is at the origin.

Why would you want to specify a point? Well, you might have additional information you want to capture, for instance, the data might be against time, and you know at time zero the y value should be zero.

It turns out that the linear regression equation we have learned so far is just a special case of the general equation that goes through any point that you specify. And that is good news because it means we only need a small modification to our knowledge to know how to put the regression line through any point we choose.

The equations we have used so far to calculate the slope of the regression

Note the $\bar{x}$ and $\bar{y}$

$$r = \frac{\sum ((x - \bar{x}) * (y - \bar{y}))}{(n - 1) * s_x * s_y}$$

line

$$s_x = \sqrt{\frac{\sum (x - \bar{x})^2}{(n - 1)}}$$

$$s_y = \sqrt{\frac{\sum (y - \bar{y})^2}{(n - 1)}}$$


$$b = r * \frac{s_y}{s_x}$$

In these equations, the $\bar{x}$ and $\bar{y}$ represent the average x and the average y. By specifying those points, we are forcing the line to go through $\bar{x}$ and $\bar{y}$. That is good news for regression, because the line needs to go through $(\bar{x}, \bar{y})$ to have the minimum R^2 .

However, the point that we specify doesn't have to be the average x and average y. We can, in fact, specify any x and y and force the regression line to go through that point instead of going through $\bar{x}$ and $\bar{y}$. So instead of $(\bar{x}, \bar{y})$ we can specify (x_0, y_0) where x_0 and y_0 represent any generic location that we

choose. When we do that, the slope equations become

$$r = \frac{\sum (x - x_0) * (y - y_0)}{(n - 1) * s_x * s_y}$$

$$s_x = \sqrt{\frac{\sum (x - x_0)^2}{(n - 1)}}$$

$$s_y = \sqrt{\frac{\sum (y - y_0)^2}{(n - 1)}}$$

For instance, if we want to force a y-intercept of 10, we would use $x = 0$ and $y = 10$ and set the equations to be

$$r = \frac{\sum((x - 0) * (y - 10))}{(n - 1) * s_x * s_y}$$

$$s_x = \sqrt{\frac{\sum (x - 0)^2}{(n - 1)}}$$

$$s_y = \sqrt{\frac{\sum (y - 10)^2}{(n - 1)}}$$

The most common point to force the regression line through, other than the mean, is probably the origin, (0,0). If you put (0,0) into those equations they

$$r = \frac{\sum ((x - 0) * (y - 0))}{(n - 1) * s_x * s_y}$$

$$s_x = \sqrt{\frac{\sum (x - 0)^2}{(n - 1)}}$$

$$s_y = \sqrt{\frac{\sum (y - 0)^2}{(n - 1)}}$$

become

In this case, the equations simplify down significantly and become

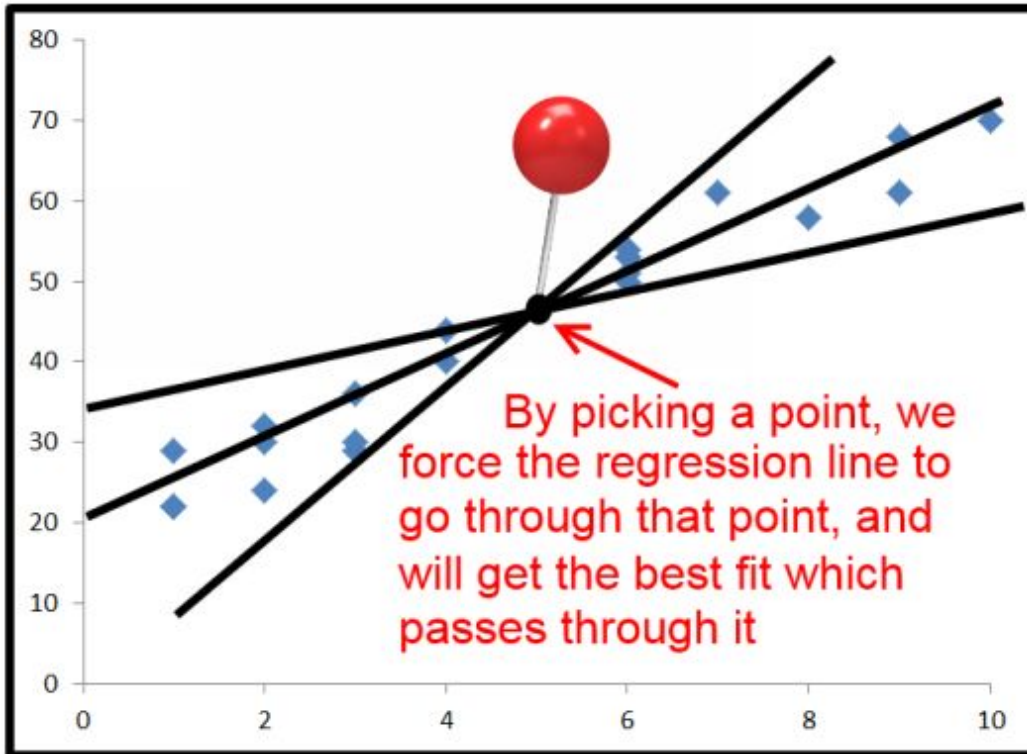
$$r = \frac{\sum xy}{(n - 1) * s_x * s_y}$$

$$s_x = \sqrt{\frac{\sum x^2}{(n - 1)}}$$

$$s_y = \sqrt{\frac{\sum y^2}{(n - 1)}}$$

And we have already seen these simplified equations before, when we gave an example where we modified the data so $\bar{x}$, $\bar{y}$ were the origin. That shows us that one way to think of what we are doing is centering the data on the point we want the regression line to go through.

Effectively, what is going on is that by specifying x and y , you are placing a pin in a graph that the regression line will pivot around



It will pivot to give the best possible R^2 while going through the pinned location. You can choose to place your pin at the average x and average y , $(\bar{x}, \bar{y})$, which will give the best R^2 overall. You could place your pin at the origin or some x or y intercept if you have specific knowledge about where you want to force your line to be, or you could place the pin at any other x, y coordinate. The rest of the equation will operate to give the best R^2 given the constraint you have placed on it.

We listed some modified equations to get the slope of a regression line which goes through an arbitrary point. Additionally, the equation to solve for the intercept has changed too. Previously we used

$$y = a + bx$$

And since we knew the regression line went through $(\bar{x}, \bar{y})$ we could substitute those values in for x and y , then solve for a .

$$\bar{y} = a + b\bar{x}$$

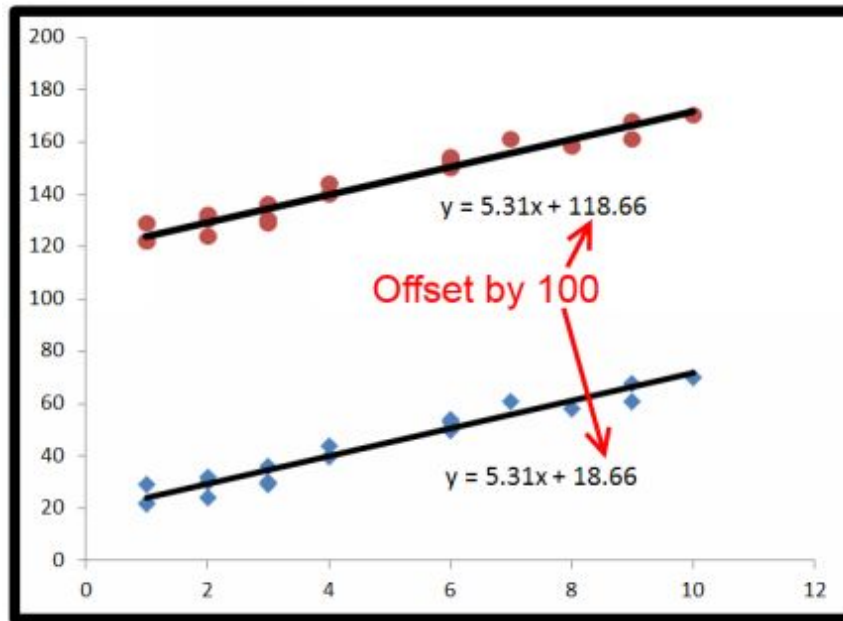
$$a = \bar{y} - b\bar{x}$$

However, that only worked because we were forcing the line to go through $(\bar{x}, \bar{y})$. Now by specifying a different point (x_0, y_0) , that different point is the only location that we know the regression line passed through. We can use that point to calculate our intercept value. The equation simply becomes

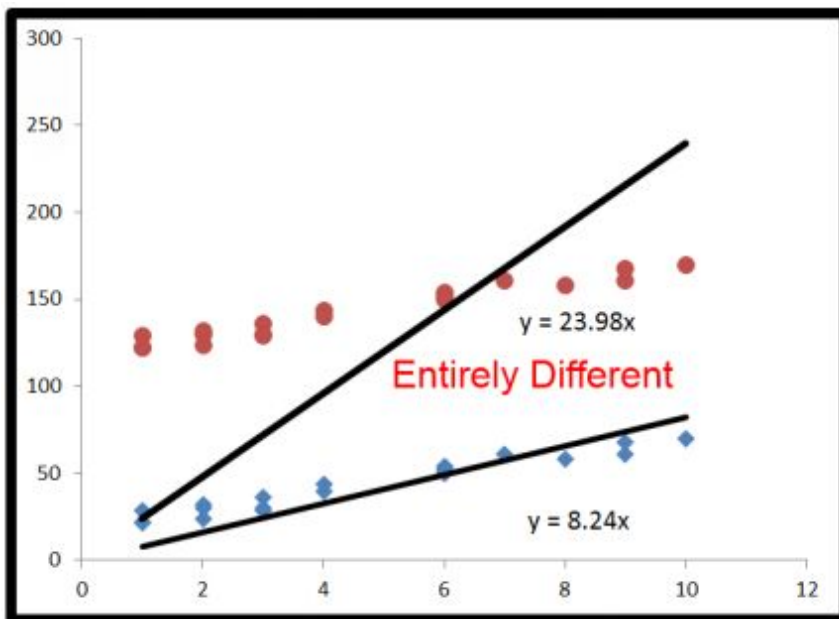
$$a = y_0 - bx_0$$

Something to note if you are specifying a point like this is that the R^2 value will always be less than the default of using average x and average y . (Unless the point you are specifying happens to be on the line which also passes through $(\bar{x}, \bar{y})$). In fact, by specifying a point that is not average x and y , this could be one of the situations where you could get a negative R^2 . Sometimes a worse R^2 is acceptable because it is more important to anchor the line to a specific point than to have the best possible fit. However you should be aware of the effect offset can have on your regression line if you are not using $(\bar{x}, \bar{y})$.

If you have two sets of data that are merely offset by 100 from each other in y , the default linear regression line will be the same for both sets of data, with the offset reflected in the intercept of the line equation.



But if you force the line to go through the origin, you will get an entirely different curve fit for the data sets with and without the offset



We saw one example of an offset before, in the Moore's law example. Initially, the data was centered on years A.D., so the data went from 1970 – 2016. However, I offset the data to make it years after 1970 in order to remove that constant term from the equation. Removing the constant term made the equation a lot nicer (it had a bigger effect in that exponential regression than it would in linear regression) but it didn't actually change the

curve fit. If however, I had forced the regression line to go through the origin at (0,0), then whether I set my origin at year 0, or at year 1970 would have made a big difference.

Multiple Regression

This is the point in the book where we will throw caution to the wind a bit and just show how multiple regression works without going very deep into the checks you might want to do when actually using it. As an overview, things that you might want to be aware of when doing multiple regression are

- Each new variable should explain some reasonable portion of the dependent variable. Don't add a bunch of new independent variables without a reason
- Avoid using variables that are highly correlated to other variables that you already have. Specifically, avoid any variable that is a linear combination of other variables. i.e. you wouldn't want to use variable x_3 if

$$x_3 = 2 * x_1 + 5 * x_2$$

$$x_3 = x_1^2 * x_2$$

But you might still want to use x_3 if
Since that isn't a linear combination.

Resources that you might checkout in order to see what checks to do when doing multiple regression include

- [A Youtube video series on multiple regression](#) – with a focus on data preparation
- This page has some good examples of what you should be aware of when doing multiple regression
<http://www.statsoft.com/Textbook/Multiple-Regression>

There are several different methods that can be used for multiple linear regression. We are going to start by demonstrating one method that does multiple regression as a sequence of simple linear regressions. This method has the advantage of building on what we already know and being

understandable. However, it is more difficult and time-consuming, especially for large problems.

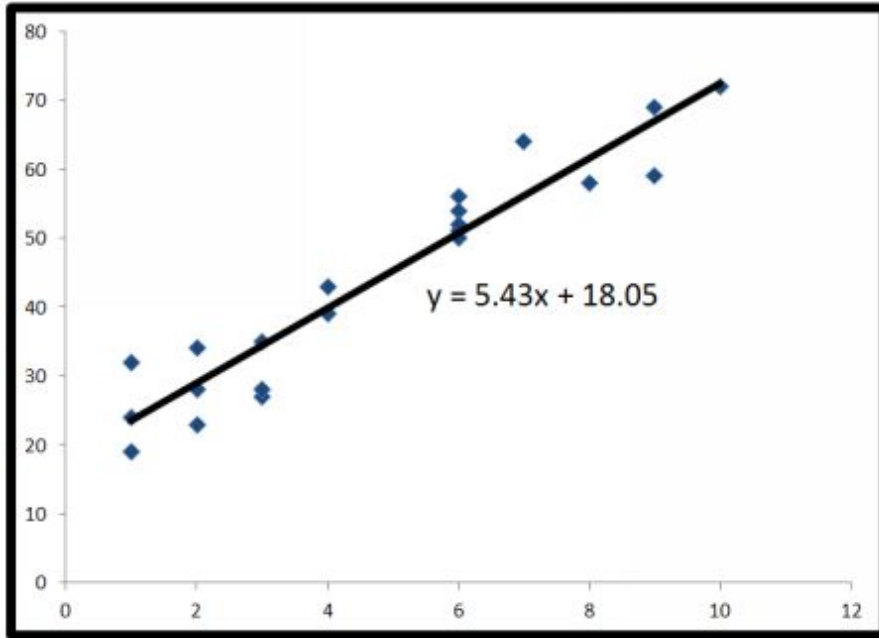
The other method we will show is the typical method used in most software packages. It is called the Moore-Penrose Pseudo-Inverse. We will show how to do Moore-Penrose Pseudo Inverse, but not attempt to derive it or prove it. This method is completely matrix math based, which is nice because there are a lot of good algorithms for matrix math, however, the insights into exactly what is happening inside the matrix math process are difficult to extract.

Multiple Regression Overview The first way we are going to do multiple regression in this book is as a series of single linear regressions. This uses all the same equations for correlation, standard deviation, and slope that we have used before. The only difference is we will have to do the equation multiple times in a certain order and keep track of what we do.

This series of single regressions isn't the only way to do multiple regression. In fact, it isn't the most numerically stable. However, it is completely understandable based on what we already know.

Using The Residual Of Linear Regression

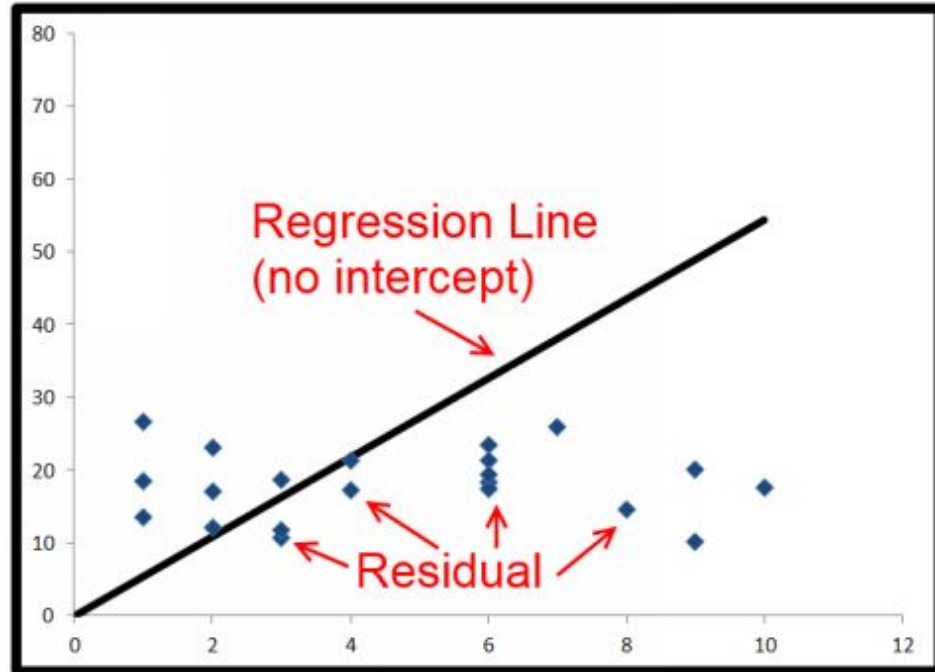
Let's take a look at what is left over after we do a single regression. If we had this data and this regression line



Then the regression line has accounted for some of the variation in y , but not all of it. We can, in fact, subtract out the amount of y that the slope of the regression line accounts for. When we do that, we are left with the residual which has both the error and the intercept

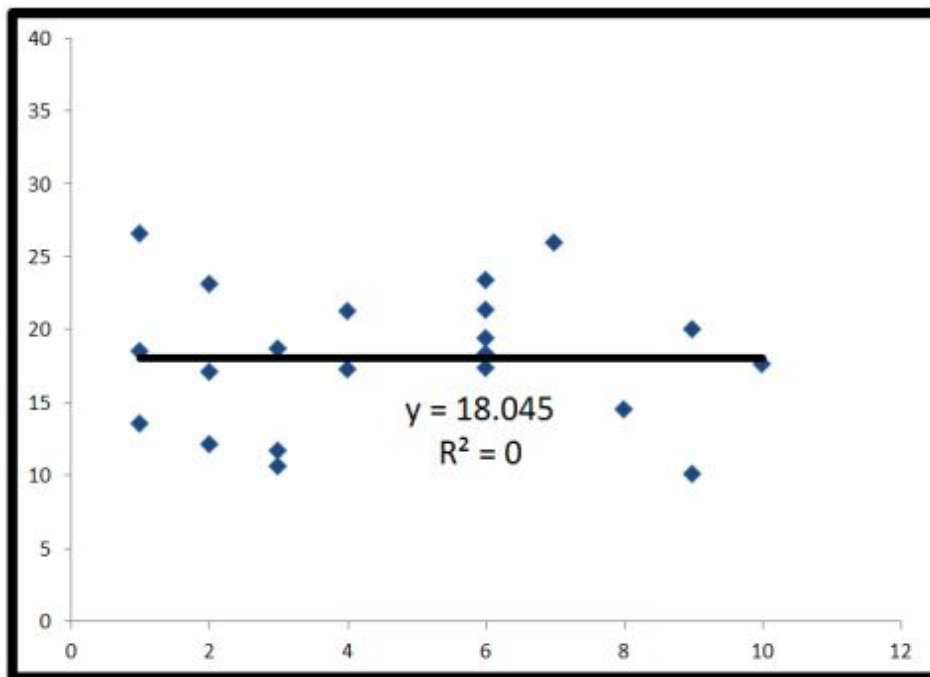
$$y - bx = y_{residual}$$

When doing this, we have broken y down into two parts, the regression line, and the remaining residual variation the regression doesn't account for. That



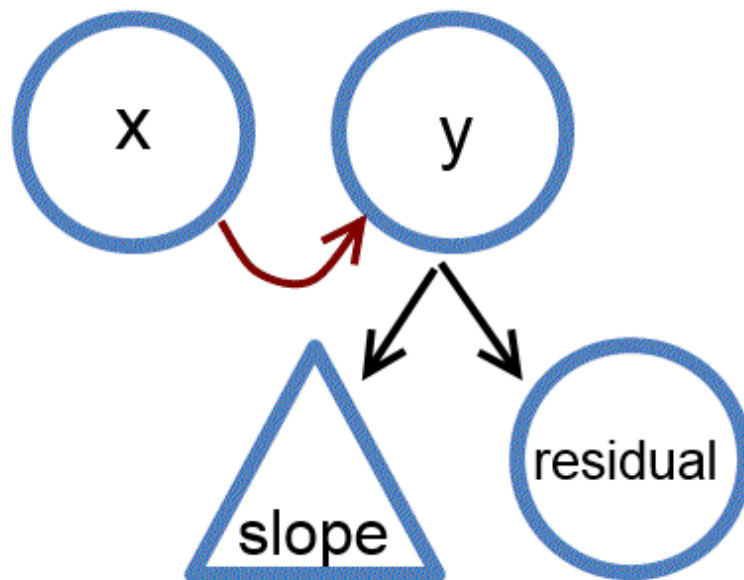
is shown below

Notice that for the residual points there is no way to do any regression on them. All of the data that correlates between x and y has been removed. As a result, any regression would be a horizontal line with an R^2 of zero. That is shown below.



More accurately, there is no way to do an additional correlation to the regression data using the independent variable we already used. So with a single linear regression, the residual was the error (and the intercept). But with multiple regression, we will use the residual from one regression as the starting point for the regression with another variable. So just because there is no way to do the correlation with x_1 against the residual, doesn't mean we can't do the correlation with x_2 against the residual.

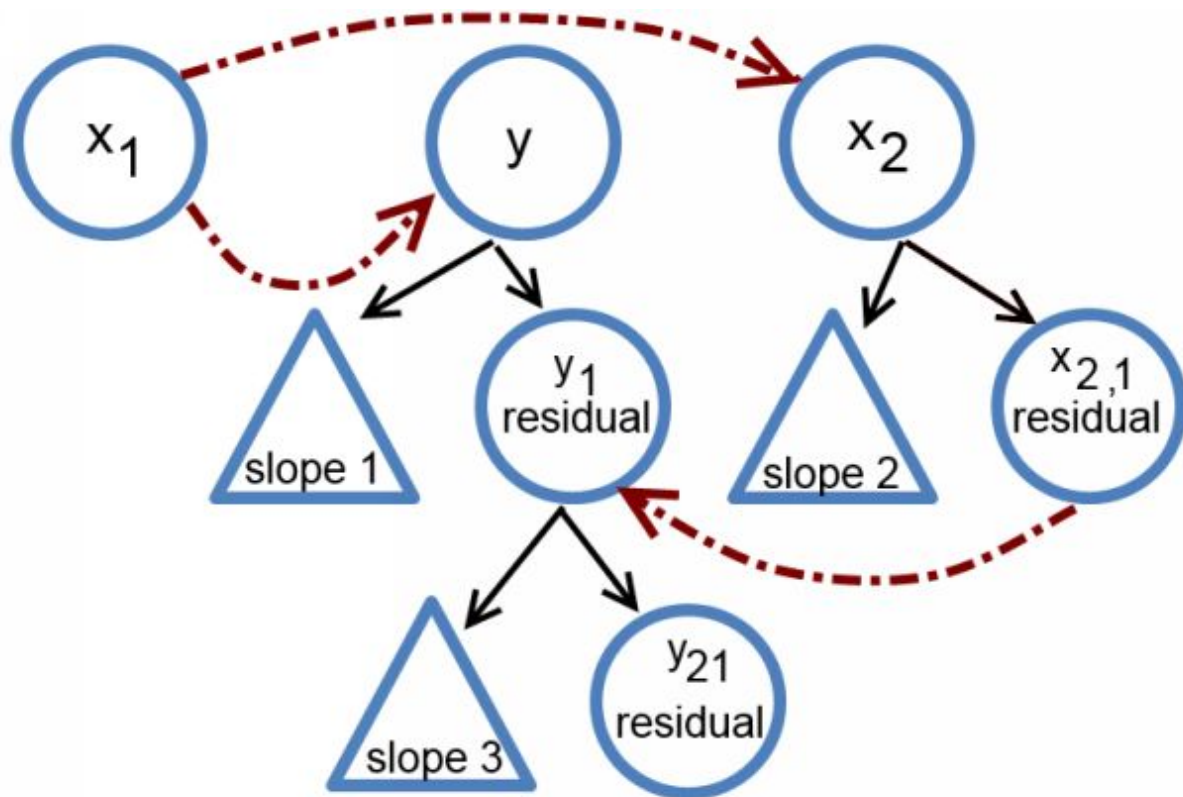
So with one independent variable, what we did was use the independent variable to get out one slope and one residual (although all we ended up



using was the slope)

The final result that we actually used was the slope, which is shown as a triangle above.

With two independent variables, what we will do is use the first independent variable to get a slope and residual out of each of the other two variables. And then use the residual from the second independent variable to get a slope and residual out of the residual of the dependent variable.



This chart is a little bit busy, and there is no need to memorize it. There are really only two key points to this chart

- The process starts with one independent variable, which does a regression against each of the other variables
- In the next step, you use the residual values and do another regression, removing one variable at a time, until you have no more independent variables.

The naming that we are using is that $x_{2,1}$ is x_2 with x_1 removed. y_{21} is y with x_2 and x_1 removed.

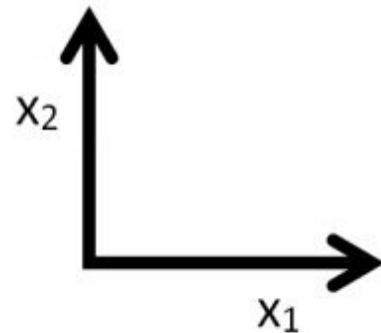
It seems to make sense to do a regression of x_1 vs. y , and then do a regression of x_2 against the residual of y . (i.e. the variation in y that x_1 doesn't account for) But why do we need to do a regression of x_2 against x_1 ? And why do we use that residual to do the regression against y residual as opposed to using x_2 ?

A Multiple Regression Analogy To Coordinate Systems

The reason we are doing a regression of x_2 against x_1 is to make sure that the portion of x_2 that we use in the next step is completely independent of x_1 .

One good way of understanding this is with an analogy to coordinate systems.

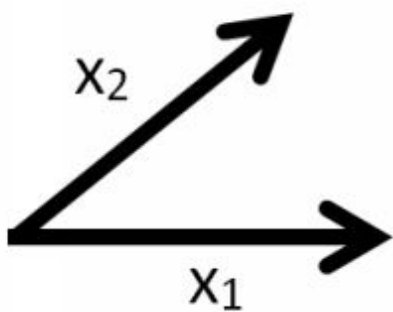
Imagine you want to specify point y as a location on a coordinate system labeled x_1 and x_2 direction. How you would normally expect to do it would



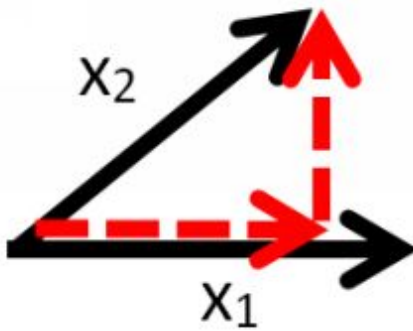
be to have a coordinate system that looks like this

What you have here is an x_2 direction that is completely orthogonal to x_1 .

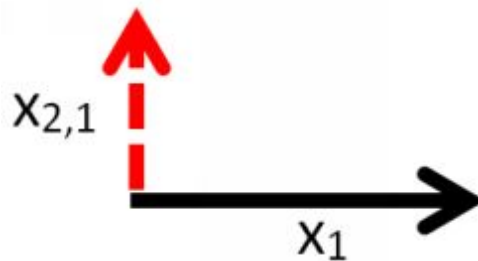
Your location in the x_1 direction tells you nothing about your location in the x_2 . This is what we want to have. But what we actually have with our two independent variables is analogous to a coordinate system that looks like this



Where x_1 and x_2 are related. I.e. if we have a high value in the x_1 direction, we probably have a high value in the x_2 direction. Now, if we were dealing with coordinate systems, what we would do is break x_2 down into two parts, one that was parallel to x_1 , and one that was orthogonal (i.e. at a right angle) to x_1 .



We could then throw away the part that was parallel, and measure the location using the unmodified x_1 as well as the orthogonal part of x_2 .



This is exactly what we are doing when we do the regression of x_2 against x_1 . If we were using coordinate systems, we could find the parallel portion of x_2 with dot products. With this data we are using regression, finding the portion of x_2 that x_1 can explain, and subtracting it out.

After we have a location in x_1 and $x_{2,1}$ coordinates we would use the relationship between x_2 and $x_{2,1}$ to convert back to our original vectors.

To summarize how we obtain the portion of x_2 that x_1 can and cannot explain

- Take the regression of x_2 against x_1
- The slope of the regression line is the part of x_2 that x_1 explains
- If you multiply that slope by x_1 and subtract it from x_2 , that is the residual
- The residual is the portion of x_2 that the regression cannot explain. Hence, that is the portion that we want to use for the next round of regression analyses

We should note here that if you don't have any residual for the independent variables against each other, then that means that they are not independent of each other. One of the variables is a linear combination of one or more of the other variables, so it would need to be discarded.

Multiple Regression Equations

In order to calculate the slope at any given step, we will use the standard equations that we know and love

$$r = \frac{\sum((x - \bar{x}) * (y - \bar{y}))}{(n - 1) * s_x * s_y}$$

$$s_x = \sqrt{\frac{\sum (x - \bar{x})^2}{(n - 1)}}$$

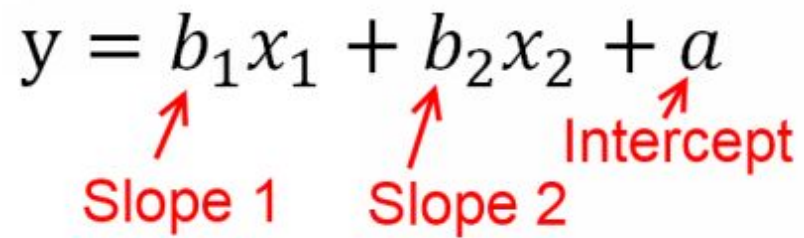
$$s_y = \sqrt{\frac{\sum (y - \bar{y})^2}{(n - 1)}}$$

$$b = r * \frac{s_y}{s_x}$$

Except that the x and y variables might change, for instance sometimes x could be x_1 , and y could be x_2 .

After we have done all of the linear regressions, we will combine them to get

$$y = b_1x_1 + b_2x_2 + a$$


Slope 1 Slope 2 Intercept

an equation of the form

The regression analysis that we do at each individual step doesn't directly provide the b_1 or b_2 values that we want for the equation above. The example below shows how to get those from the regression steps that we do.

Multiple Regression Example On Simple Data

We have 10 points of data, each of which has an x_1 , x_2 , and y . Can we find the regression equation relating y to x_1 and x_2 ?

x_1	x_2	y
1	8	43
3	3	24
10	10	80
11	9	78
8	12	84
7	6	51
4	7	47
0	2	10
15	10	95
18	14	124

For this example (although you wouldn't know this for most data sets)

- X_1 was generated as random integers between 0 and 20
- X_2 was generated as $.5 * x_1 + .5 * (\text{random integer between 0 and 20})$
- Y was generated as $y = 3 * x_1 + 5 * x_2$

Note that this ensured that there was some correlation between x_1 and x_2

Step 1 – Remove x_1 from y Here we will find the correlation and slope of the x_1 and y relationship as we have in the past. One difference here is that we will call the resulting slopes between single variables lambda, which is the symbol λ . We will call these lambdas since these are intermediate results. We will reserve the symbol 'b' for the final slopes.

As shown below, the slope of the first regression that we calculate, which we will call λ_1 , is 5.536. This is the correlation value of .946 multiplied by the ratio of the standard deviations, 34.68 / 5.93.

			Remove x1 y vs x1	
			Correlation	0.946
			Lamda	Lamda 1 5.536
Average	7.70	8.10	63.60	
Std. Deviation	5.93	3.75	34.68	

x1	x2	y	Residual y1
1	8	43	37.46
3	3	24	7.39
10	10	80	24.64
11	9	78	17.11
8	12	84	39.72
7	6	51	12.25
4	7	47	24.86
0	2	10	10.00
15	10	95	11.97
18	14	124	24.36

By finding that slope, we can calculate the residual value y_1 . This residual value shows us how much of y was not based on that x_1 value.

When we did this regression, we got this equation

$$\begin{array}{c} \text{Dependent} \\ \text{Variable} \end{array} \begin{array}{c} \text{Independent} \\ \text{Variable} \end{array}$$

$$\begin{array}{c} \text{y} \\ \text{Slope} \end{array} = \begin{array}{c} \lambda_1 \\ \text{Residual} \end{array} x_1 + y_1$$

Where y_1 is the residual value. We can rearrange the equation to calculate those residual values. (Recall that the residual values are an array of

numbers that are the same length as the other variables)

$$y_1 = y - \lambda_1 x_1$$

We are using the variable y_1 to denote y with the independent variable x_1 removed. The results are these residual values.

				Remove x1	
				y vs x1	
				Correlation	0.946
				Lamda 1	
				Lamda	5.536
Average	7.70	8.10	63.60		
Std. Deviation	5.93	3.75	34.68		

x1	x2	y
1	8	43
3	3	24
10	10	80
11	9	78
8	12	84
7	6	51
4	7	47
0	2	10
15	10	95
18	14	124

Residual
y1
37.46
7.39
24.64
17.11
39.72
12.25
24.86
10.00
11.97
24.36

Now, what does it mean when we say that x_1 has been removed from y ? Previously the correlation between y and x_1 was .946. However, the correlation between the residual value, y_1 , and x_1 is zero.

Correlation 0.00

x1	y1
1	37.46
3	7.39
10	24.64
11	17.11
8	39.72
7	12.25
4	24.86
0	10.00
15	11.97
18	24.36

So the new variable, y_1 , which we have created, is completely independent of x_1 .

Step2 – Remove x_1 from x_2

Each step is going to be the same, just with different variables. This is because we are repeating the same single linear regression in order to do the multiple regression. In this step, we remove x_1 from x_2 . We need to find the correlation between x_1 and x_2 , use that correlation to get a regression slope that we will call λ_2 , and then calculate a residual value to determine how much of x_2 was not based on x_1 .

The equation that we will use to get the resulting residual is

$$x_{2,1} = x_2 - \lambda_2 x_1$$

Here, we are using the variable $x_{2,1}$ to denote x_2 with the independent variable x_1 removed. This is the same as the previous equation, except with x_2 values instead of y values. The resulting lambda and residuals are shown below.

				Remove x1	
				y vs x1	x2 vs x1
Correlation				0.946	0.800
				Lamda 1 Lamda 2	
Lamda				5.536	0.50712
Average	7.70	8.10	63.60		
Std. Deviation	5.93	3.75	34.68		
				Residual	
				y1	x2,1
x1	x2	y			
1	8	43		37.46	7.49
3	3	24		7.39	1.48
10	10	80		24.64	4.93
11	9	78		17.11	3.42
8	12	84		39.72	7.94
7	6	51		12.25	2.45
4	7	47		24.86	4.97
0	2	10		10.00	2.00
15	10	95		11.97	2.39
18	14	124		24.36	4.87

Once again, this new $x_{2,1}$ variable has zero correlation with x_1 .

Step 3 – Remove $x_{2,1}$ from y_1

We now have two new variables, $x_{2,1}$ and y_1 . We need to do a regression analysis to find the relationship between those variables. The important thing about this process is that we are using the two residual variables, not any of the three initial variables. Both of those two variables that we are using have the influence of x_1 removed, so we will only find the relationship between those variables that x_1 does not account for.

The key thing that we get from this is λ_3

	Remove x1			Remove x2,1	
	y vs x1	x2 vs x1		y ₁ vs x2,1	
Correlation	0.946	0.800		1.000	
Lamda	Lamda 1	Lamda 2		Lamda 3	
	5.536	0.50712		5.000	
Average	20.98	4.20			
Std. Deviation	11.26	2.25			
	Residual			Residual	
	y1	x2,1		y21	
	37.46	7.49		0.00	
	7.39	1.48		0.00	
	24.64	4.93		0.00	
	17.11	3.42		0.00	
	39.72	7.94		0.00	
	12.25	2.45		0.00	
	24.86	4.97		0.00	
	10.00	2.00		0.00	
	11.97	2.39		0.00	
	24.36	4.87		0.00	

We also get another residual y_{21} , which is y with both the influence of x_1 and x_2 removed. However, we don't care about that residual because we have already done a regression on all of our variables, so we don't need to keep the residual for another step.

Using the Lambda Values to Get Slopes We now have 3 matches which represent slopes between individual variables, lamda1, lambda 2, and lambda 3. We want to get an equation that relates all the independent variables to y . So we have to relate the lambdas to the slopes in this equation

$$y = b_1x_1 + b_2x_2$$

When we matched y with x_1 , we had the equation

$$y = \lambda_1x_1 + y_1$$

If we can convert that to the form above, we can pair b_1 , b_2 with the resulting coefficients in front of x_1 and x_2 . To do that we need to get y_1 out of the equation. These are the steps we did when we removed the variables

$$y = \lambda_1 x_1 + y_1 \quad \text{Step 1 Equation}$$

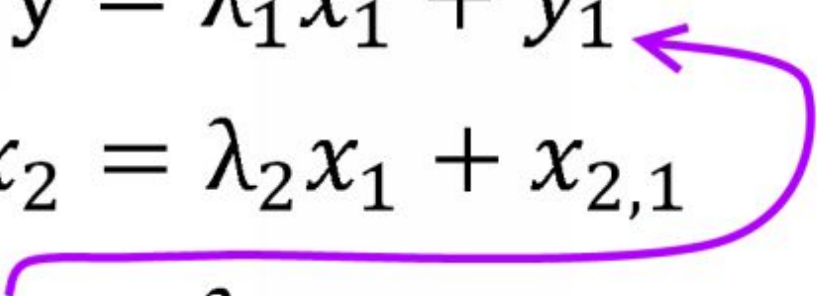
$$x_2 = \lambda_2 x_1 + x_{2,1} \quad \text{Step 2 Equation}$$

$$y_1 = \lambda_3 x_{2,1} + y_{21} \quad \text{Step 3 Equation}$$

We can combine all three of those equations to get the equation in the form that we want, which is

$$y = b_1 x_1 + b_2 x_2 + y_{\text{residual}}$$

The way we will combine those equations is first by substituting the y_1 from the third equation into the first equation.

$$\begin{aligned} y &= \lambda_1 x_1 + y_1 \\ x_2 &= \lambda_2 x_1 + x_{2,1} \\ y_1 &= \lambda_3 x_{2,1} + y_{21} \end{aligned}$$


That gets rid of the third equation, and we end up with a modified first equation, as well as the original second equation as shown below.

$$y = \lambda_1 x_1 + \lambda_3 x_{2,1} + y_{21}$$

$$x_2 = \lambda_2 x_1 + x_{2,1}$$

Need to
remove $x_{2,1}$

So we have combined 3 equations into 2 equations, and are making progress. The only remaining problem is the $x_{2,1}$ that is now in the new first equation since that was not one of the original variables. We can get rid of that by solving for $x_{2,1}$ in the second equation and then substituting it in. When we solve the second equation for $x_{2,1}$, we get

$$x_{2,1} = x_2 - \lambda_2 x_1$$

And when we substitute that into the first equation, what we get is

$$y = \lambda_1 x_1 + \lambda_3 (x_2 - \lambda_2 x_1) + y_{21}$$

Now, remember the x 's and y 's are variables. The lambdas are actually constants, which are the slopes of each of our individual regressions. So if we rearrange this equation to collect all of the variables together (basically combine x_1 terms), we get

$$y = \lambda_1 x_1 - \lambda_3 \lambda_2 x_1 + \lambda_3 x_2 + y_{21}$$

And with one final simplification of pulling the x_1 term out, we get

$$y = (\lambda_1 - \lambda_3 \lambda_2) x_1 + \lambda_3 x_2 + y_{21}$$

And this is our final answer. Remember, our objective was to get an equation of this form

$$y = b_1x_1 + b_2x_2 + y_{residual}$$

Which is what we have. So if we pair up the coefficients in front of the

$$b_1 = (\lambda_1 - \lambda_3\lambda_2)$$

$$b_2 = \lambda_3$$

$$y_{residual} = y_{2,1}$$

variables, we see that

We previously solved for a lambda 1 value of 5.536, a lambda 2 value of .5071 and a lambda 3 value of 5.0 when we plug that into our equations we

$$b_1 = (5.536 - .5071 * 5) = 3$$

$$b_2 = 5.0$$

get

That gives us

$$y = 3x_1 + 5x_2$$

In this case, our intercept is 0. But if we didn't know that, we could find it using a slight modification of the intercept equation we had before

$$\bar{y} = 3\bar{x}_1 + 5\bar{x}_2 + a$$

Note this assumes you are using the regression equation through the mean as shown above. If you want to force the regression to go through a specific

point, you can do that by modifying the slope equations as we saw before. However the easiest way to force the line through a specific point with multiple regression is to “center” the data around that point at the start of the problem, so that point is the origin. Then you can solve all of the steps using zero in place of the mean x or mean y , and “un-center” the data in the final regression equation. (I.e. if you solved centered your data by subtracting 5 from x_1 , and 10 from x_2 , then make sure you add 5 to the x_1 variable and 10 to the x_2 variable in the final regression equation) The average values that we had were 7.7 for x_1 , 8.1 for x_2 and 63.6 for y .

Average	7.70	8.10	63.60
Std. Deviation	5.93	3.75	34.68

x1	x2	y
1	8	43
3	3	24
10	10	80
11	9	78
8	12	84
7	6	51
4	7	47
0	2	10
15	10	95
18	14	124

If we plug those values into our equation and solve for the intercept, we determine that there is a zero value for a , the intercept for this set of data.

$$63.6 = 3 * 7.7 + 5 * 8.1 + a$$

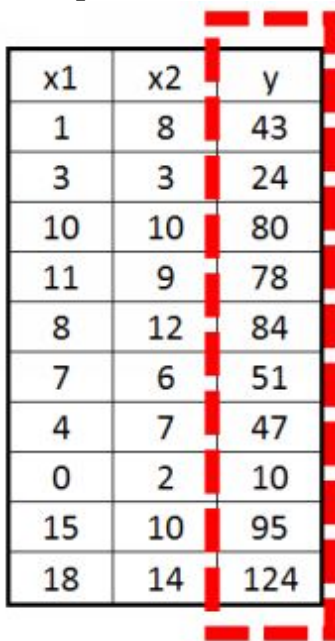
$$a = 0$$

Moore-Penrose Pseudo-Inverse

Let's do the same problem again using the matrix math method. The data that we have is a matrix of numbers, multiplied by a matrix of coefficients set equal to our results matrix. In general terms, we would call this

$$[A] [b] = [y]$$

The $[A]$ matrix is made up of the coefficients in front of the x terms, the $[b]$ matrix is the actual slope and intercept values we are trying to calculate, and the $[y]$ matrix is our resulting y values. For this specific problem, with this example data, then the rightmost column is our $[y]$ matrix



x1	x2	y
1	8	43
3	3	24
10	10	80
11	9	78
8	12	84
7	6	51
4	7	47
0	2	10
15	10	95
18	14	124

Now if this was a typical linear algebra problem where we had the same number of inputs as unknowns, we could just multiply both sides of the equation by an inverted $[A]$ matrix and get a result for $[b]$, which is what we

are trying to solve for

$$[A]^{-1} [A] [b] = [A]^{-1} [y]$$

$$[b] = [A]^{-1} [y]$$

However, for regression problems, we have a different number of unknowns compared to inputs. We typically have many more inputs than unknowns. This is an over-constrained problem, which is why we are solving it using least squares regression. Least squares regression gives us a solution that is as close as possible to all of the points but does not force the result to exactly match every point.

To put it a different way, you can only do a matrix inversion on a square matrix (and not always then). With a typical regression problem $[A]$ is rectangular, not square, and hence we cannot do a standard matrix inversion.

That, of course, is why we will do a Pseudo Inverse instead. This will act like a standard matrix inversion for our purposes, but it will inherently incorporate a least squares regression, which means it will work on a rectangular matrix.

The symbol for a standard matrix inverse is -1 . i.e. $[A]^{-1}$ is the inverse of $[A]$.

The symbol for Moore-Penrose Pseudo Inverse is a dagger. So our final solution will be

$$[A]^\dagger [A] [b] = [A]^\dagger [A] [y]$$

Dagger Symbol

$$[b] = [A]^\dagger [y]$$

Sometimes things such as a plus sign or an elongated plus sign are used instead of the dagger symbol since it is kind of unusual.

The Equation For Moore-Penrose Pseudo-Inverse To generate A-dagger we need to use this equation

$$[A]^\dagger = ([A]' [A])^{-1} [A]'$$

The equation shows that we have to

- Multiply A transpose by A
- Take the inverse of that
- Multiply that by A transpose

Notice that we are still taking an inverse in this process. However, we are taking the inverse during the second step, after we have generated a square matrix using the product of [A] transpose and [A].

Plugging In The Data And Solving The Problem Now let's do the Pseudo-Inverse for the example data we already saw. First, let's make our matrices. Recall that this is the data that we are attempting to do the regression on.

x1	x2	y
1	8	43
3	3	24
10	10	80
11	9	78
8	12	84
7	6	51
4	7	47
0	2	10
15	10	95
18	14	124

The [A] matrix is shown below. It is column based and each column corresponds to the coefficients of one unknown variable.

Matrix

A =

x1	x2	a
1	8	1
3	3	1
10	10	1
11	9	1
8	12	1
7	6	1
4	7	1
0	2	1
15	10	1
18	14	1

The first column corresponds to the unknown coefficient in front of x_1 , and the second column corresponds to the coefficient in front of x_2 . Additionally, and importantly, notice the column of 1's in the matrix. We need the column

of all 1's in order to capture the 'y' intercept. Without that column of 1's the process will be doing a least squares regression through the origin ($y=0$). With the column of 1's, we will do the regression analysis we have seen before where we can extract an intercept.

The [b] matrix is a single column that has the coefficients we are trying to

$$\begin{bmatrix} b_1 \\ b_2 \\ a \end{bmatrix}$$

find as well as the intercept. In this case, we have

The [y] matrix is a single column of the 'y' results. For this example it is

y
43
24
80
78
84
51
47
10
95
124

Once we generate the Pseudo-Inverse using the [A] matrix, the result of that gets multiplied by the [y] matrix.

Pseudo-Inverse First Step

The first step to calculate the Pseudo-Inverse is to matrix multiply the transpose of the [A] matrix by the [A] matrix. As an equation it is

$$[A]' [A] =$$

In this case, the transpose of the [A] matrix is

Matrix transpose

A' =

x1	1	3	10	11	8	7	4	0	15	18
x2	8	3	10	9	12	6	7	2	10	14
a	1	1	1	1	1	1	1	1	1	1

This is a 3 by 10 (3 rows, 10 columns) matrix multiplied by a 10 by 3 [A] matrix. When doing matrix math you can only do multiplication if the two middle terms are the same. In this case, they are both 10. The result is an 3

$$[3 \times 10] [10 \times 3] = [3 \times 3]$$

by 3 matrix

The resulting matrix is

Transpose * Matrix

$$A' * A =$$

909	784	77
784	783	81
77	81	10

Matrix Inverse

The next step is to find the inverse of that matrix. This book is only going to show the result and not the actual process of finding a matrix inverse since

there are a lot of good resources that show how to do it. The inverse of

909	784	77
784	783	81
77	81	10

Is

Inverse

$$(A' * A)^{-1}$$

0.009	-0.01	0.022
-0.01	0.022	-0.09
0.022	-0.09	0.673

As a side note, remember back at the beginning of this section on multiple linear regression when we said you should avoid rows that are linear combinations of other rows? This matrix inverse is the reason why. In the matrix product, two rows that are exactly the same or rows that are linear sums of other rows give a singular matrix that is non-invertible. Rows that are too similar make the matrix ill-conditioned. (This wasn't a problem in our other method using a sequence of linear regressions)

Multiply That Inverse By [A] Transpose The next step is to multiply that inverse by the transposed [A] matrix. When we do that we get this matrix, which is the Pseudo-Inverse of the original [A] matrix

Moore Penrose Pseudo Inverse

$$(A' * A)^{-1} * A'$$

-0.05786	0.01534	-0.00088	0.01904	-0.04072	0.01719	-0.02034	4.9E-05	0.04313	0.02506
0.0723	-0.05956	0.01609	-0.01696	0.08217	-0.03826	0.01702	-0.04813	-0.03951	0.01484
-0.04012	0.46436	-0.02351	0.09076	-0.25204	0.27756	0.11872	0.48948	0.08792	-0.21314

Final Step

The final step is to multiply the inverse matrix by the [y] matrix. Recall that the [b] matrix is the product of those two terms, and the [b] matrix contains all of the coefficients and the intercept that we are trying to calculate.

When we do the multiplication of the 3 x 10 pseudo inverse by the 10 x 1 [y] matrix, we get a 3 x 1 resulting matrix that contains the two 'b' coefficients, and the 'a' intercept. That result is shown below.

Final Result

$$((A' * A)^{-1} * A') * y$$

3	b1
5	b2
0	a

This is the same result that we got when we did the regression as a series of linear regressions. Notice that in this case, the 'a' intercept is zero. That means the regression line is going through the origin. So we would have gotten the same result whether or not we included the column of ones in our [A] matrix for this set of data. (That is not the case when the intercept is non-zero)

Time Complexity Of This Solution This Moore-Penrose Pseudo-Inverse had some obvious advantages over the other method of a regression that we showed, which was a sequence of linear regressions. One large advantage is that this Pseudo-Inverse process will be the same no matter the size of the problem. We could have used this Pseudo-Inverse with one unknown slope, and we can use it with any number of unknown slopes. In a later example, we will show this process again where we have three slopes and an intercept to calculate, and that example will not be significantly more difficult than

this one. (This is in contrast to what we will see with the sequence of linear regressions, which will have a lot more steps.) Truthfully, the Pseudo-Inverse process does get more difficult as we add more variables. However, that difficulty is hidden in the matrix multiplication and matrix inverse steps that we do. The time complexity of matrix multiplication and matrix inversions grows with the size of the matrices. However, there has been a lot of mathematical work done to generate optimized algorithms for those processes, so the fact that they are more difficult as the matrices grow larger is somewhat hidden. This Wikipedia page shows the time complexity of matrix operations,

https://en.wikipedia.org/wiki/Computational_complexity_of_mathematical_operations

For our purposes, in this book, the biggest drawback of the Moore-Penrose Pseudo-Inverse is that while it is easy to utilize the algorithm, it is difficult to understand exactly how the algorithm does the regression.

What Is Next?

We just saw two different methods for how to do multiple regression with two independent variables. There are two more examples that we will show with multiple regression. The first is a regression with two independent variables on the ‘Modern Family’ data. The second is a regression with three variables in order to make sure it is clear how to expand this process to larger sets of data.

Multiple Regression On The Modern Family Data

Because of your amazing work purchasing ads from ABC, you have now been promoted to Studio Executive, and you need to predict the viewership of future seasons of Modern Family in order to know if the season is worth launching. No longer do you have the luxury of waiting until the first episode of a season airs and using that data to make your predictions for the season. Now you need to decide if that first episode should even get made. Let's look again at the unmodified data for the Modern Family Viewership.

Number of Viewers Of Modern Family - Unmodified Data								
Season	1	2	3	4	5	6	7	8
Episode	Viewers (Millions)							
1	12.60	12.67	14.52	14.44	11.71	11.38	9.46	8.24
2	9.99	11.92	14.54	12.08	11.65	10.56	8.72	7.41
3	8.82	11.44	13.45	12.31	11.12	10.30	7.99	6.97
4	9.35	11.45	13.24	12.28	10.64	9.71	7.88	7.50
5	8.66	11.97	13.65	12.52	10.94	10.16	7.80	7.38
6	9.32	13.14	13.04	10.62	10.32	9.92	8.15	7.23
7	8.77	12.24	13.37	12.43	10.75	9.83	8.38	7.33
8	9.16	12.09	12.91	11.89	10.19	10.88	7.35	6.81
9	7.12	10.57	12.72	12.01	9.47	9.02	8.20	6.81
10	9.67	11.07	12.20	10.94	10.61	9.69	8.35	7.57
11	10.22	11.83	14.03	12.04	9.51	9.29	8.17	7.59
12	7.83	11.12	12.12	11.01	9.14	9.44	7.80	
13	9.83	10.98	11.89	10.83	9.59	9.87	7.48	
14	9.19	13.16	12.90	9.83	9.87	9.77	8.10	
15	9.84	10.57	11.23	10.05	8.52	8.80	8.22	
16	8.01	10.17	11.54	10.62	9.22	9.32	8.14	
17	8.95	10.11	11.63	10.53	9.39	9.61	7.69	
18	9.18	10.90	10.60	9.09	10.09	8.71	7.51	
19	9.51	9.61	10.35	10.88	9.00	9.43	7.44	
20	8.88	9.95	10.21	10.38	9.59	8.85	7.42	
21	10.01	9.90	10.06	9.64	8.39	8.00	7.16	
22	9.48	10.15	10.58	9.02	8.55	8.57	6.79	
23	10.34	10.30	9.36	10.03	9.08	8.13		
24	10.14	10.31	10.07	10.01	10.45	7.20		

The first time we worked with this data, we normalized it and then attempted to find the regression based on episode number in a season. The reason we normalized the data was to remove the effect of season number on the total

viewers. Here, since we want to find the regression of both season number and episode number, we won't normalize it, and will just use the unmodified data as viewers in millions.

Looking at just episode 1 of each season of the data, it appears that the show gained viewers from season 1 to season 2 and from season 2 to season 3. After that, it lost viewers each season. Even though this is multiple regression, it is still linear. With two independent variables, we are making a plane instead of a line, (with more we would be making a hyperplane). As a result, we would not do a very good job of capturing the effect of increasing and then decreasing the dependent variable. In order to ignore that effect, I will only use season 3-7 data in this multiple regression analysis. This ignores the growth in viewers that was experienced in the first two seasons.

So the problem at hand is: what is a regression analysis that accounts for both episode number and season number in order to predict the number of viewers in an episode of Modern Family?

For this example we will only show the multiple regression as a sequence of single linear regressions. We will not show the process using Moore-Penrose Pseudo-Inverse for this example since the process is exactly the same as we saw in the last example, and this matrix based process will not display well in this book with the large amounts of data that will be in the Modern Family data set. We will do Moore-Penrose again in the next example, where we increase the number of independent variables.

To do the regression, we will start by listing all of the data into 3 columns. There are 118 total points of data between seasons 3-7, so it isn't feasible to show tables that long in this format. As a result, all of this Modern Family data is truncated at the bottom. Like all of the other examples in this book, you can download the Excel sheet with the data here <http://www.fairlynerdy.com/linear-regression-examples/> for free.

Modern Family Data
Seasons 3-7

Season	Episode	Viewers
x1	x2	y
3	1	14.52
3	2	14.54
3	3	13.45
3	4	13.24
3	5	13.65
3	6	13.04
3	7	13.37
3	8	12.91
3	9	12.72
3	10	12.20
3	11	14.03
3	12	12.12
3	13	11.89
3	14	12.90
3	15	11.23
3	16	11.54

We will call the season number our first independent variable, x_1 , the episode number the second independent variable, x_2 , and the number of viewers the dependent variable y .

The first step is to remove the influence of x_1 from y_1 . The second step is to remove the influence of x_1 from x_2 . We do this doing the traditional way of finding the correlation between the two sets of numbers and then multiplying the correlation by the ratio of the standard deviations to get the slope. Here we are calling the slope lambda because we are reserving the words “slope”

to mean the final slope of the full regression line, and the lambdas are intermediate results.

				y vs x1		x2 vs x1	
Average	4.97	12.31	10.12	Avg	15.06	13.27	
Std Dev	1.41	6.86	1.79	Std Dev	1.11	6.85	
				Correlation	-0.784	-0.040	
					Lamda 1	Lamda 2	
					-0.994	-0.193	
Modern Family Data Seasons 3-7							
Season	Episode	Viewers		y1		x2,1	
x1	x2	y		Residual of y when correlated with x1		Residual of x2 when correlated with x1	
3	1	14.52		17.50		1.58	
3	2	14.54		17.52		2.58	
3	3	13.45		16.43		3.58	
3	4	13.24		16.22		4.58	
3	5	13.65		16.63		5.58	
3	6	13.04		16.02		6.58	
3	7	13.37		16.35		7.58	
3	8	12.91		15.89		8.58	
3	9	12.72		15.70		9.58	
3	10	12.20		15.18		10.58	
3	11	14.03		17.01		11.58	
3	12	12.12		15.10		12.58	

The value of -.784 that we get for correlation is the correlation of y vs. x_1 .

The lambda1 that we get is that correlation multiplied by the standard deviation of y, divided by the standard deviation of x_1 . So

$$\lambda_1 = -.994 = \frac{-.784 * 1.79}{1.41}$$

We then find the residual by subtracting the independent variable multiplied by the slope from the dependent variable. For instance, the first residual of y vs. x_1 is 17.5

$$17.5 = 14.52 - (-.994 * 3)$$

The first residual of x_2 vs. x_1 is 1.58.

$$1.58 = 1 - (-.193 * 3)$$

Since we had initial columns that were 118 data points long, the residuals y_1 , and $x_{2,1}$, both have columns that are 118 long as well.

The next step is to remove $x_{2,1}$ from y_1 . The process here is exactly the same as above, except we are doing it with the two residual columns instead of with the initial data.

	y vs x1	x2 vs x1		y1 vs x2,1
Avg	15.06	13.27	Avg	16.76
Std Dev	1.11	6.85	Std Dev	0.67
Correlation	-0.784	-0.040	Correlation	-0.794
	Lamda 1	Lamda 2		Lamda 3
	-0.994	-0.193		-0.128

y1	x2,1	y21
Residual of y when correlated with x1	Residual of x2 when correlated with x1	Residual of y1 when correlated with x2,1
17.50	1.58	17.71
17.52	2.58	17.85
16.43	3.58	16.89
16.22	4.58	16.81
16.63	5.58	17.35
16.02	6.58	16.87
16.35	7.58	17.33
15.89	8.58	16.99
15.70	9.58	16.93
15.18	10.58	16.54
17.01	11.58	18.50
15.10	12.58	16.72
14.87	13.58	16.62
15.88	14.58	17.75

The final result that we get is lamda3, which is the slope of the regression line of y_1 vs. $x_{2,1}$. That value is -.128, which was the correlation multiplied by the ratio of standard deviations.

$$\lambda_3 = -.128 = \frac{-.794 * 1.11}{6.85}$$

Now we have values for lamda1, lamda2, and lamda3, which are -.994, -.193 and -.128 respectively. We need to back-solve to get actual slope values in the multiple regression, so we have an equation of this form

$$y = b_1x_1 + b_2x_2 + a$$

The regression equations we solved to get this were

$$y = \lambda_1x_1 + y_1 \quad \text{Step 1 Equation}$$

$$x_2 = \lambda_2x_1 + x_{2,1} \quad \text{Step 2 Equation}$$

$$y_1 = \lambda_3x_{2,1} + y_{21} \quad \text{Step 3 Equation}$$

As we saw before in the previous example, these equations combine into one equation

$$y = (\lambda_1 - \lambda_3\lambda_2)x_1 + \lambda_3x_2 + y_{21}$$

And when we match coefficients we get the result of

$$b_1 = (\lambda_1 - \lambda_3\lambda_2)$$

$$b_2 = \lambda_3$$

And when we substitute the values we found for lambda and solve for the slopes, the results we get are

$$b_1 = -.994 - (-.193 * -.128) = -1.019$$

$$b_2 = -.128$$

What does this tell us?

The b_1 coefficient pairs with variable x_1 . x_1 is the season number. This means that every season has on average 1.019 million fewer viewers than the season before it. The b_2 coefficient pairs with x_2 , which is the episode number. The b_2 coefficient is $-.128$, which tells us that every episode in a given season has $.128$ million fewer viewers than the episode before it.

Solving For Intercept

A linear regression is defined by slopes and intercept. So far we have solved for the slopes. To get the intercept, we need to know one point that the regression plane passed through. In this case, since we did the correlation and standard deviations around the average of the data, the regression plane passes through x_1 average, x_2 average, and y average. That equation is

$$\bar{y} = -1.019\bar{x}_1 - .128\bar{x}_2 + a$$

From the initial data, these are the average values

	Season	Episode	Viewers
	x_1	x_2	y
Average	4.97	12.31	10.12
Std Dev	1.41	6.86	1.79

When we plug the average values in we get

$$10.12 = -1.019 * 4.97 - .128 * 12.31 + a$$

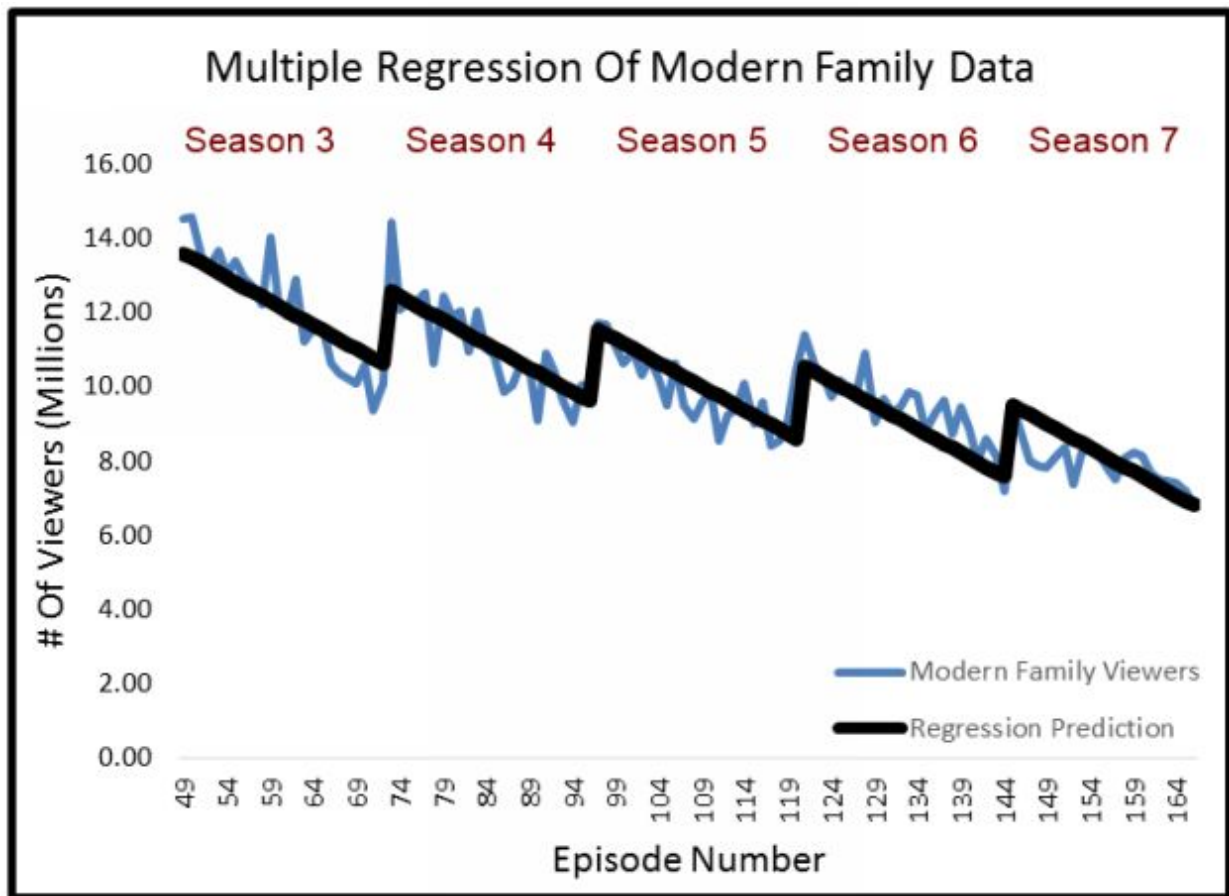
This results in an intercept of 16.76. (i.e. 16.76 million viewers). As a result, our final equation relating the number of viewers to the season number and

episode number is

$$\begin{aligned} \# \text{ Viewers} = & -1.019 * \text{Season} \# \\ & - .128 * \text{Episode} \# \\ & + 16.76 \end{aligned}$$

How Good Is This Regression?

Let's plot the regression line against the actual data and see how good it was. In actual fact, our regression is a two-dimensional plane relating viewers to season number and episode number within a season. However, three-dimensional plots tend not to show up well, so instead, I will plot viewers against episode number in the entire series. Note that because we opted to analyze the data starting with season 3, the plot starts with episode 49 in the series instead of 1.

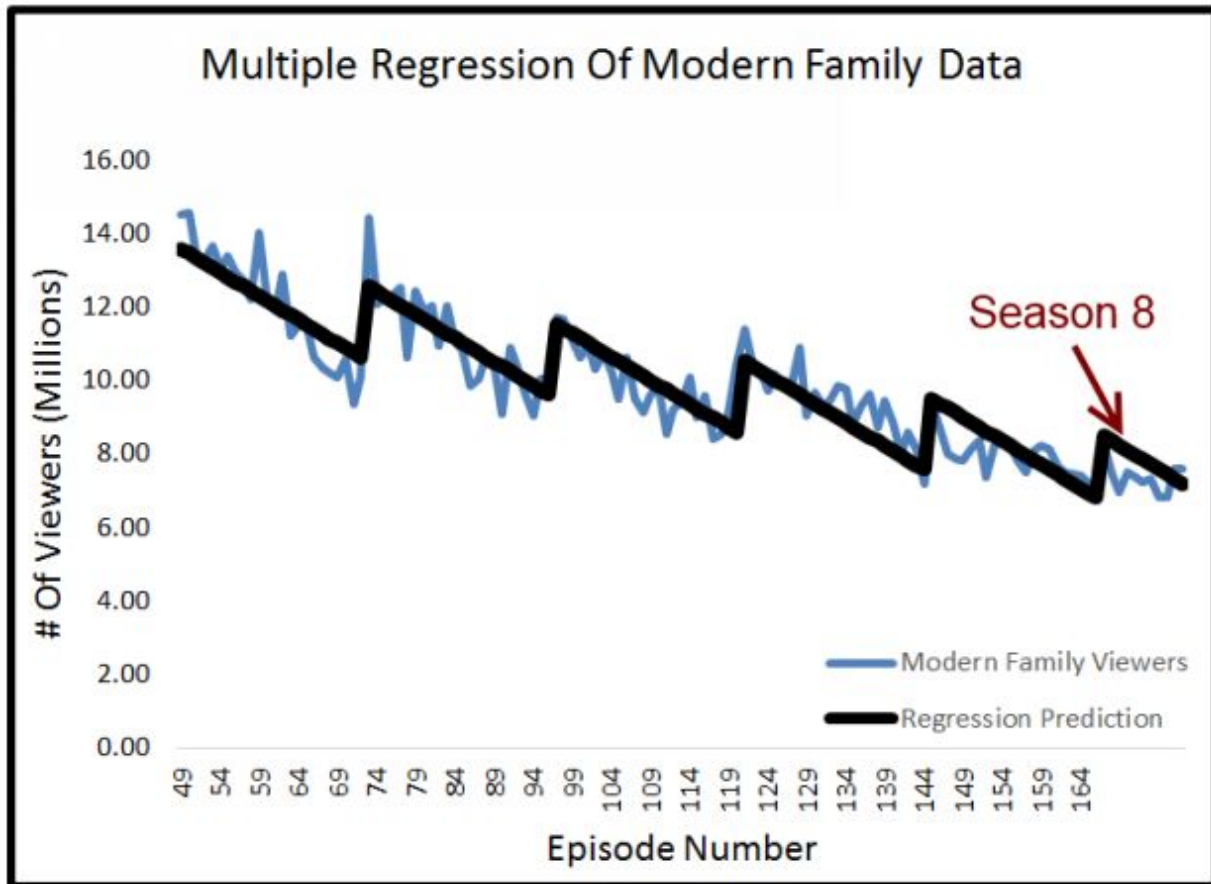


High level we see a reasonable plot for the regression. The saw tooth effect that we see in the regression line is an artifact of compressing the planar regression onto a single line. The step ups occur when we end each season and go to the next one. Basically, we are seeing that each new season has more viewers at the beginning than the previous season has at the end, but by the end of each season, there are fewer viewers at the end than there was at the end of the previous season.

Although the regression line captures the general trend of the data, we still see the same effect that we saw in the single regression, that there is episode to episode variation in the data that linear regression, even multiple linear regression does not capture.

In terms of extrapolating this regression into new data, here is a plot of the regression line against season 8 data, which was not used to generate the

regression



Season 8 appears to be running slightly under, but reasonably close to this regression.

R Squared

We can do an R^2 calculation to see what amount of the error our regression analysis accounted for relative to just using the mean value for number of viewers.

We calculate the R^2 as we did previously, by finding the regression sum squared error divided by the total sum squared error of the actual number of

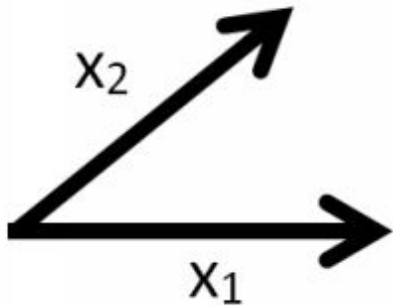
viewers compared to the average,

$$R^2 = 1 - \frac{SS_{Regression}}{SS_{Total}}$$

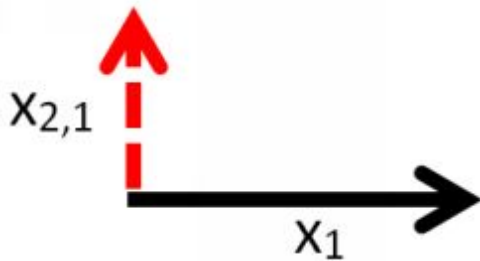
In this case, we get an R^2 value of .857. That means we have accounted for 85.7% of the total error that we would have gotten if we had just used the mean value. (We should note that this R^2 value isn't directly comparable to the .383 value we got when we did the simple linear regression on the modern family data, because when we did that analysis, we normalized the viewership by the episode 1 viewers in each season, effectively creating a different data set than we used here).

Going Back To The Coordinate System Analogy

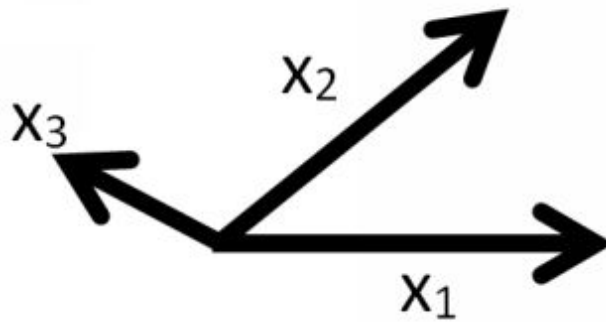
Previously we made an analogy of multiple regression to coordinate systems. With two independent variables what we did was similar to taking two vectors that could be pointed partially in the same direction



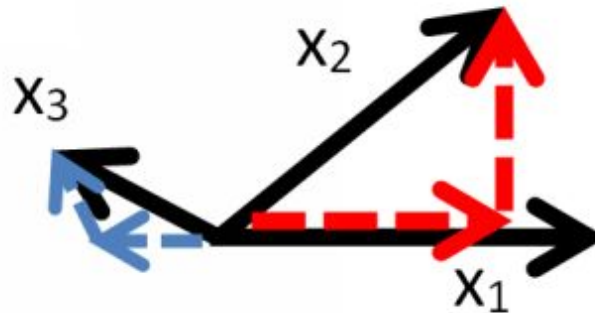
And turning them into two perpendicular vectors



The next section looks at multiple regression with three or more independent variables. So before diving into the equations, let's extend this analogy to three different vectors. The three vectors we have here are an x_1 , x_2 , and x_3 . In this example x_1 points right, x_2 points right and up, x_3 points left and up and out of the page.

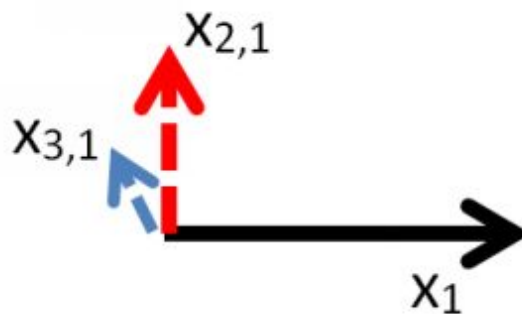


The first thing that we do is separately remove x_1 from x_2 and remove x_1 from x_3 . We do that the same as before, by turning x_2 into two vectors, one that is parallel to x_1 and one that is perpendicular to x_1 . We also turn x_3 into two vectors, one that is parallel to x_1 , and one that is perpendicular to x_1 . (Note, bear with this analogy, we aren't actually showing the math of how to do this calculation for vectors but we will show how to do it for the

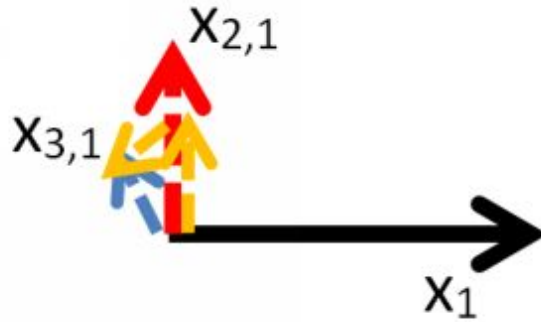


independent variables.)

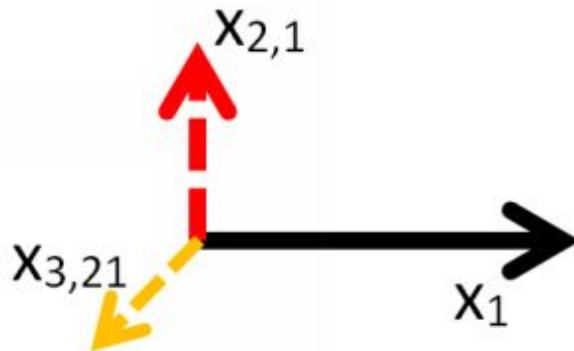
We discard the portions that were parallel to x_1 and only keep the residual.



The two residual vectors are now both perpendicular to x_1 , but they are not necessarily perpendicular to each other. So we have to go through the exercise again and turn $x_{3,1}$ into two new vectors, one that is parallel to $x_{2,1}$ and one that is perpendicular to $x_{2,1}$.



Then we throw away the parallel vector and are left with 3 vectors that are all perpendicular to each other.



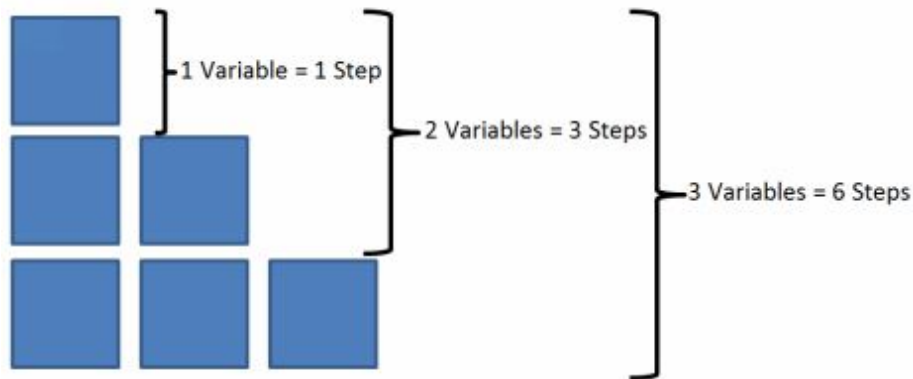
That was the process we would follow if we were dealing with vectors and coordinate systems. We will follow a very similar process with the independent variables. We will go through each independent variable in turn and remove the part that has any correlation with any of the remaining independent variables. When we are done all the residual vectors will have zero correlation to each of the other vectors. We will then use the regression slopes that we generated at each individual step, both between each independent variable and between them and the dependent variables and generate a regression equation for the overall data set.

3 Variable Multiple Regression As A Series Of Single Regressions

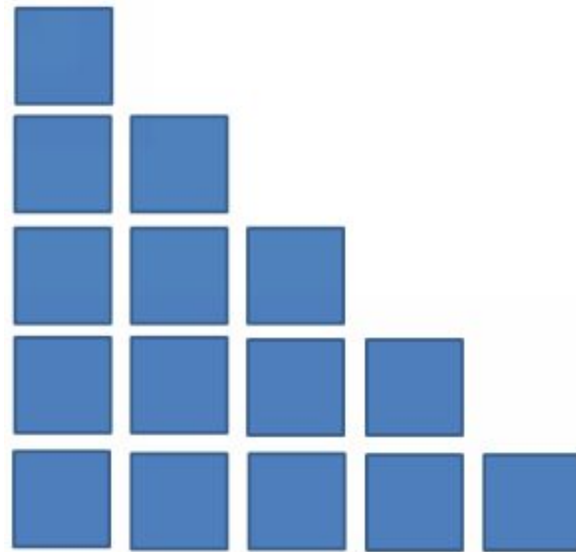
When we did the multiple regression as a series of single linear regressions, the regression with 2 independent variables used the same equations as the regression with 1 independent variable, except with more steps. Now with 3 independent variables, there are steps than the regression with 2 independent variables.

- With 1 independent variable, we solved the regression in 1 step
- With 2 variables we needed 3 steps
- With 3 variables we need 6 steps

This is shaped like a staircase



With 4 variables we would need 10 steps total, and with 5 we would need 15.



5 Independent Variables
Would Require 15 Steps

However, using 3 independent variables is sufficient to demonstrate the process without getting too tedious. That same process can be followed with additional variables, except it takes more bookkeeping of the equations.

With this method, the multiple regression is an order n squared $O(n^2)$ process. This is a computer science term that means that if we double the number of independent variables, we will multiply the required steps by approximately 4. That makes this exact process unsuitable for very large problems because the amount of work needed to solve the problem expands faster than the size of the input data. However, this process is suitable for small problems, and for understanding how multiple regression works, which is why we will continue with it.

With 3 independent variables, we will have 6 equations that we will need to keep track of, so it will be more bookkeeping than in the previous examples. The important thing to keep in mind is that we are sequentially removing the influence of each variable from all the remaining ones.

With 3 independent variables, we will start with an x_1 , x_2 , x_3 variables, and a y variable.

The first three steps will be to remove the x_1 variable from each of the other three variables, resulting in three residuals.

- Step 1 - remove x_1 from $y \rightarrow y_1$
- Step 2 - remove x_1 from $x_2 \rightarrow x_{2,1}$
- Step 3 - remove x_1 from $x_3 \rightarrow x_{3,1}$

The next two steps remove the influence of x_2 on the remaining variables.

Since the starting variable x_2 might have some x_1 in it, which we don't want, we do the removal of x_2 via the $x_{2,1}$ residual since this is x_2 with x_1 removed.

This results in two new residuals that have both x_1 and x_2 removed.

- Step 4 - remove $x_{2,1}$ from $y_1 \rightarrow y_{21}$
- Step 5 - remove $x_{2,1}$ from $x_{3,1} \rightarrow x_{3,21}$

The final step is to remove the influence of x_3 from the dependent variable.

This is done using $x_{3,21}$.

- Step 6 - remove $x_{3,21}$ from $y_{21} \rightarrow y_{3,21}$

Our objective with three independent variables is to get an equation relating each of them to the dependent variable y . That equation would have the form

$$y = b_1x_1 + b_2x_2 + b_3x_3 + a$$

After doing the 6 individual reductions, the equations that we have are

$$y = \lambda_1 x_1 + y_1$$

$$x_2 = \lambda_2 x_1 + x_{2,1}$$

$$x_3 = \lambda_3 x_1 + x_{3,1}$$

$$y_1 = \lambda_4 x_{2,1} + y_{21}$$

$$x_{3,1} = \lambda_5 x_{2,1} + x_{3,21}$$

$$y_{21} = \lambda_6 x_{3,21} + y_{321}$$

We need to rearrange and combine these six equations so that we are left with a single equation that only has the terms

- y = on the left side of the equation
- y_{321} because that is the final residual
- x_1 , x_2 , and x_3 , because those are the independent variables that we have.
- Any lambda is ok because those are constants that we have already solved for.

We need to get rid of all the intermediate x and y variables. Looking at those 6 equations below, I have highlighted the terms that we need to keep.

$$\boxed{y} = \lambda_1 \boxed{x_1} + y_1$$

$$\boxed{x_2} = \lambda_2 \boxed{x_1} + x_{2,1}$$

$$\boxed{x_3} = \lambda_3 \boxed{x_1} + x_{3,1}$$

$$y_1 = \lambda_4 x_{2,1} + y_{21}$$

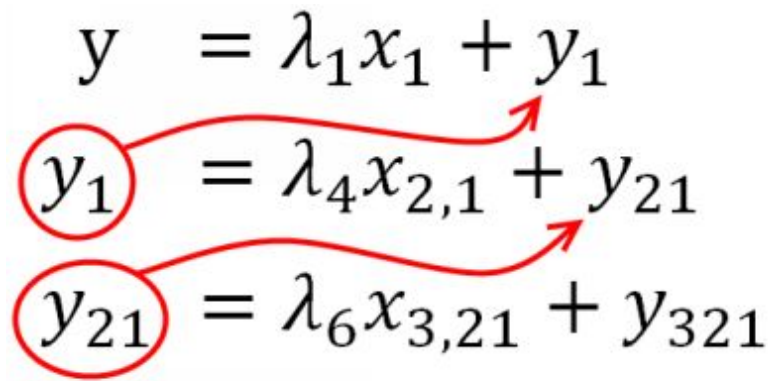
$$x_{3,1} = \lambda_5 x_{2,1} + x_{3,21}$$

$$y_{21} = \lambda_6 x_{3,21} + \boxed{y_{321}}$$

We are going to have to touch all 6 equations to solve this problem. In general, all that we will be doing is substituting less complicated variables in for more complicated ones and in general unwinding the problem. The initial thing that we see is that the first equation starts with a 'y =' in it. This is a good thing because that is what we want out of our final answer. So we use the first equation as our base and substitute into it.

There are many different paths we could take, and what I will show is only intended to be illustrative. You could do the substitutions in a different order.

The first thing I did was rearrange the equations so that all the equations with a 'y' on the left side were together. Then I substituted the y_1 in for the y_1 term on the right side of the first equation, and the y_{21} term in for the y_{21} term on the right side of a different equation, as shown below.

$$\begin{aligned}
 y &= \lambda_1 x_1 + y_1 \\
 \textcircled{y_1} &= \lambda_4 x_{2,1} + y_{21} \\
 \textcircled{y_{21}} &= \lambda_6 x_{3,21} + y_{321}
 \end{aligned}$$


$$x_{3,1} = \lambda_5 x_{2,1} + x_{3,21}$$

$$x_3 = \lambda_3 x_1 + x_{3,1}$$

$$x_2 = \lambda_2 x_1 + x_{2,1}$$

After we have done this, we have reduced the 6 equations down to 4 equations, so we have made progress. Those 4 equations are shown below.

$$y = \lambda_1 x_1 + \lambda_4 x_{2,1} + \lambda_6 x_{3,21} + y_{321}$$

$$x_{3,1} = \lambda_5 x_{2,1} + x_{3,21}$$

$$x_3 = \lambda_3 x_1 + x_{3,1}$$

$$x_2 = \lambda_2 x_1 + x_{2,1}$$

In the first equation, the x_1 and y_{321} terms can remain, since they are an independent variable and the final residual. However, the $x_{2,1}$ and $x_{3,21}$ terms need to be removed.

For the last 3 equations, we need to rearrange each of them in order to isolate a variable that we want to get rid of. Of those three, the first equation has two variables we need to get rid of, $x_{2,1}$ and $x_{3,21}$. Therefore, when we rearrange that equation to get rid of $x_{3,21}$, we use $x_{3,1}$, so we will need to get rid of that too. Fortunately, we can isolate one of those variables in each of the last 3 equations. If we subtract to get $x_{3,21}$ isolated in the first equation, $x_{3,1}$ isolated in the second equation, and $x_{2,1}$ isolated in the third equation, what we get is

$$y = \lambda_1 x_1 + \lambda_4 x_{2,1} + \lambda_6 x_{3,21} + y_{321}$$

$$x_{3,21} = x_{3,1} - \lambda_5 x_{2,1}$$

$$x_{3,1} = x_3 - \lambda_3 x_1$$

$$x_{2,1} = x_2 - \lambda_2 x_1$$

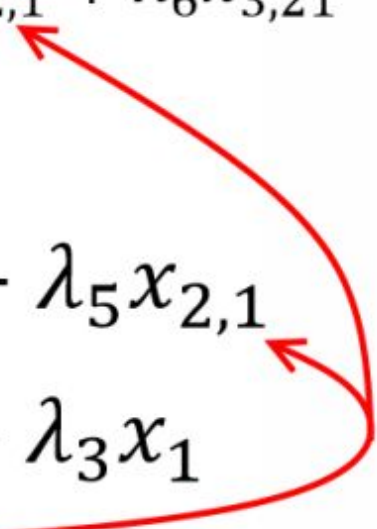
Now we just need to substitute in order to get rid of the variables that we don't need. The next substitution I will do is to get rid of $x_{2,1}$. There are two locations that the variable needs to be substituted in

$$y = \lambda_1 x_1 + \lambda_4 x_{2,1} + \lambda_6 x_{3,21} + y_{321}$$

$$x_{3,21} = x_{3,1} - \lambda_5 x_{2,1}$$

$$x_{3,1} = x_3 - \lambda_3 x_1$$

$$\textcircled{x_{2,1}} = x_2 - \lambda_2 x_1$$



That reduces the total number of equations by one. And substituting $x_{3,1}$ out of the next step will reduce it by another one.

$$y = \lambda_1 x_1 + \lambda_4 (x_2 - \lambda_2 x_1) + \lambda_6 x_{3,21} + y_{321}$$

$$x_{3,21} = x_{3,1} - \lambda_5 (x_2 - \lambda_2 x_1)$$

$$\textcircled{x_{3,1}} = x_3 - \lambda_3 x_1$$


The final substitution that we need to make is for $x_{3,21}$

$$y = \lambda_1 x_1 + \lambda_4 (x_2 - \lambda_2 x_1) + \lambda_6 x_{3,21} + y_{321}$$

$$\textcircled{x_{3,21}} = x_3 - \lambda_3 x_1 - \lambda_5 (x_2 - \lambda_2 x_1)$$


And the end result is this single equation

$$y = \lambda_1 x_1 + \lambda_4 (x_2 - \lambda_2 x_1) + \lambda_6 (x_3 - \lambda_3 x_1 - \lambda_5 (x_2 - \lambda_2 x_1)) + y_{321}$$

We are getting close to being done; this is nearly the equation that we need. We have a linear equation that relates our dependent variable y to the independent variables x_1, x_2, x_3 . We still need to resolve y_{321} into a constant intercept. First, however, we should clean up the equation above by grouping the constants (lambdas) based on what variable they are multiplied by. I.e. rearrange the equation so that there is only a single x_1 , a single x_2 , and a single x_3 each multiplied by some arrangement of lambdas.

When we multiply out all of the parentheses, what we get is

$$y = \lambda_1 x_1 + \lambda_4 x_2 - \lambda_4 \lambda_2 x_1 + \lambda_6 x_3 - \lambda_6 \lambda_3 x_1 - \lambda_6 \lambda_5 x_2 + \lambda_6 \lambda_5 \lambda_2 x_1 + y_{321}$$

And then when we group all of the coefficients for the respective x_1, x_2 , and x_3 terms together, what we get is

$$y = (\lambda_1 - \lambda_4 \lambda_2 - \lambda_6 \lambda_3 + \lambda_6 \lambda_5 \lambda_2) x_1 + (\lambda_4 - \lambda_6 \lambda_5) x_2 + \lambda_6 x_3 + y_{321}$$

Remember that our objective was to get an equation of this form

$$y = b_1x_1 + b_2x_2 + b_3x_3 + a$$

So if we match up the coefficients in front of the x_1 , x_2 , and x_3 terms for each variable, what we get is

$$b_1 = (\lambda_1 - \lambda_4\lambda_2 - \lambda_6\lambda_3 + \lambda_6\lambda_5\lambda_2)$$

$$b_2 = (\lambda_4 - \lambda_6\lambda_5)$$

$$b_3 = \lambda_6$$

An Example Of 3 Variable Regression

At this point, we have had several pages of processing the equations, so let's look at 3 variable regression for some example data. Here I've generated three short strings of random numbers, an x_1 , x_2 , and x_3 .

								Average	Std Dev
Random 0-> 20	x1	20	18	4	15	20	14	15.17	6.01
X1 * .5 + .5 * Rand(0-20)	x2	13	9	4	10	11	9	9.33	3.01
X1 * .333 + X2 * .333 + .333 * Rand(0-20)	x3	14	11	8	6	13	12	10.67	3.08
y = 2 * x1 + 3 * x2 - 5 * x3 + 7	y	16	15	-13	37	15	2	12.00	16.64

The x_1 is a random number between 0 and 20. The x_2 is partly a different random number between 0 and 20, and partly drawn from x_1 . The x_3 is partly a third different random number between 0 and 20, and partly drawn from x_1 and x_2 . Exactly how those strings of numbers were generated isn't all that significant, what is important is that the numbers have some correlation, but are not completely correlated.

The value of y is set as $y = 2 * x_1 + 3 * x_2 - 5 * x_3 + 7$, although you would not typically know this at the start of the problem. Our objective is going to be to derive those constants in order to recreate our 'y=' equation. The first step is to do a linear regression of the x_1 variable against each of the other 3 variables. This will result in our λ_1 , λ_2 , and λ_3 . As we saw during the section on single linear regression, those values are the correlation of the two strings of numbers, multiplied by the ratio of their standard deviations. Basically, those values are the slope values that we would have gotten if we were only doing simple linear regression instead of multiple linear regression.

When we remove the x_1 variable from the other variables, we get the lambda's, and we also get a set of residuals that we will use for the next step.

Random 0-> 20
 $X1 * .5 + .5 * \text{Rand}(0-20)$
 $X1 * .333 + X2 * .333 + .333 * \text{Rand}(0-20)$
 $y = 2 * x1 + 3 * x2 - 5 * x3 + 7$

							Average	Std Dev
x1	20	18	4	15	20	14	15.17	6.01
x2	13	9	4	10	11	9	9.33	3.01
x3	14	11	8	6	13	12	10.67	3.08
y	16	15	-13	37	15	2	12.00	16.64

		Correlation	Lamda		Residual Values						Average	Std Dev	
Remove x1		0.658	1.819	λ_1	y1	-20.39	-17.75	-20.28	9.71	-21.39	-23.47	-15.59	12.53
		0.935	0.468	λ_2	x2,1	3.64	0.57	2.13	2.98	1.64	2.45	2.23	1.07
		0.620	0.317	λ_3	x3,1	7.66	5.29	6.73	1.24	6.66	7.56	5.86	2.41

For the next step, we don't use the initial x_1 , x_2 , x_3 , y values at all. We just use the residual values that we created, the y_1 , $x_{2,1}$ and $x_{3,1}$ values. Using those, we remove the $x_{2,1}$ term from the other two values, and in doing so generate two more lambdas, and to more sets of residuals.

							Average	Std Dev	
Random 0-> 20	x1	20	18	4	15	20	14	15.17	6.01
X1 * .5 + .5 * Rand(0-20)	x2	13	9	4	10	11	9	9.33	3.01
X1 * .333 + X2 * .333 + .333 * Rand(0-20)	x3	14	11	8	6	13	12	10.67	3.08
y = 2 * x1 + 3 * x2 - 5 * x3 + 7	y	16	15	-13	37	15	2	12.00	16.64

Correlation		Lamda								Average	Std Dev	
Remove x1	0.658	1.819	λ1	y1	-20.39	-17.75	-20.28	9.71	-21.39	-23.47	-15.59	12.53
	0.935	0.468	λ2	x2,1	3.64	0.57	2.13	2.98	1.64	2.45	2.23	1.07
	0.620	0.317	λ3	x3,1	7.66	5.29	6.73	1.24	6.66	7.56	5.86	2.41

Correlation		Lamda								Average	Std Dev	
Remove x2,1	0.269	3.155	λ4	y21	-31.86	-19.55	-26.99	0.32	-26.55	-31.19	-22.64	12.07
	-0.014	-0.031	λ5	x3,21	7.77	5.31	6.80	1.34	6.71	7.64	5.93	2.41

Unsurprisingly, the next step will be to remove the $x_{3,21}$ residual values from the y_{21} residual values. Once we do that, we end up with the six lambdas'

that we need

Random 0-> 20										Average	Std Dev
X1 * .5 + .5 * Rand(0-20)	x1	20	18	4	15	20	14			15.17	6.01
X1 * .333 + X2 * .333 + .333 * Rand(0-20)	x2	13	9	4	10	11	9			9.33	3.01
	x3	14	11	8	6	13	12			10.67	3.08
y = 2 * x1 + 3 * x2 - 5 * x3 + 7	y	16	15	-13	37	15	2			12.00	16.64

Remove x1	Correlation	Lamda								Average	Std Dev	
	0.658	1.819	λ1	y1	-20.39	-17.75	-20.28	9.71	-21.39	-23.47	-15.59	12.53
	0.935	0.468	λ2	x2,1	3.64	0.57	2.13	2.98	1.64	2.45	2.23	1.07
	0.620	0.317	λ3	x3,1	7.66	5.29	6.73	1.24	6.66	7.56	5.86	2.41

Remove x2,1	Correlation	Lamda								Average	Std Dev	
	0.269	3.155	λ4	y21	-31.86	-19.55	-26.99	0.32	-26.55	-31.19	-22.64	12.07
	-0.014	-0.031	λ5	x3,21	7.77	5.31	6.80	1.34	6.71	7.64	5.93	2.41

Remove x3,21	Correlation	Lamda									
	-1.000	-5.000	λ6	y321	7	7	7	7	7	7	

Now that we have the 6 lambdas, i.e. the 6 slope relationships between individual variables, we can use the equations we derived earlier to get the global slope values.

$$b_1 = (\lambda_1 - \lambda_4 \lambda_2 - \lambda_6 \lambda_3 + \lambda_6 \lambda_5 \lambda_2)$$

$$b_2 = (\lambda_4 - \lambda_6 \lambda_5)$$

$$b_3 = \lambda_6$$

When we plug in these lambda values

	Correlation	Lamda	
Remove x1	0.658	1.819	λ_1
	0.935	0.468	λ_2
	0.620	0.317	λ_3

	Correlation	Lamda	
Remove x2,1	0.269	3.155	λ_4
	-0.014	-0.031	λ_5

	Correlation	Lamda	
Remove x3,21	-1.000	-5.000	λ_6

The b values that we get are

b1	2.0
b2	3.0
b3	-5.0

The result we get is a b_1 coefficient of 2, a b_2 coefficient of 3, and a b_3 coefficient of -5. Those are the values that we initially used when we generated our y values from our x values, which shows that we correctly solved the problem.

At this point, we have this solution for our regression

$$y = 2x_1 + 3x_2 - 5x_3 + a$$

All that we have left to do is solve for the intercept, a.

Solving For the Intercept

Solving for the intercept in multiple linear regression turns out to be nearly identical to what we did for simple linear regression. Since we know all the slopes, all we need to know is a single point that this regression hyperplane passes through. The only difference is that this point is defined by 4 coordinates (x_1 , x_2 , x_3 , y) instead of 2. Since we used the default regression equation at each step, the point that we pinned the hyperplane around was the mean value for each of the 4 variables. That means our equation is

$$\bar{y} = 2\bar{x}_1 + 3\bar{x}_2 - 5\bar{x}_3 + a$$

We know our average values from the initial data

								Average	Std Dev
Random 0-> 20	x1	20	18	4	15	20	14	15.17	6.01
X1 * .5 + .5 * Rand(0-20)	x2	13	9	4	10	11	9	9.33	3.01
X1 * .333 + X2 * .333 + .333 * Rand(0-20)	x3	14	11	8	6	13	12	10.67	3.08
y = 2 * x1 + 3 * x2 - 5 * x3 + 7	y	16	15	-13	37	15	2	12.00	16.64

Which means that our equation is

$$12.0 = 2 * 15.17 + 3 * 9.33 - 5 * 10.67 + a$$

Solving that equation for 'a' gives an intercept of 7, which matches the value we used when we generated the data.

The final result is that our regression equation for this data is

$$y = 2x_1 + 3x_2 - 5x_3 + 7$$

We have now successfully completed the multiple regression with 3 independent variables.

Multiple Regression With Even More Independent Variables We won't show an example with more than 3 independent variables using the series of single linear regression process because the process would be the same. The

only difference is that there will be an increasing number of steps with more variables and equations to keep track of.

The Same Example Using Moore-Penrose Pseudo-Inverse

Let's do the same example using the other process we know for multiple linear regression. Recall that what we are doing is solving for our intercept and coefficients in the [b] matrix by calculating the Pseudo-Inverse and multiplying it by the dependent variable matrix [y]. As an equation it is

$$[b] = [A]^{\dagger} [y]$$

The equation we use to calculate the Pseudo-Inverse is

$$[A]^{\dagger} = ([A]' [A])^{-1} [A]'$$

As a step by step process, it is

- Multiply A transpose by A
- Take the inverse of that
- Multiply that by A transpose

With this set of data

	Random 0-> 20					
	X1 * .5 + .5 * Rand(0-20)					
	X1 * .333 + X2 * .333 + .333 * Rand(0-20)					
	y = 2 * x1 + 3 * x2 -5*x3 + 7					
x1	20	18	4	15	20	14
x2	13	9	4	10	11	9
x3	14	11	8	6	13	12
y	16	15	-13	37	15	2

That makes our [A] matrix

A			
x1	x2	x3	a
20	13	14	1
18	9	11	1
4	4	8	1
15	10	6	1
20	11	13	1
14	9	12	1

Notice again the column of 1's that was included in the [A] matrix. For this example there is a non-zero intercept, so that column is required to get the same slopes and intercept as we used to generate the data. Without the column of 1's, we would be doing a least squares regression through the

y
16
15
-13
37
15
2

origin. The [y] matrix of the dependent data is

[A] transpose is

A'						
x1	20	18	4	15	20	14
x2	13	9	4	10	11	9
x3	14	11	8	6	13	12
a	1	1	1	1	1	1

[A] transpose is a 4 by 6 matrix, and [A] is a 6 by 4 matrix. When we multiply them we will get a 4 by 4 matrix. That result is

Transpose * Matrix

$$A' * A =$$

1561	934	1028	91
934	568	624	56
1028	624	730	64
91	56	64	6

The next step is to calculate the inverse of that matrix result. That inverse is

Inverse

$$(A' * A)^{-1}$$

0.048	-0.083	-0.011	0.167
-0.083	0.176	0.001	-0.398
-0.011	0.001	0.034	-0.203
0.167	-0.398	-0.203	3.519

When the inverse is multiplied by [A] transpose, we get the Pseudo-Inverse, which is a 4 by 6 matrix in this case.

Moore Penrose Pseudo Inverse

$$(A' * A)^{-1} * A'$$

-0.110	0.159	-0.063	-0.010	0.067	-0.043
0.249	-0.292	-0.018	0.126	-0.104	0.039
0.063	-0.021	0.030	-0.157	0.027	0.059
-1.164	0.705	0.967	0.822	-0.164	-0.167

The final step is to multiply that matrix by the [y] matrix of our dependent values. When we do that we get the regression slopes and intercepts.

Final Result

$$((A' * A)^{-1} * A') * y$$

2.00	b1
3.00	b2
-5.00	b3
7.00	a

The order that those values are in matches the order of the columns in the [A] matrix. I.e. b_1 is the first result because the first column of the [A] matrix was the coefficients in front of x_1 . The 'a' intercept is the last result because the 1's column was on the right side of the [A] matrix.

These are the same values that we got when we did this calculation as a series of single linear regressions. One difference, however, is that this Pseudo-Inverse process did not get substantially more difficult as we increased the number of independent variables, which makes it much more useful for large-scale problems than the sequence of single linear regressions.

Adjusted R^2

We started this book with R^2 , and we are going to end it with R^2 , specifically some tweaks to R^2 to make it more applicable to multiple regression. These tweaks generate something called “adjusted R^2 ”. The reason we have an adjusted R^2 is to help us know if we should or should not include additional independent variables in a regression.

Let’s say that we have 5 independent variables, x_1 , x_2 , x_3 , x_4 , and x_5 , as well as the dependent variable y . I might know that y is highly correlated to x_1 , x_2 , and x_3 , but am unsure if I should include x_4 or x_5 in the regression or not. After all, you don’t want to include extra independent variables that are not influencing since that can cause you to overfit your data.

If I just used R^2 as my metric for the quality of the regression fit, then I have a problem. Namely that adding more independent variables will never decrease R^2 . Even if the variables that I add are random noise, the basic R^2 will never decrease. As a result, it is difficult to use R^2 to spot overfitting.

Adjusted R^2 addresses this problem by penalizing the R^2 value for each additional independent variable used in the regression. The equation for adjusted R^2 is.

$$adj\ R^2 = R^2 - (1 - R^2) * \frac{k}{(n - k - 1)}$$

Where

- n is the number of data points
- k is the number of independent variables
- R^2 is the same R^2 that we have seen throughout the book

I have also seen the adjusted R^2 equation written as

$$adj R^2 = 1 - (1 - R^2) * \frac{(n - 1)}{(n - k - 1)}$$

Both of those equations give the same results, so take your pick on which one to use. Personally, I like this one

$$adj R^2 = R^2 - (1 - R^2) * \frac{k}{(n - k - 1)}$$

because it is obvious that you are starting with the traditional R^2 and subtracting away from it. To get R^2 we use the traditional equation we saw at the beginning.

$$R^2 = 1 - \frac{SS_{Regression}}{SS_{Total}}$$

And the variables n and k in the adjusted R^2 equation can just be counted.

So what is happening in this equation?

We start with R^2 and then subtract from it

$$adj R^2 = \overset{\text{Start With } R^2}{R^2} - \overset{\text{Then Subtract}}{(1 - R^2) * \frac{k}{(n - k - 1)}}$$

The more we subtract, the lower the resulting adjusted R^2 , and hence the worse the result is. The value we subtract is the product of two terms which

$$\overset{\text{Term 1}}{\left| (1 - R^2) \right|} * \overset{\text{Term 2}}{\left| \frac{k}{(n - k - 1)} \right|}$$

move in opposing directions

As you increase the number of independent variables, theoretically R^2 goes up (it can't go down but it could be unchanged) which decreases the first term. However, as k increases the numerator on the second term gets bigger AND the denominator gets smaller. So the second term increases as the number of independent variables go up.

Which Effect Is Larger?

Well, that depends on your data. If the independent variable that you added improved R^2 , then you could see an increase in your adjusted R^2 . If it didn't have much of an impact, then adding an additional variable could decrease the adjusted R^2 .

The denominator on the second term has some interesting properties as well

$$\frac{k}{\boxed{(n - k - 1)}}$$

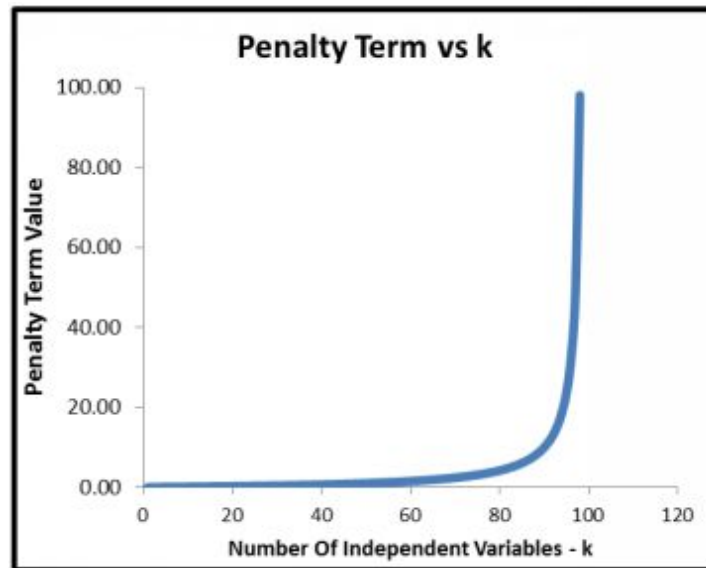
The n term is the number of data points. That shows us that the number of data points compared to the number of independent variables is important. The reason is that as the number of independent variables approaches the number of data points, it is very easy to overfit. As a result, the adjusted R^2 starts heavily penalizing as k approaches n .

Let's say that we have 100 data points. As we increase the number of independent variables from 1 to 98, this part of the penalty term in the

$$\frac{k}{(n - k - 1)}$$

adjusted R^2 equation

has these values



Obviously, this goes asymptotic as the number of independent variables approaches 100, which is the number of data points. If you had 99 independent variables, the resulting penalty term is undefined.

Interestingly, if you have more independent variables than number of data points, then this part of the equation turns negative

$$\frac{k}{(n - k - 1)}$$

This would make adjusted R^2 greater than R^2 , which is not good. You should not have more independent variables than the number of data points. In fact, a good rule of thumb is to have at least 10 times more data points than the number of independent variables.

Adjusted R^2 Conclusion The end result is that you can use the adjusted R^2 equation to determine if you should or shouldn't include certain independent variables in the regression equation. Run the regression both ways, and see which result gives the higher adjusted R^2 .

If You Found Errors Or Omissions

We put some effort into trying to make this book as bug-free as possible, and including what we thought was the most important information. However, if you have found some errors or significant omissions that we should address please email us here

ImFairlyNerdy (at) gmail (dot) com

And let us know. If you do, then let us know if you would like free copies of our future books. Also, a big thank you!

More Books

If you liked this book, you may be interested in checking out some of my other books such as

- [Bayes Theorem Examples](#) – Which walks through how to update your probability estimates as you get new information about things. It gives half a dozen easy to understand examples on how to use Bayes Theorem
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- [Hypothesis Testing: A Visual Introduction To Statistical Significance](#) – Which demonstrates how to tell the difference between events that have occurred by random chance, and outcomes that are driven by an outside event. This book contains examples of all the major types of statistical significance tests, including the Z test and the 5 different variations of a T-test.

Thank You

Before you go, I'd like to say thank you for purchasing my eBook. I know you have a lot of options online to learn this kind of information. So a big thank you for downloading this book and reading all the way to the end.

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But it's often better to have one-on-one conversations. So I encourage you to reach out over email with any questions you have or just to say hi!

Simply write here:

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~ Scott Hartshorn